Configuration Summary

```
FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-09-24 11:14:07
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
```

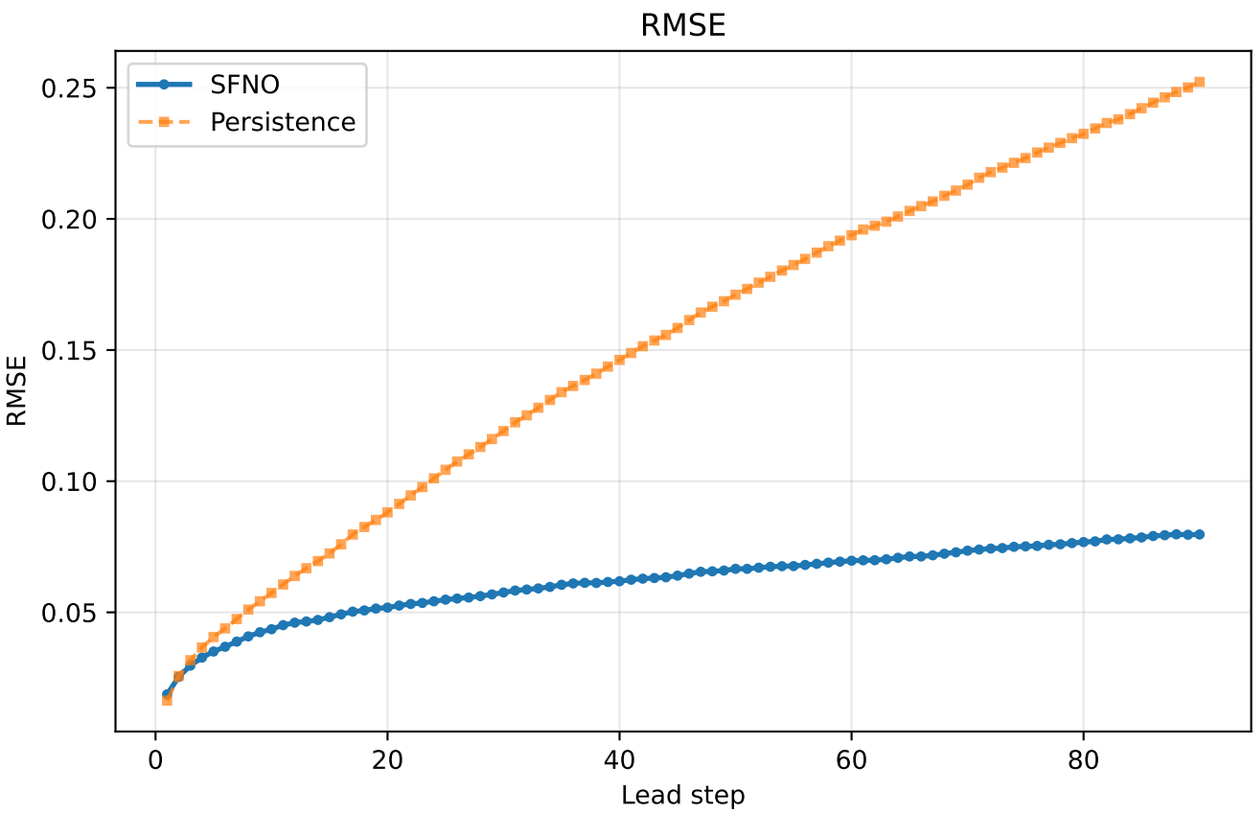
```
norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 0.0001
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
```

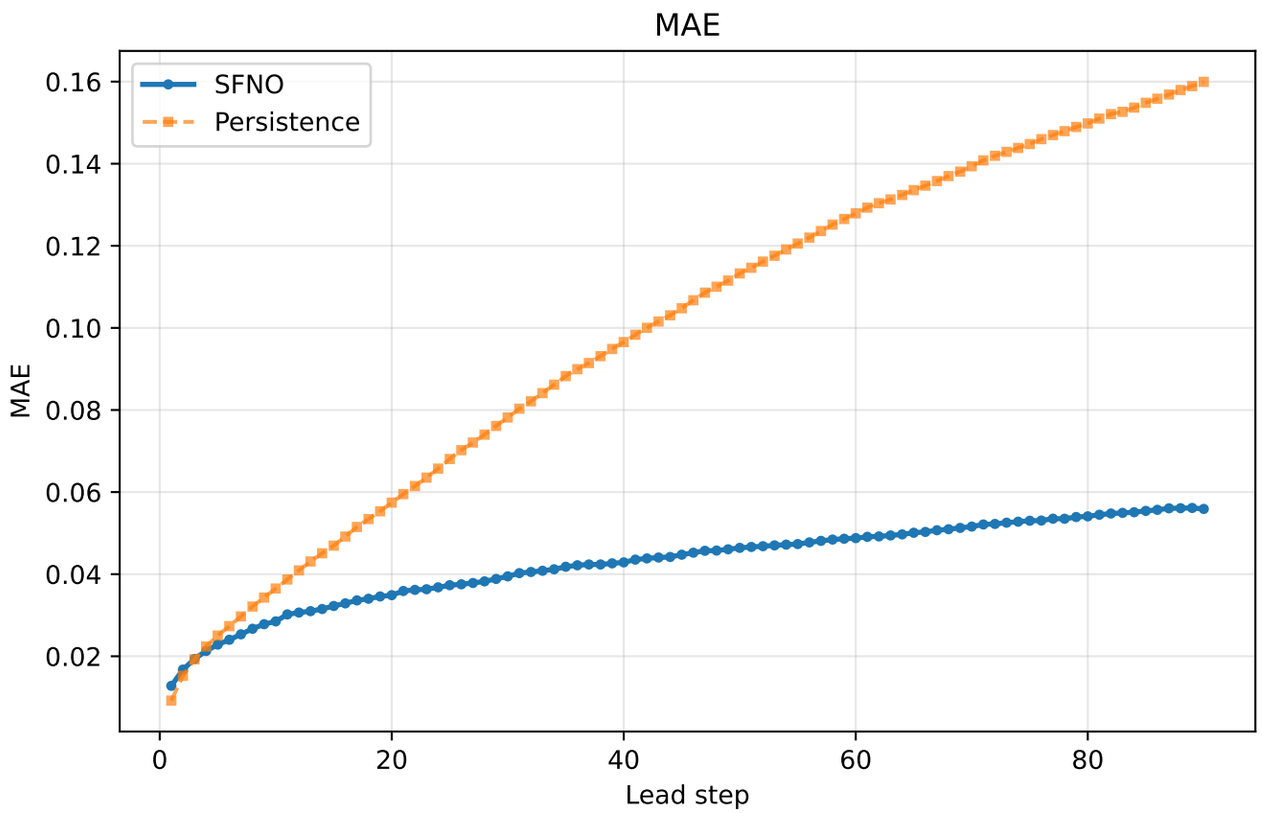
```
val_predictions_path..................... results_64x128/predictions_val.nc
test_predictions_path.................... results_64x128/predictions_test.nc
processed_data_path...................... data/processed_64x128/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed_64x128/std_params.nc
trend_params_path........................ data/processed_64x128/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
compute_dhw.............................. True
dhw_baseline_years....................... (1985, 2012)
dhw_output_val........................... dhw_val.nc
dhw_output_test.......................... dhw_test.nc
output_dir............................... results_64x128/
report_file.............................. results_64x128/analysis_report.txt
metrics_csv.............................. results_64x128/metrics_by_lead_step.csv
figures_dir.............................. results_64x128/figures/
analysis_pdf............................. results_64x128/forecast_analysis.pdf
generate_plots........................... True
```

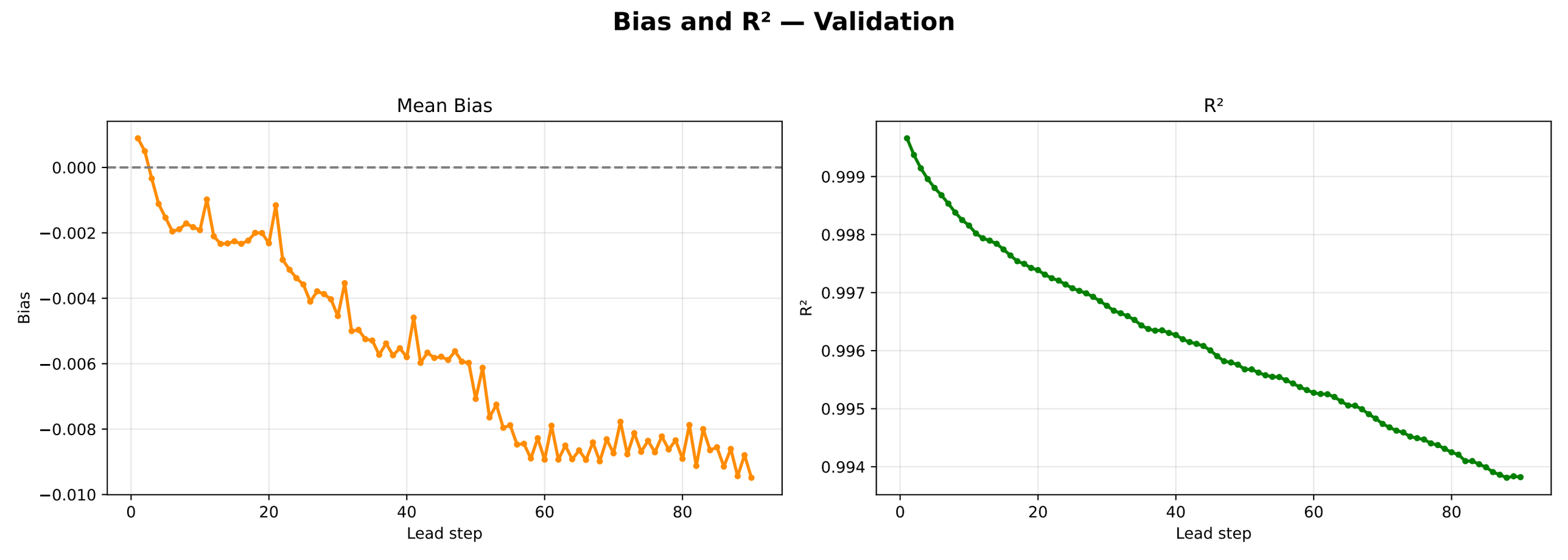
Metrics by Lead Step — Validation

Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0187	0.0164	-14.6%	0.0128	0.0009	0.9506	0.9592	0.9997
2	0.0254	0.0257	1.2%	0.0168	0.0005	0.9032	0.8991	0.9994
3	0.0297	0.0318	6.5%	0.0194	-0.0003	0.8622	0.8481	0.9991
4	0.0327	0.0366	10.5%	0.0213	-0.0011	0.8290	0.8008	0.9990
5	0.0351	0.0406	13.6%	0.0228	-0.0015	0.7989	0.7574	0.9988
6	0.0369	0.0439	15.9%	0.0240	-0.0020	0.7741	0.7202	0.9987
7	0.0388	0.0474	18.1%	0.0253	-0.0019	0.7491	0.6815	0.9985
8	0.0408	0.0510	20.0%	0.0267	-0.0017	0.7233	0.6425	0.9984
9	0.0424	0.0542	21.8%	0.0278	-0.0018	0.6990	0.6072	0.9983
10	0.0436	0.0574	24.1%	0.0285	-0.0019	0.6787	0.5705	0.9982
11	0.0451	0.0606	25.6%	0.0302	-0.0010	0.6537	0.5345	0.9980
12	0.0461	0.0639	27.9%	0.0307	-0.0021	0.6341	0.4984	0.9979
13	0.0465	0.0668	30.4%	0.0310	-0.0023	0.6176	0.4642	0.9979
14	0.0471	0.0695	32.3%	0.0315	-0.0023	0.6008	0.4363	0.9978
15	0.0482	0.0725	33.5%	0.0322	-0.0023	0.5828	0.4112	0.9977
16	0.0492	0.0759	35.1%	0.0329	-0.0023	0.5650	0.3812	0.9976
17	0.0503	0.0797	36.9%	0.0336	-0.0022	0.5478	0.3482	0.9975
18	0.0507	0.0825	38.5%	0.0340	-0.0020	0.5302	0.3242	0.9975
19	0.0514	0.0852	39.6%	0.0346	-0.0020	0.5108	0.3058	0.9974
20	0.0518	0.0881	41.2%	0.0349	-0.0023	0.4933	0.2821	0.9974
21	0.0526	0.0913	42.4%	0.0359	-0.0012	0.4781	0.2589	0.9973
22	0.0532	0.0946	43.8%	0.0362	-0.0028	0.4661	0.2377	0.9972
23	0.0536	0.0978	45.2%	0.0363	-0.0031	0.4568	0.2157	0.9972
24	0.0542	0.1011	46.4%	0.0368	-0.0034	0.4391	0.1932	0.9971
25	0.0548	0.1043	47.5%	0.0373	-0.0036	0.4231	0.1725	0.9971
26	0.0553	0.1075	48.6%	0.0375	-0.0041	0.4087	0.1551	0.9970
27	0.0556	0.1103	49.5%	0.0379	-0.0038	0.3932	0.1418	0.9970
28	0.0562	0.1130	50.3%	0.0383	-0.0039	0.3769	0.1313	0.9969
29	0.0568	0.1161	51.0%	0.0388	-0.0040	0.3571	0.1164	0.9969
30	0.0576	0.1191	51.7%	0.0395	-0.0045	0.3386	0.1038	0.9968
31	0.0583	0.1225	52.4%	0.0403	-0.0035	0.3239	0.0889	0.9967
32	0.0587	0.1251	53.1%	0.0405	-0.0050	0.3118	0.0821	0.9966
33	0.0591	0.1280	53.8%	0.0408	-0.0050	0.3000	0.0694	0.9966
34	0.0597	0.1310	54.4%	0.0412	-0.0053	0.2898	0.0582	0.9965
35	0.0605	0.1339	54.8%	0.0418	-0.0053	0.2788	0.0464	0.9964
36	0.0610	0.1363	55.2%	0.0422	-0.0057	0.2643	0.0400	0.9964
37	0.0612	0.1386	55.8%	0.0424	-0.0054	0.2542	0.0378	0.9963
38	0.0612	0.1410	56.6%	0.0424	-0.0057	0.2439	0.0334	0.9964
39	0.0616	0.1437	57.2%	0.0426	-0.0055	0.2315	0.0265	0.9963
40	0.0618	0.1462	57.7%	0.0429	-0.0058	0.2183	0.0227	0.9963
41	0.0624	0.1489	58.1%	0.0435	-0.0046	0.2043	0.0178	0.9962
42	0.0628	0.1514	58.5%	0.0438	-0.0060	0.1967	0.0118	0.9961
43	0.0631	0.1536	58.9%	0.0441	-0.0057	0.1867	0.0084	0.9961
44	0.0634	0.1558	59.3%	0.0442	-0.0058	0.1759	0.0082	0.9961
45	0.0640	0.1584	59.6%	0.0447	-0.0058	0.1642	-0.0016	0.9960
46	0.0648	0.1614	59.9%	0.0452	-0.0059	0.1531	-0.0150	0.9959
47	0.0655	0.1643	60.2%	0.0457	-0.0056	0.1425	-0.0262	0.9958
48	0.0656	0.1665	60.6%	0.0458	-0.0059	0.1381	-0.0296	0.9958
49	0.0659	0.1686	60.9%	0.0461	-0.0060	0.1303	-0.0316	0.9958
50	0.0666	0.1711	61.1%	0.0464	-0.0071	0.1221	-0.0359	0.9957
51	0.0666	0.1733	61.6%	0.0466	-0.0061	0.1148	-0.0392	0.9957
52	0.0670	0.1757	61.9%	0.0468	-0.0076	0.1042	-0.0454	0.9956
53	0.0674	0.1779	62.1%	0.0470	-0.0073	0.0972	-0.0488	0.9956
54	0.0676	0.1803	62.5%	0.0472	-0.0080	0.0968	-0.0525	0.9955
55	0.0676	0.1824	62.9%	0.0473	-0.0079	0.0975	-0.0540	0.9955
56	0.0681	0.1847	63.2%	0.0477	-0.0085	0.0911	-0.0585	0.9955
57	0.0685	0.1871	63.4%	0.0481	-0.0084	0.0885	-0.0631	0.9954
58	0.0689	0.1896	63.6%	0.0484	-0.0089	0.0826	-0.0696	0.9954
59	0.0693	0.1917	63.8%	0.0486	-0.0083	0.0825	-0.0708	0.9953
60	0.0697	0.1938	64.0%	0.0488	-0.0089	0.0823	-0.0716	0.9953
61	0.0698	0.1959	64.4%	0.0491	-0.0079	0.0843	-0.0755	0.9953
62	0.0699	0.1974	64.6%	0.0492	-0.0089	0.0797	-0.0678	0.9952
63	0.0702	0.1989	64.7%	0.0494	-0.0085	0.0798	-0.0586	0.9952
64	0.0708	0.2009	64.8%	0.0497	-0.0089	0.0689	-0.0567	0.9951
65	0.0713	0.2030	64.9%	0.0501	-0.0086	0.0635	-0.0588	0.9951
66	0.0713	0.2049	65.2%	0.0503	-0.0089	0.0574	-0.0583	0.9951
67	0.0717	0.2067	65.3%	0.0507	-0.0084	0.0532	-0.0556	0.9950
68	0.0724	0.2088	65.4%	0.0509	-0.0090	0.0457	-0.0542	0.9949
69	0.0729	0.2108	65.4%	0.0513	-0.0083	0.0433	-0.0508	0.9948
70	0.0735	0.2131	65.5%	0.0516	-0.0087	0.0361	-0.0529	0.9947
71	0.0739	0.2157	65.7%	0.0521	-0.0078	0.0342	-0.0598	0.9947
72	0.0743	0.2178	65.9%	0.0522	-0.0088	0.0264	-0.0609	0.9946
73	0.0745	0.2196	66.1%	0.0525	-0.0081	0.0303	-0.0564	0.9946
74	0.0750	0.2214	66.1%	0.0528	-0.0087	0.0245	-0.0508	0.9945
75	0.0752	0.2232	66.3%	0.0530	-0.0084	0.0235	-0.0483	0.9945
76	0.0753	0.2253	66.6%	0.0531	-0.0087	0.0190	-0.0507	0.9945
77	0.0758	0.2272	66.7%	0.0535	-0.0082	0.0150	-0.0482	0.9944
78	0.0759	0.2290	66.8%	0.0535	-0.0086	0.0081	-0.0477	0.9944
79	0.0764	0.2308	66.9%	0.0539	-0.0083	0.0039	-0.0476	0.9943
80	0.0768	0.2324	67.0%	0.0541	-0.0089	-0.0012	-0.0445	0.9942
81	0.0771	0.2345	67.1%	0.0545	-0.0079	0.0015	-0.0448	0.9942
82	0.0778	0.2366	67.1%	0.0548	-0.0091	-0.0063	-0.0443	0.9941
83	0.0778	0.2380	67.3%	0.0549	-0.0080	-0.0065	-0.0426	0.9941
84	0.0782	0.2399	67.4%	0.0551	-0.0086	-0.0128	-0.0439	0.9940
85	0.0785	0.2422	67.6%	0.0554	-0.0086	-0.0086	-0.0456	0.9940
86	0.0791	0.2443	67.6%	0.0557	-0.0091	-0.0124	-0.0445	0.9939
87	0.0794	0.2464	67.8%	0.0560	-0.0086	-0.0103	-0.0446	0.9939
88	0.0797	0.2484	67.9%	0.0561	-0.0094	-0.0141	-0.0447	0.9938
89	0.0796	0.2501	68.2%	0.0561	-0.0088	-0.0110	-0.0431	0.9938
90	0.0797	0.2522	68.4%	0.0559	-0.0095	-0.0090	-0.0441	0.9938

Forecast Skill vs Lead Step — Validation

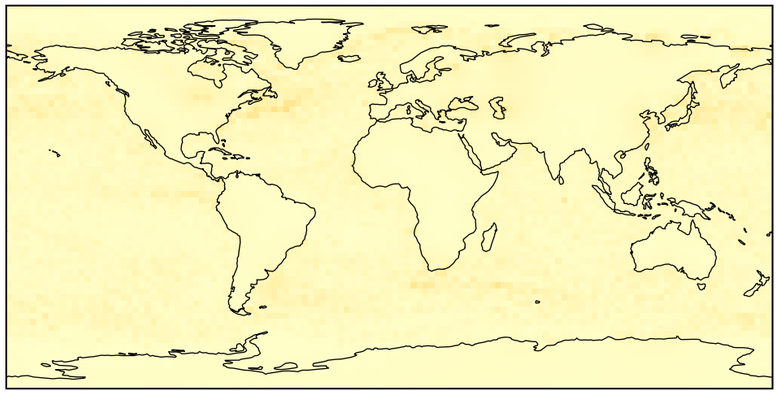




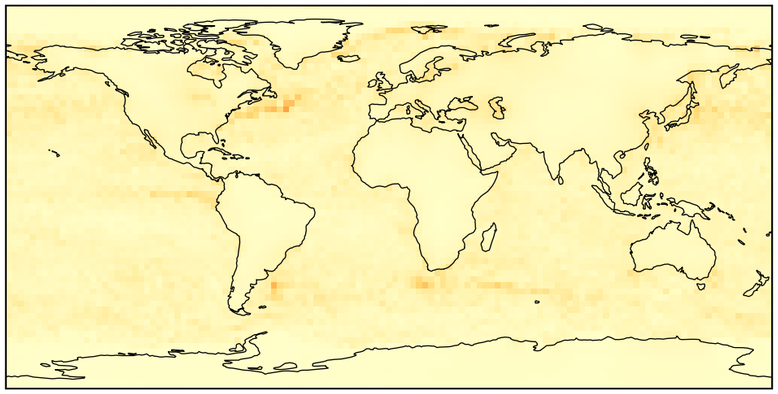

Bias and R² — Validation
Mean Bias
Bias
0.000
−0.002
−0.004
−0.006
−0.008
−0.010
0
20
40
60
80
Lead step
R²
R²
0.999
0.998
0.997
0.996
0.995
0.994
0
20
40
60
80
Lead step

Spatial RMSE (all grid) — Validation

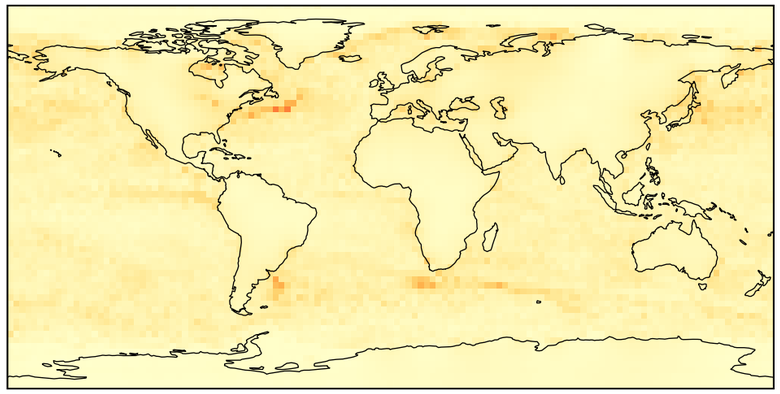
Step 1



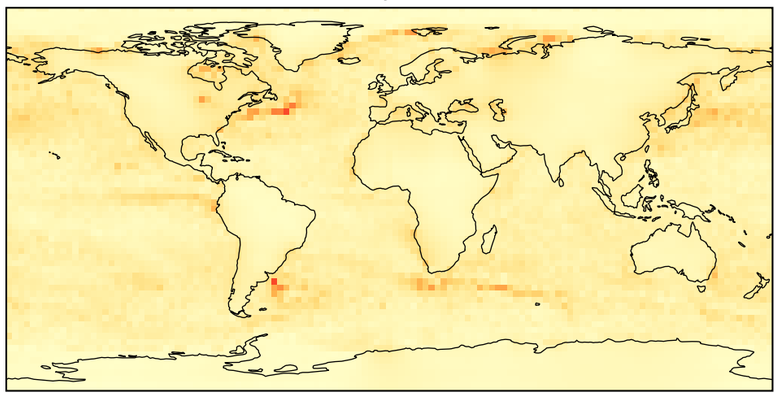
Step 3



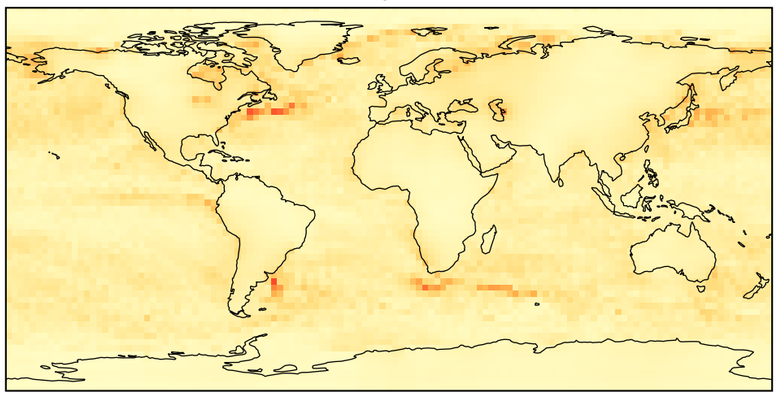
Step 7



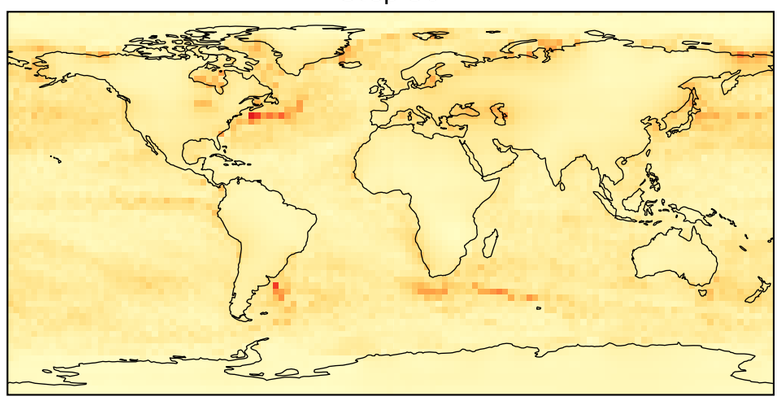
Step 14



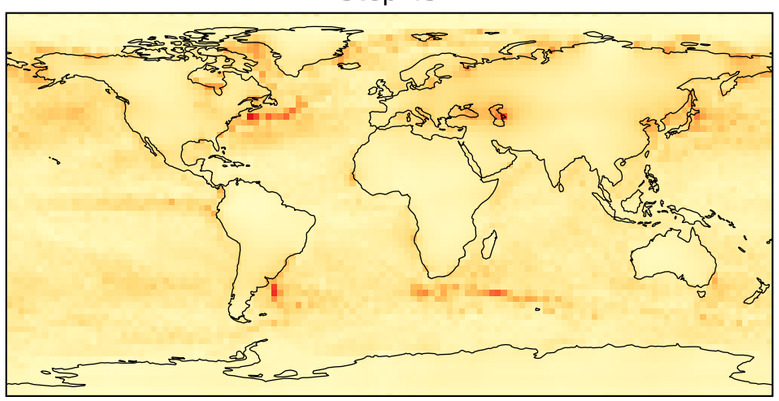
Step 21



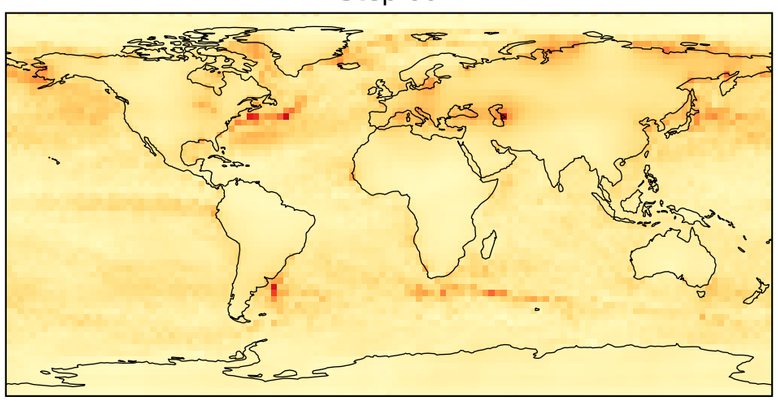
Step 30



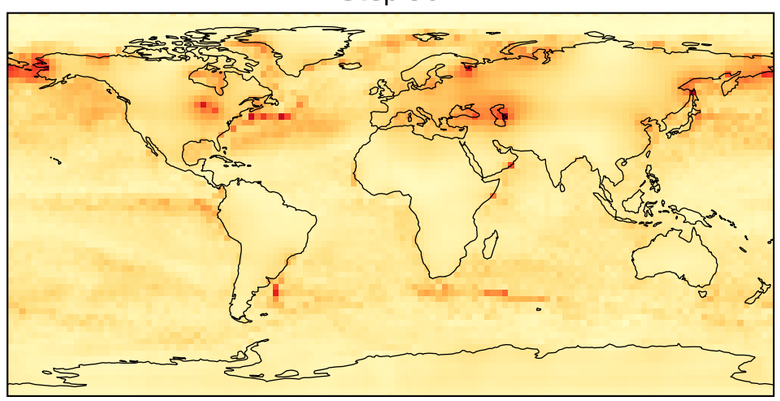
Step 45

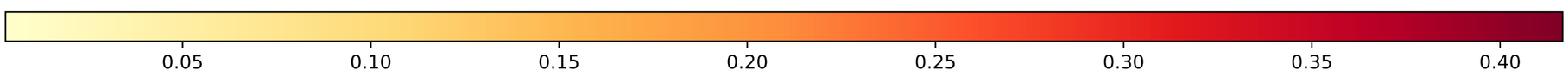


Step 60



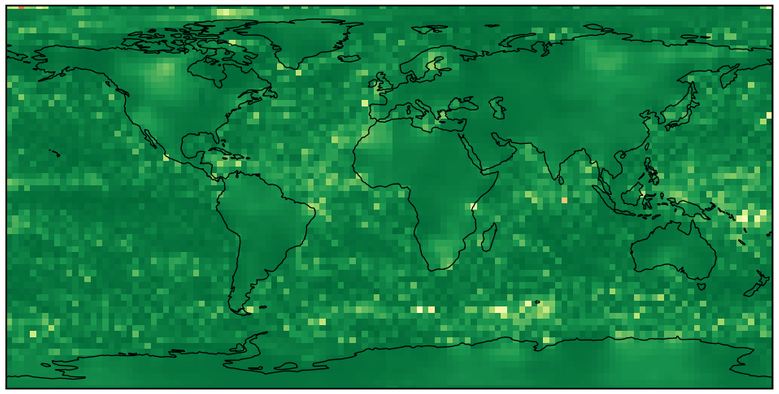
Step 90



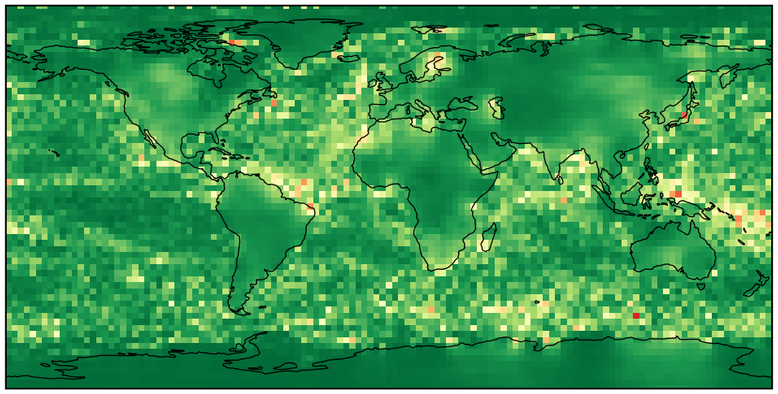


Spatial ACC (all grid) — Validation

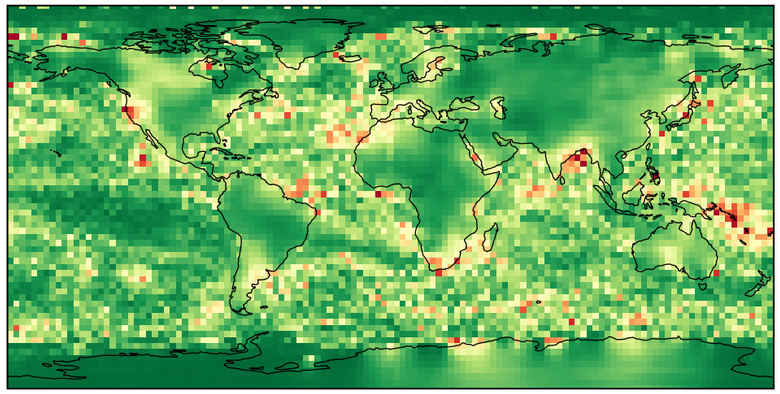
Step 1



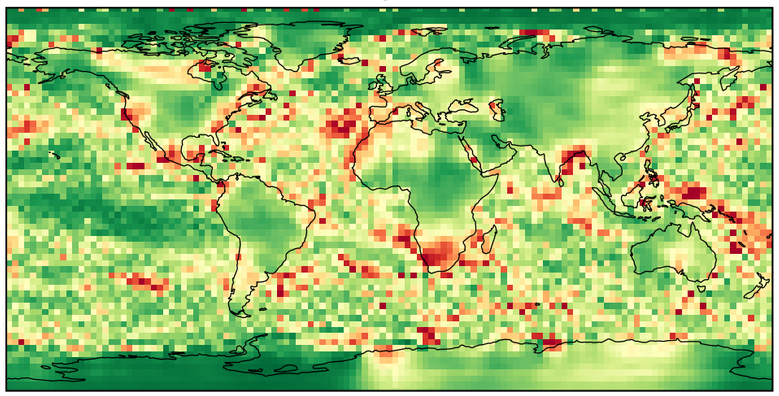
Step 3



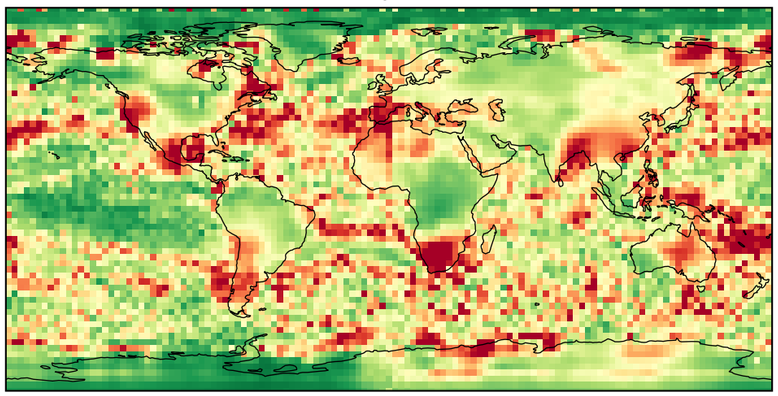
Step 7



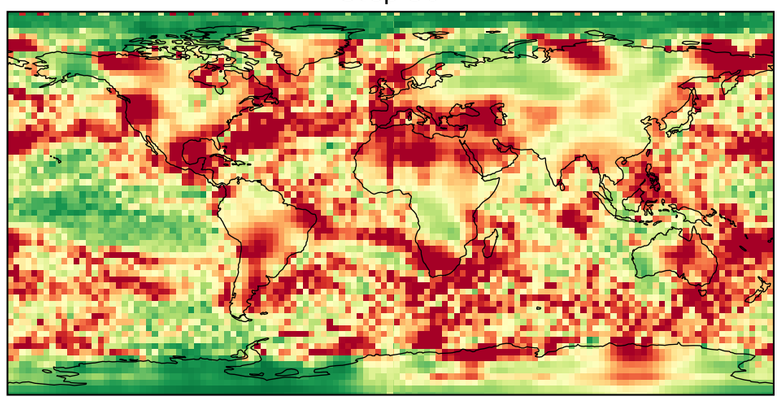
Step 14



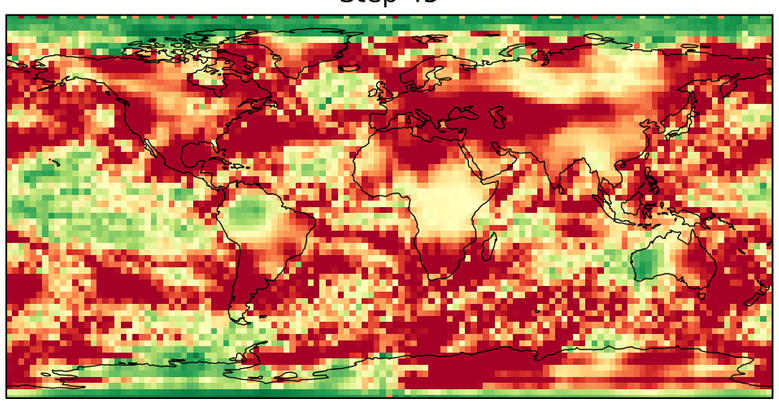
Step 21



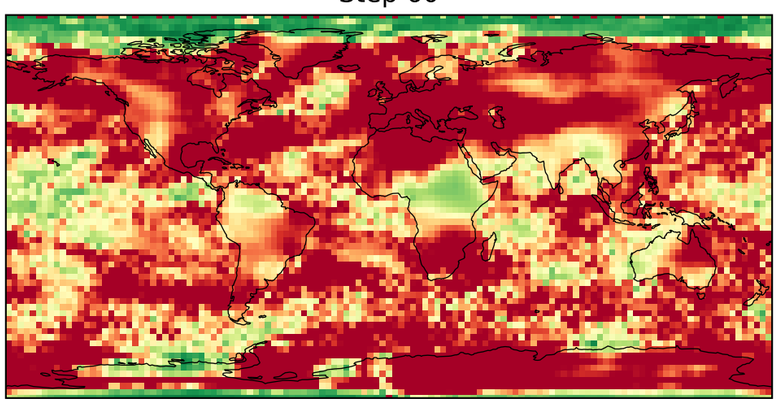
Step 30



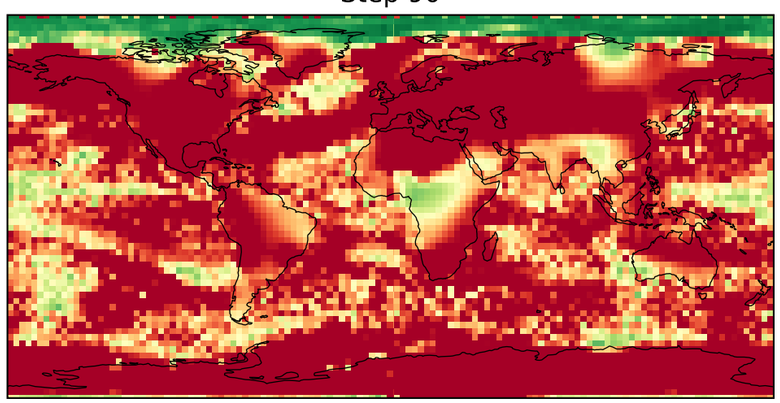
Step 45

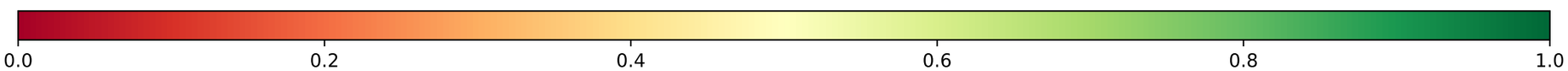


Step 60



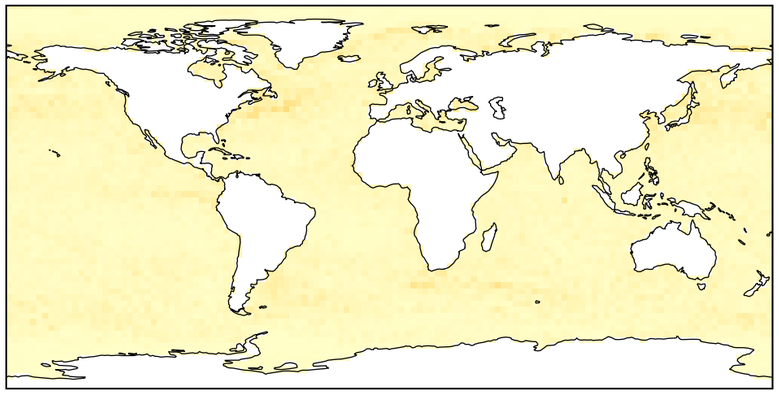
Step 90



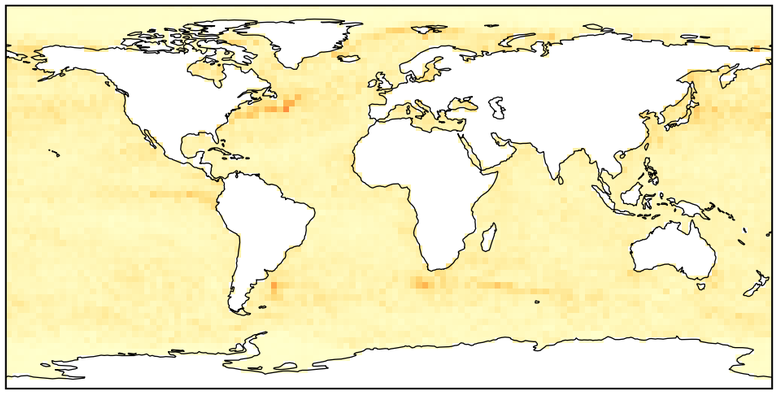


Spatial RMSE (ocean only) — Validation

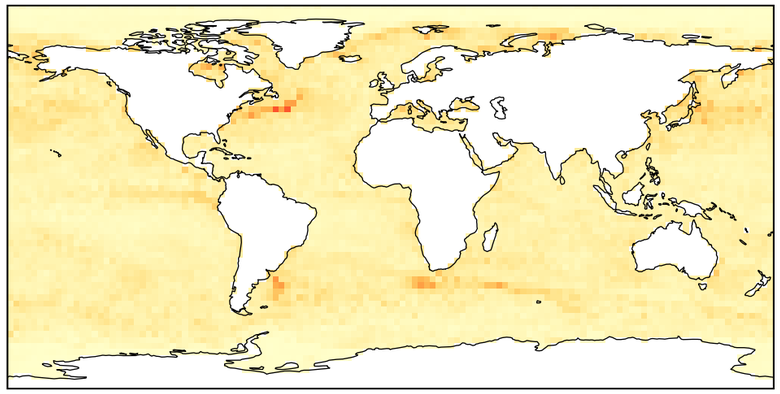
Step 1



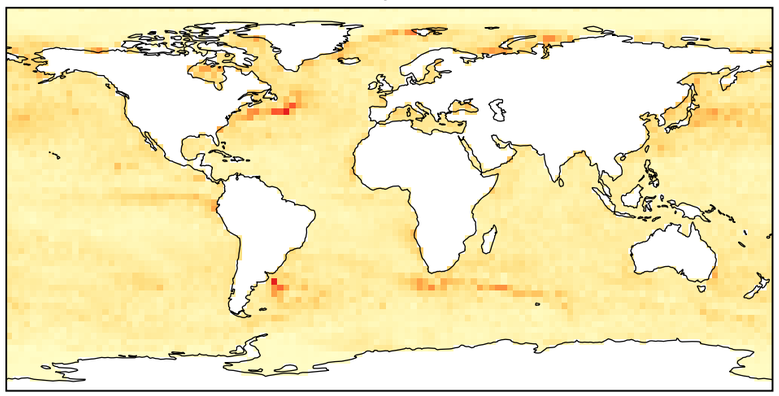
Step 3



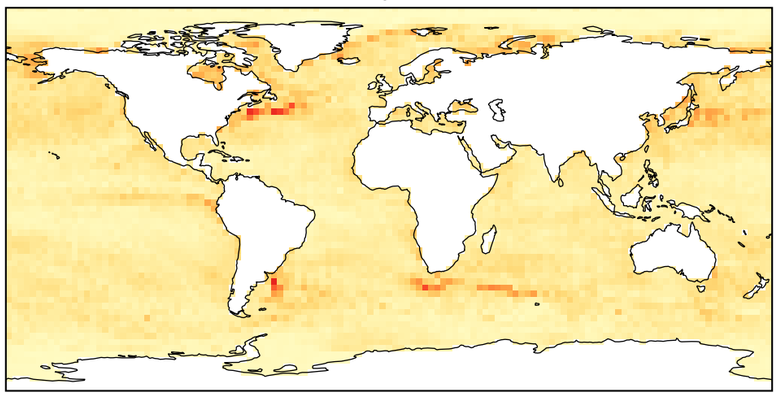
Step 7



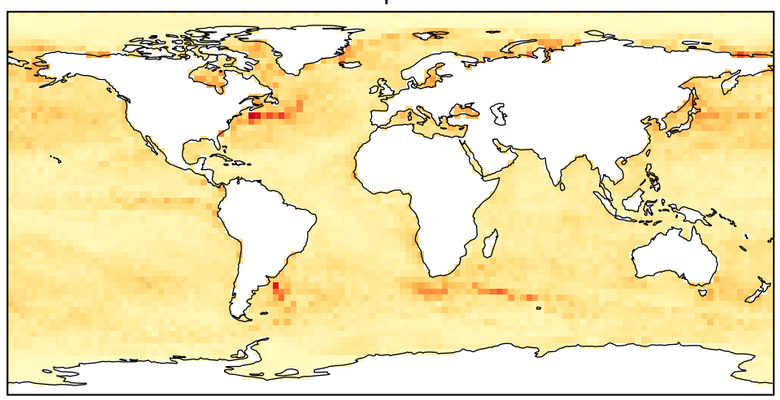
Step 14



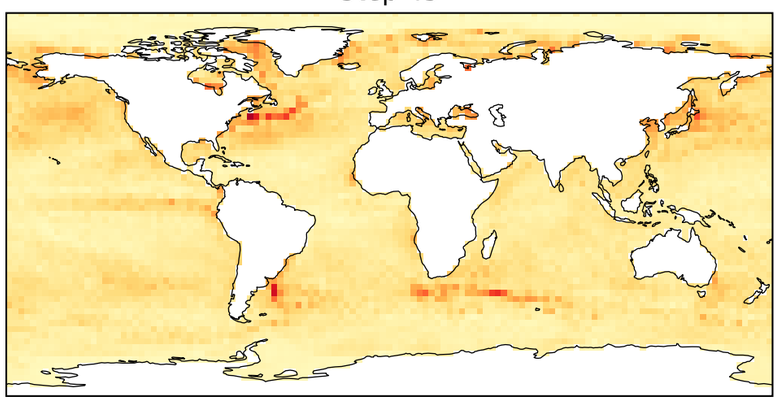
Step 21



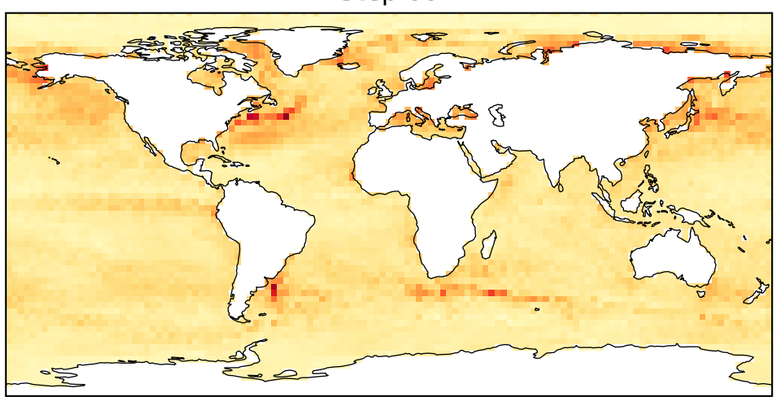
Step 30



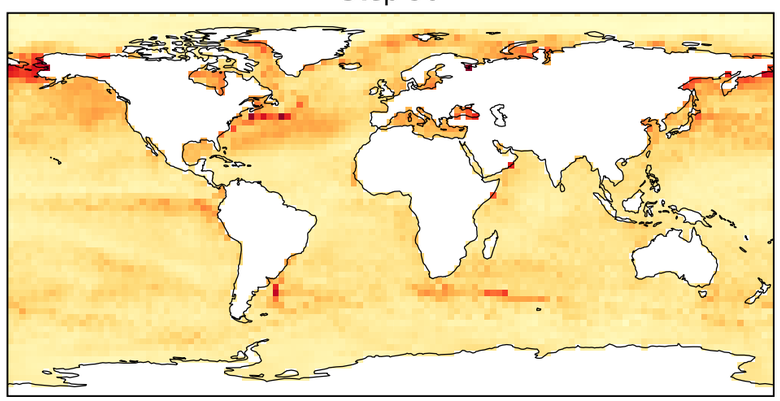
Step 45



Step 60

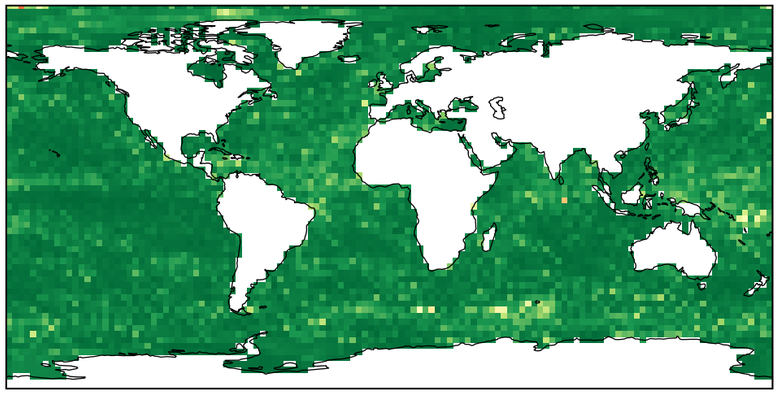


Step 90

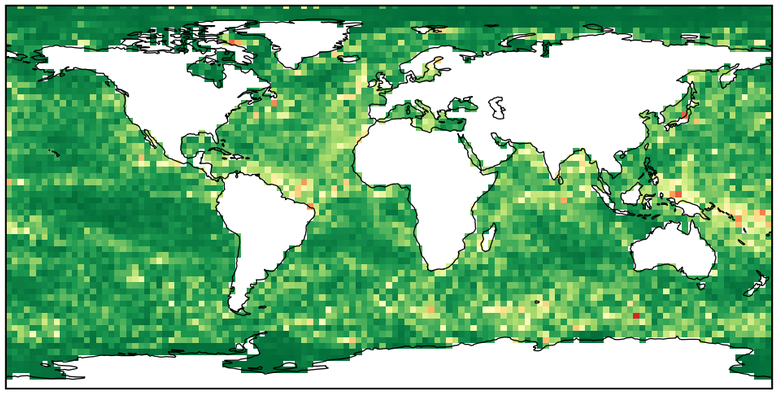


Spatial ACC (ocean only) — Validation

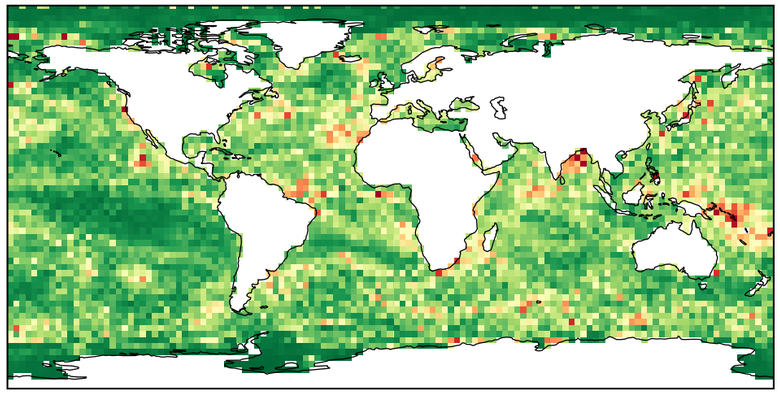
Step 1



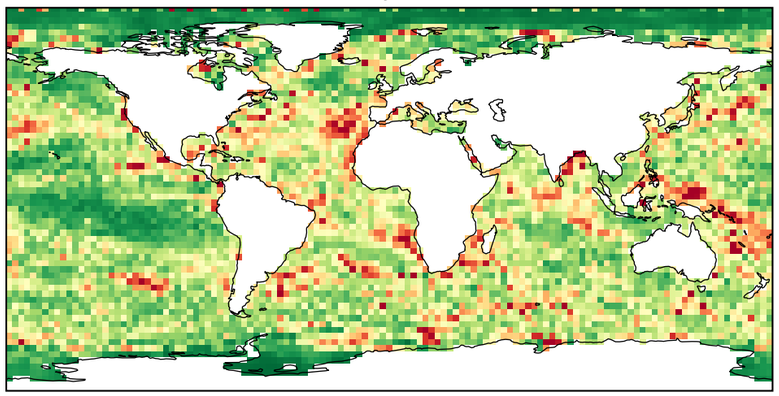
Step 3



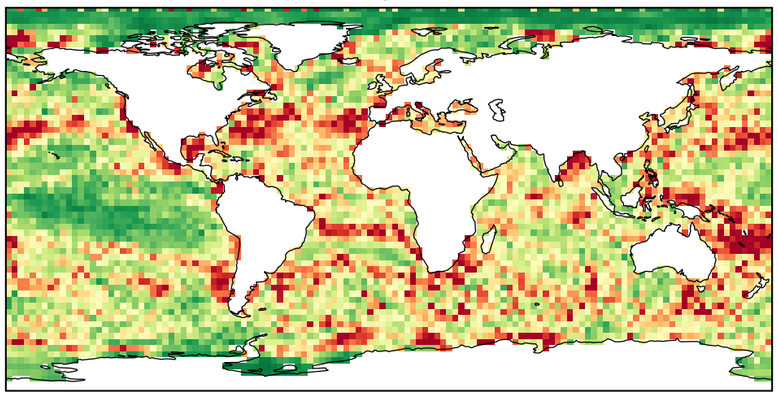
Step 7



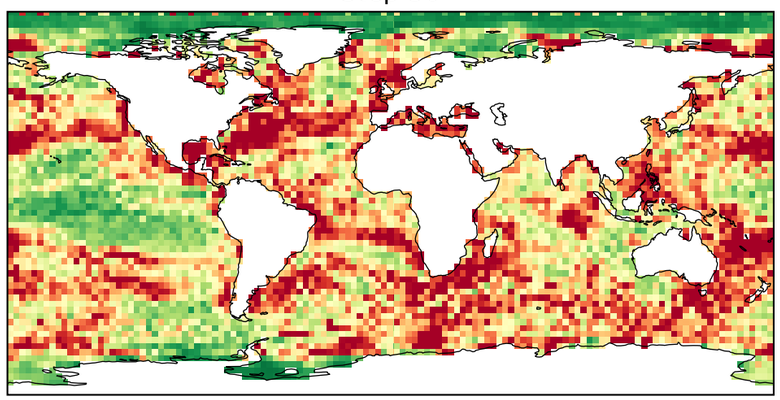
Step 14



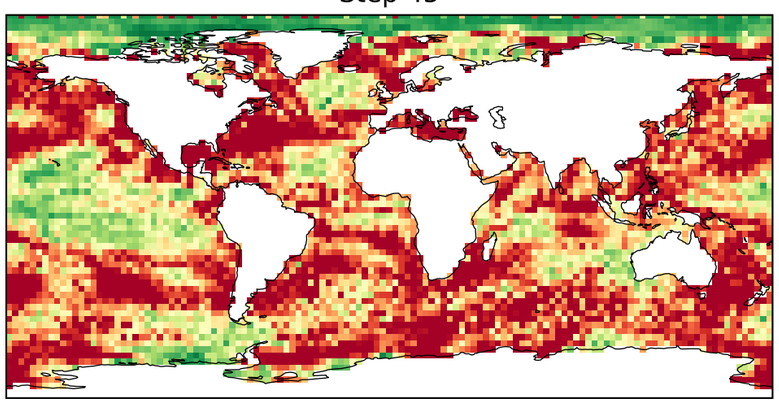
Step 21



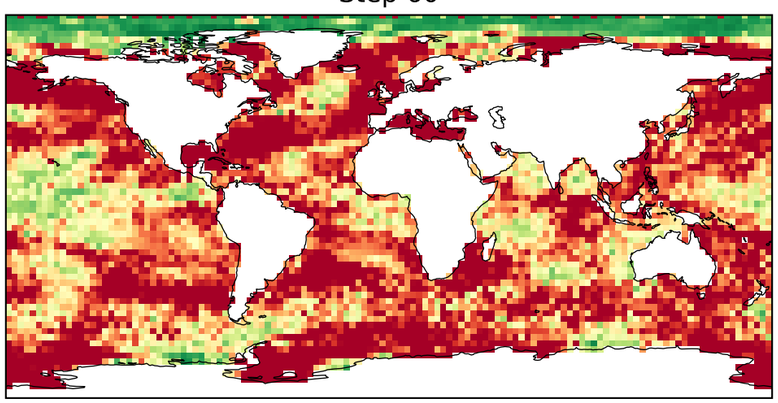
Step 30



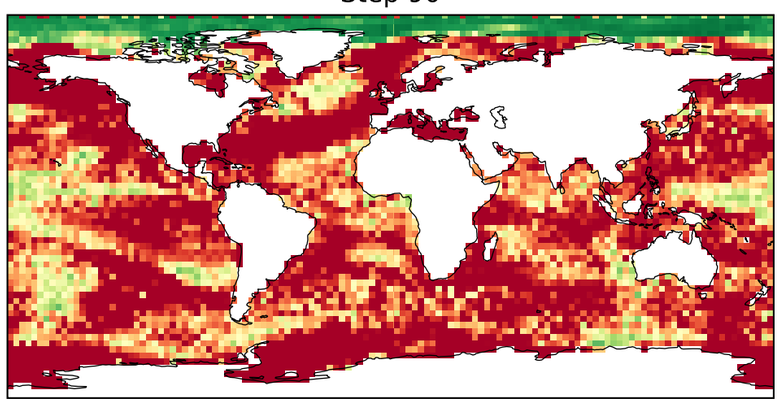
Step 45

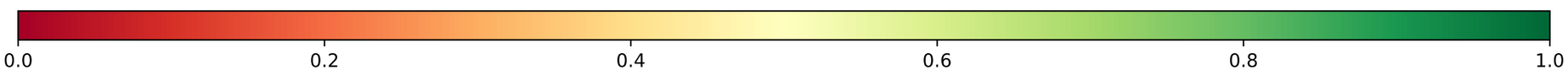


Step 60



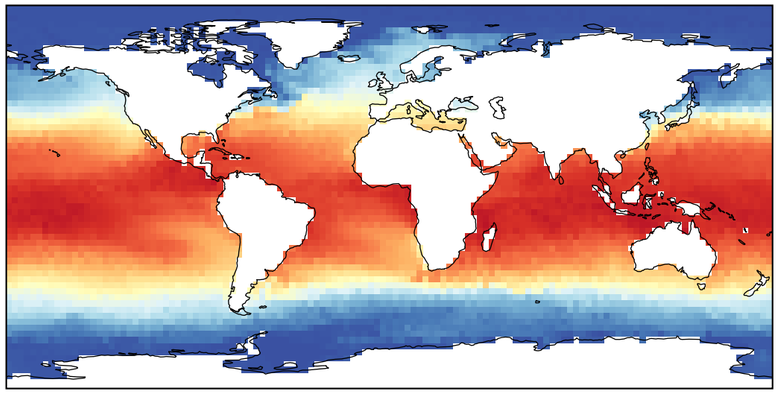
Step 90



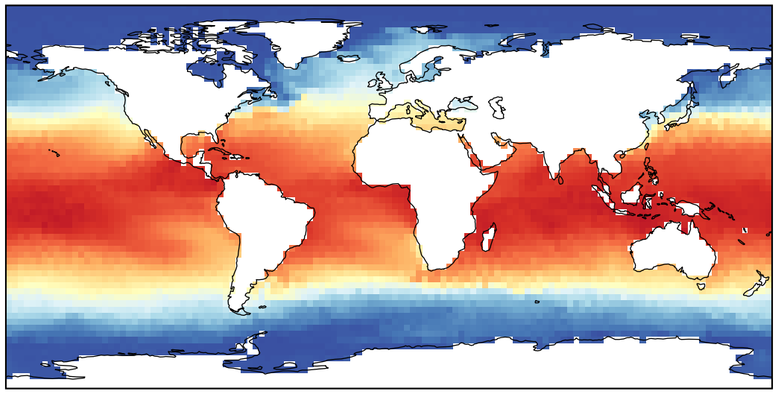


Chain init: 2016-01-11 — Validation (°C)

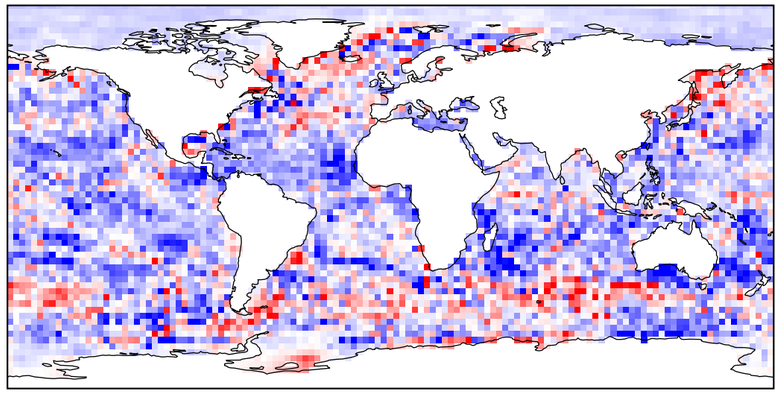
Actual — Step 1 (°C)



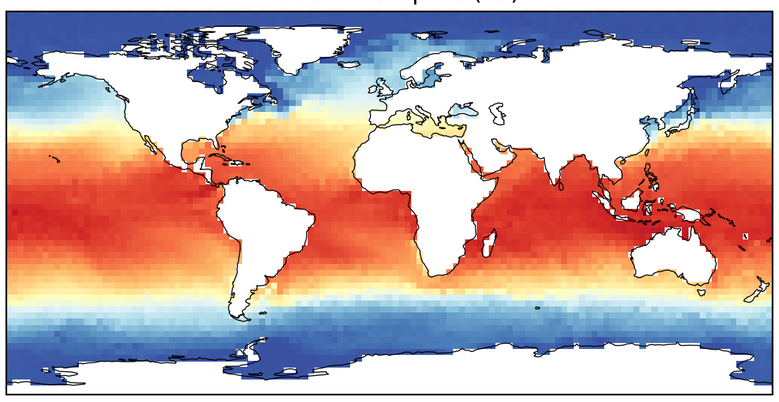
SFNO — Step 1 (°C)



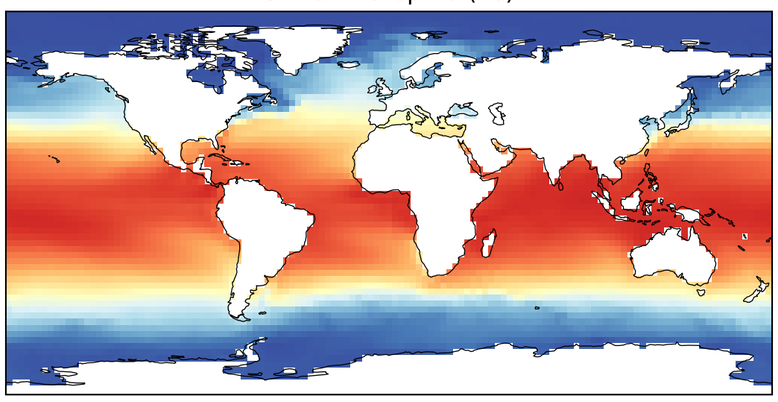
Error — Step 1 (°C)



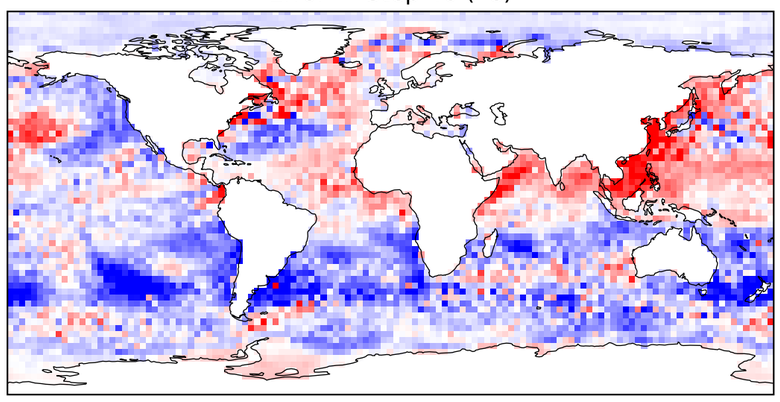
Actual — Step 46 (°C)



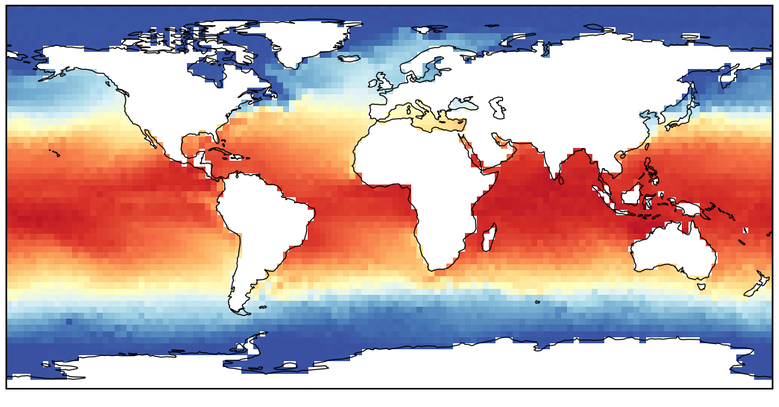
SFNO — Step 46 (°C)



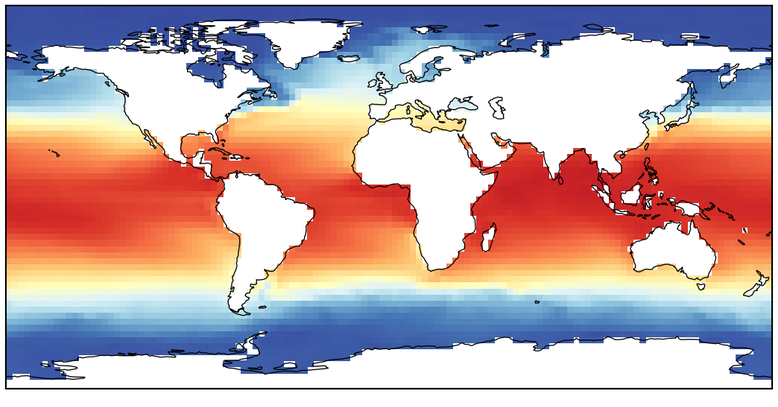
Error — Step 46 (°C)



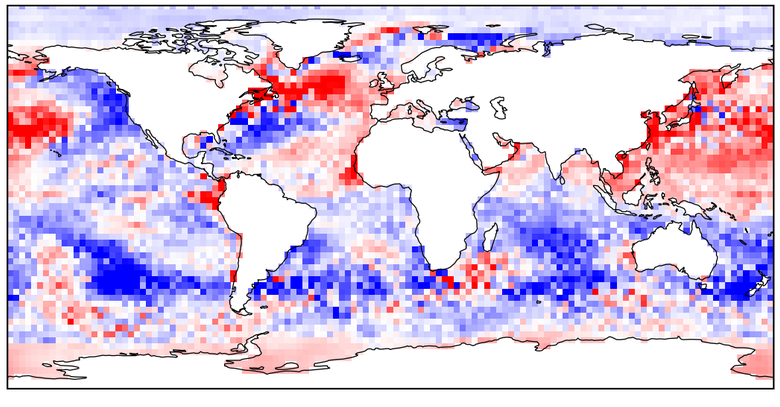
Actual — Step 90 (°C)

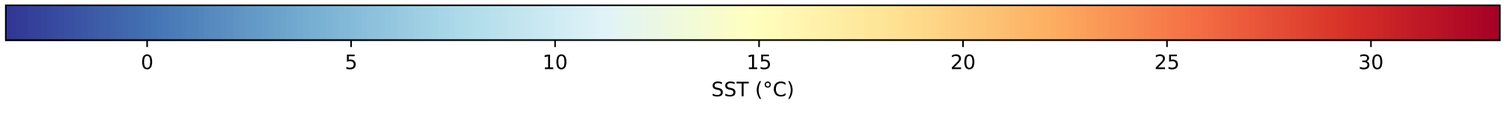


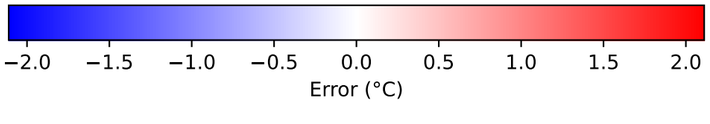
SFNO — Step 90 (°C)



Error — Step 90 (°C)

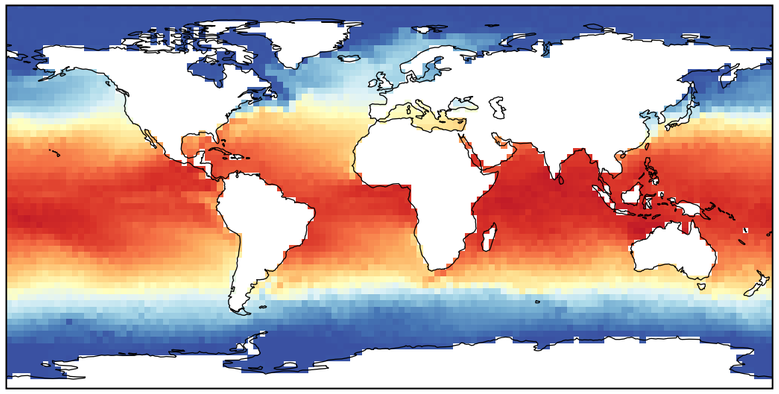




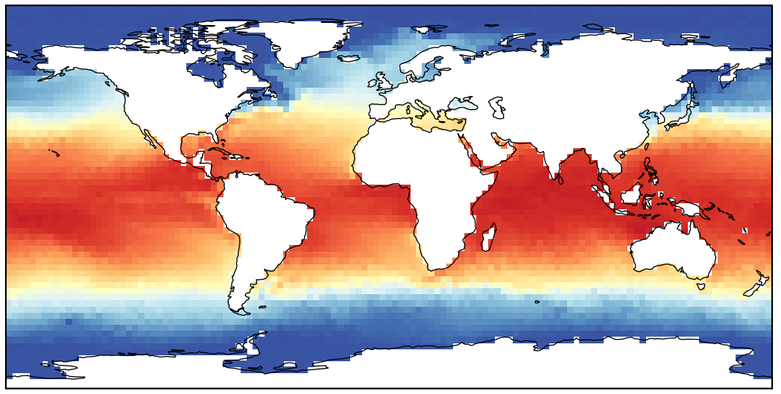


Chain init: 2016-04-10 — Validation (°C)

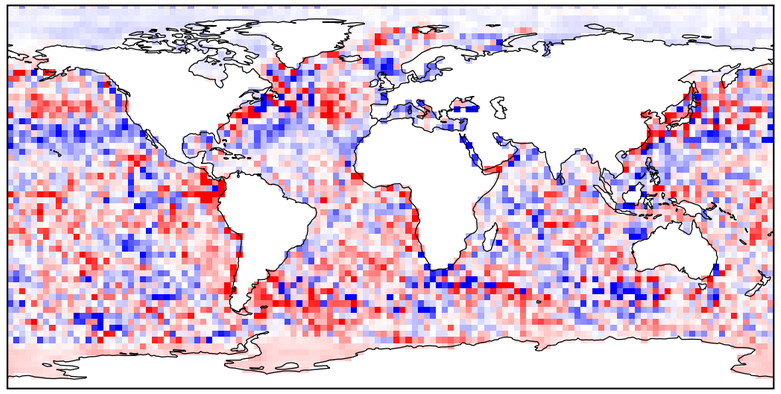
Actual — Step 1 (°C)



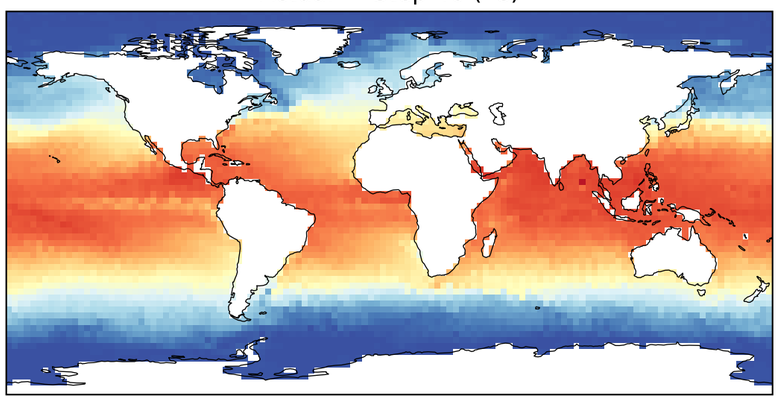
SFNO — Step 1 (°C)



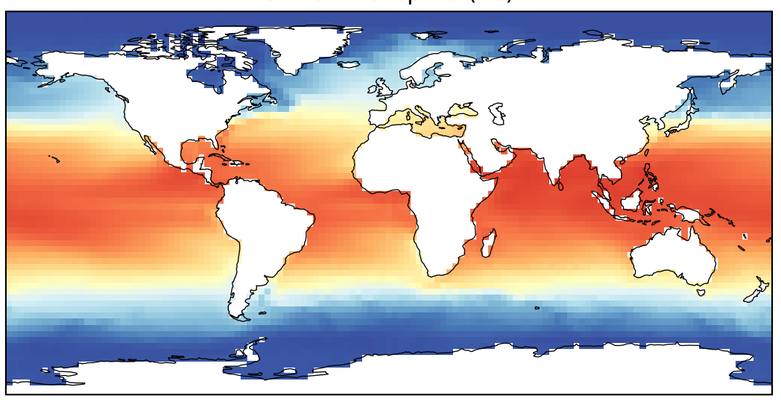
Error — Step 1 (°C)



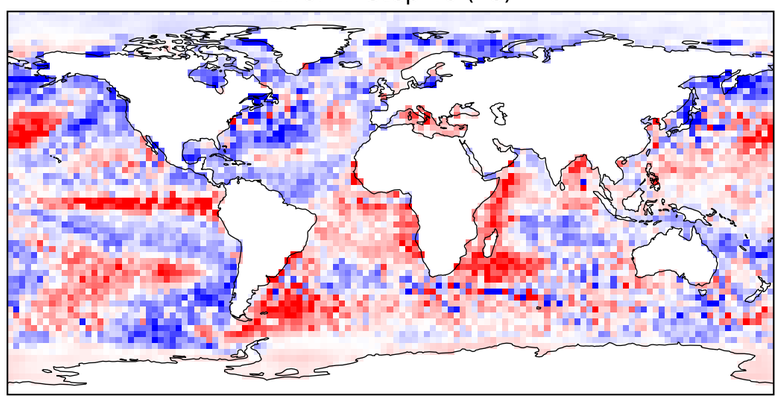
Actual — Step 46 (°C)



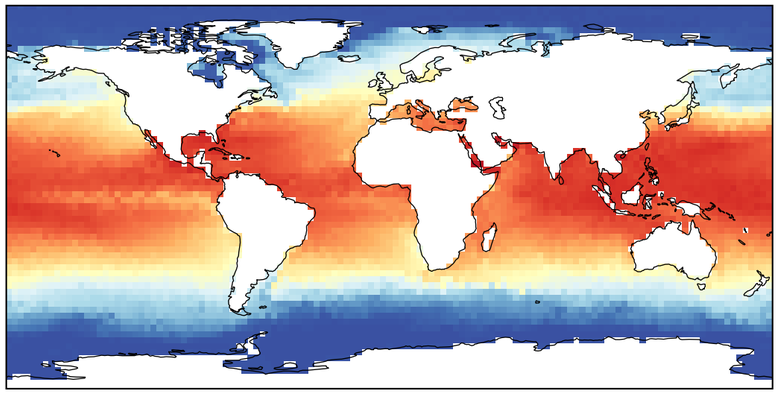
SFNO — Step 46 (°C)



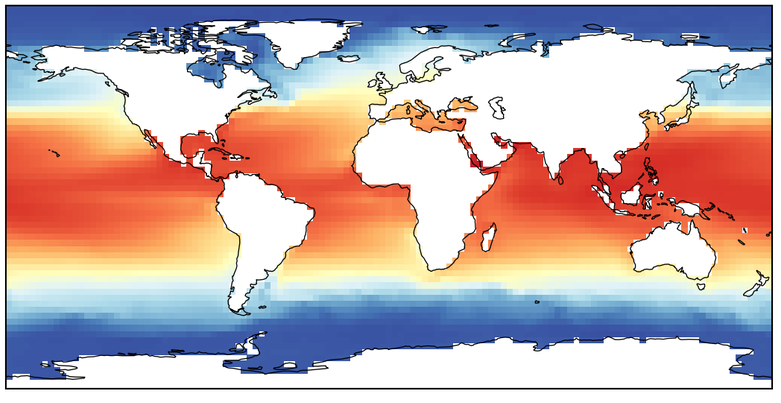
Error — Step 46 (°C)



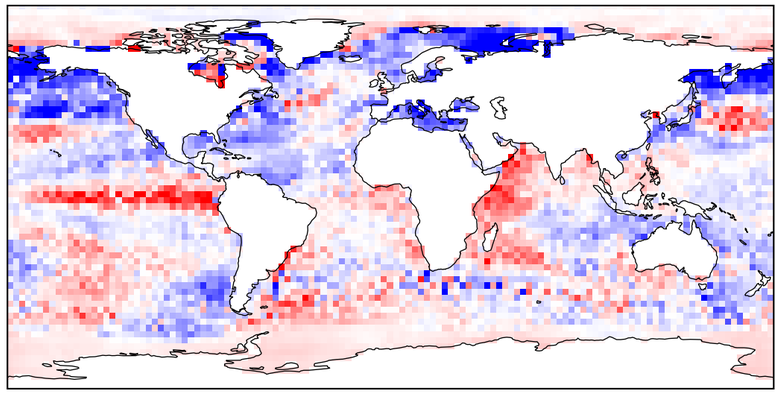
Actual — Step 90 (°C)

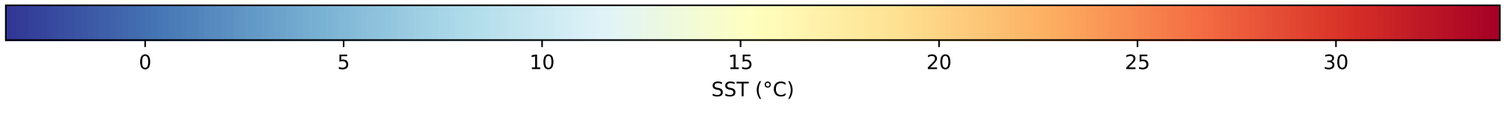


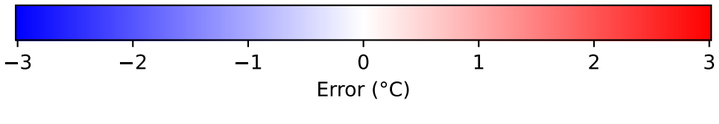
SFNO — Step 90 (°C)



Error — Step 90 (°C)

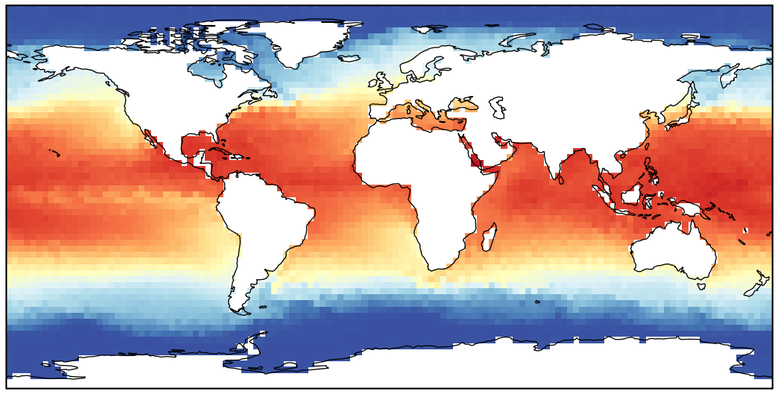




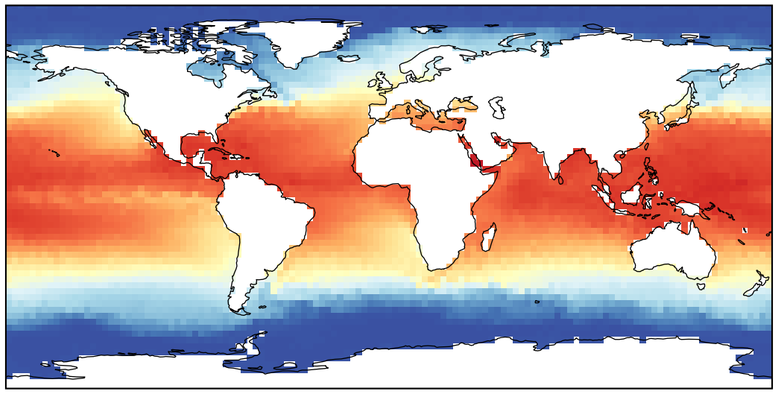


Chain init: 2016-10-07 — Validation (°C)

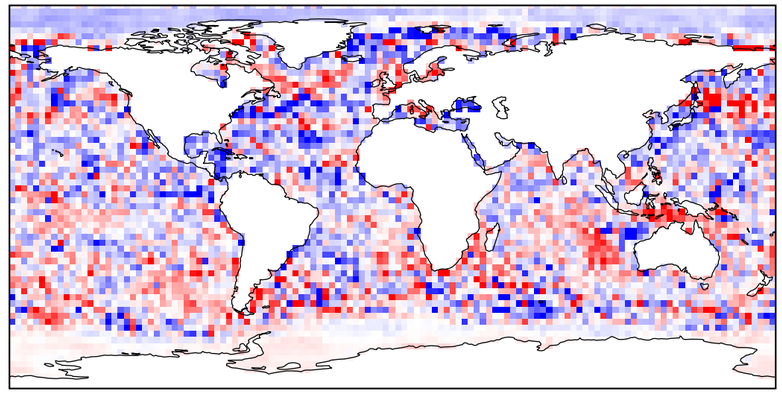
Actual — Step 1 (°C)



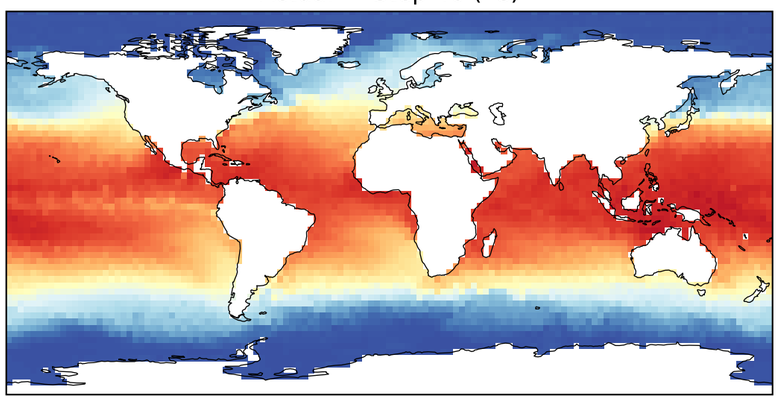
SFNO — Step 1 (°C)



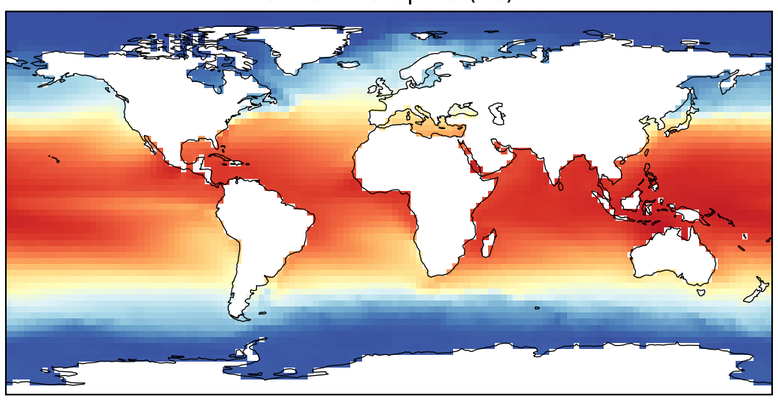
Error — Step 1 (°C)



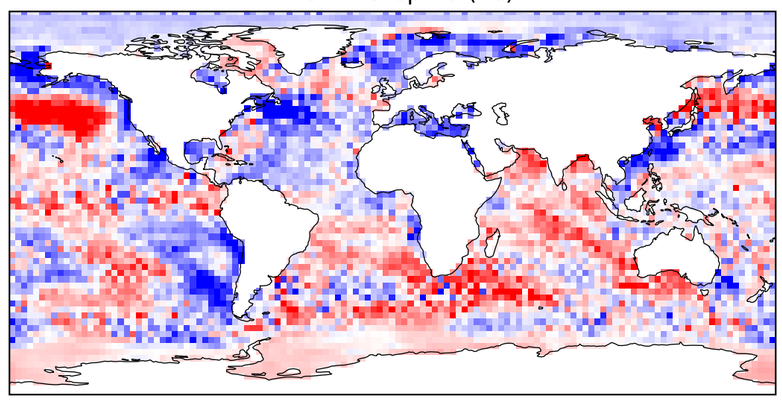
Actual — Step 46 (°C)



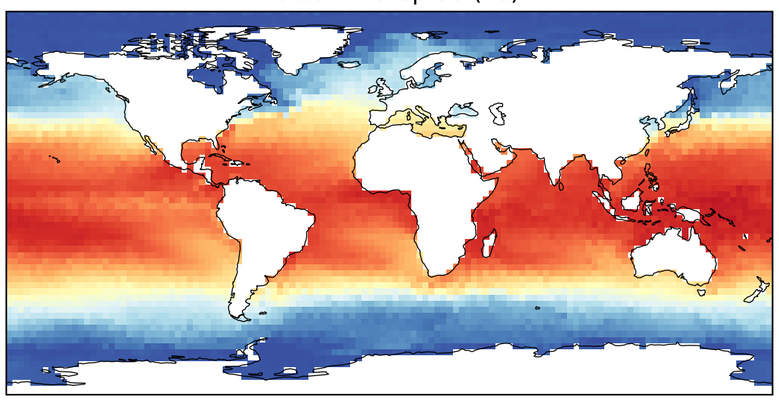
SFNO — Step 46 (°C)



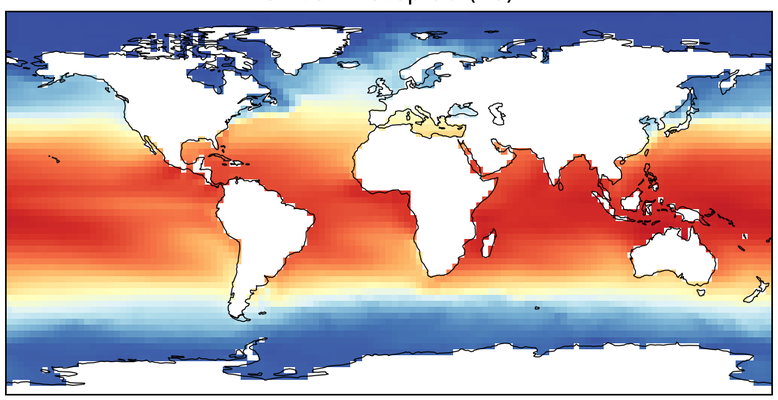
Error — Step 46 (°C)



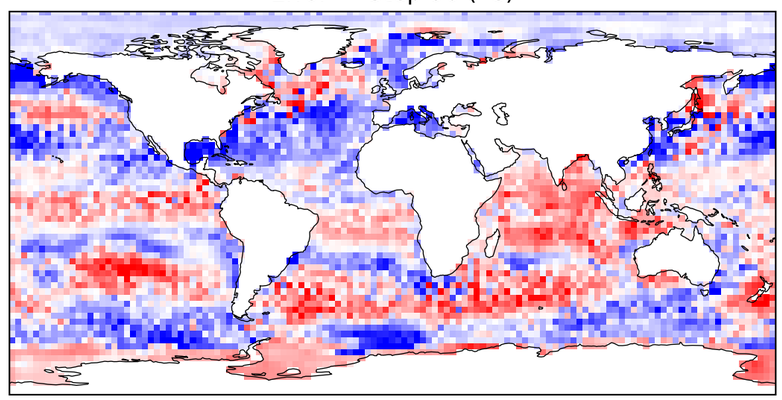
Actual — Step 90 (°C)

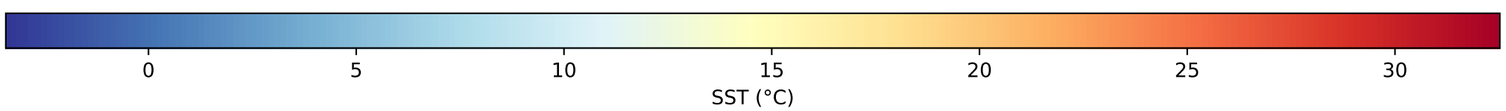


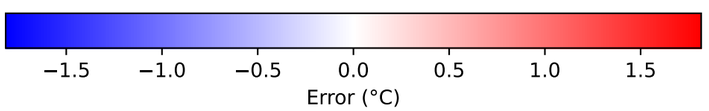
SFNO — Step 90 (°C)



Error — Step 90 (°C)

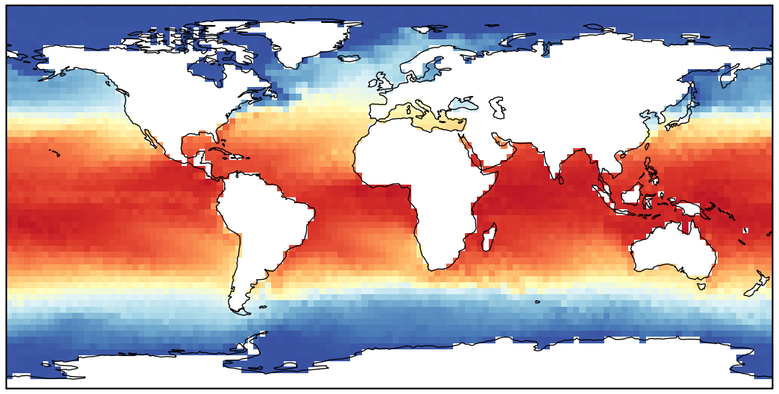




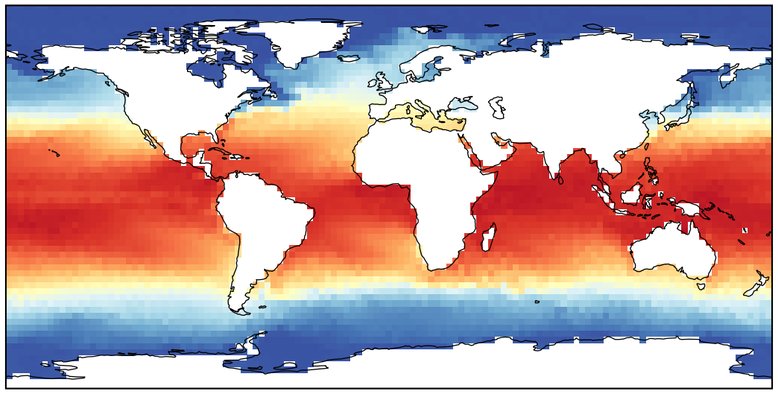


Chain init: 2017-04-05 — Validation (°C)

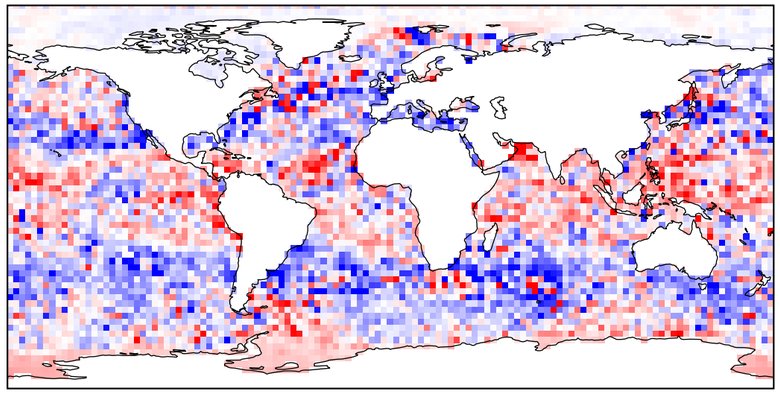
Actual — Step 1 (°C)



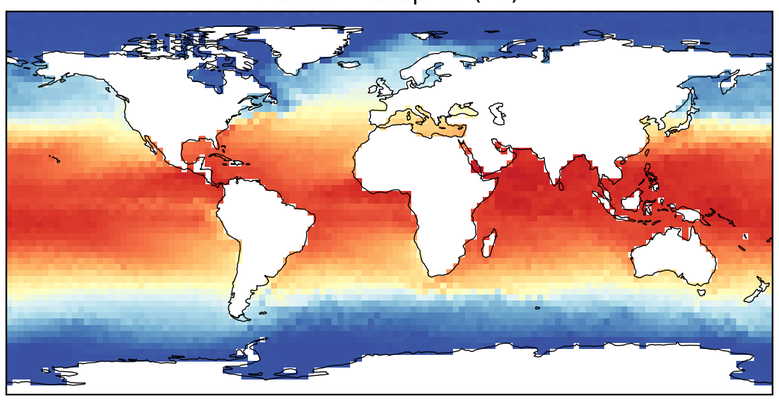
SFNO — Step 1 (°C)



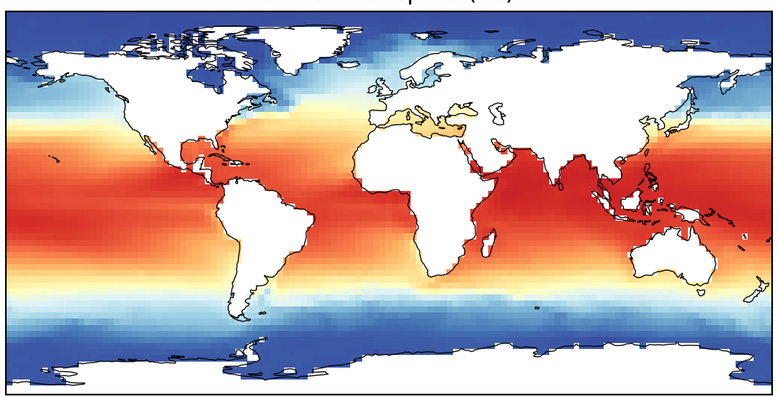
Error — Step 1 (°C)



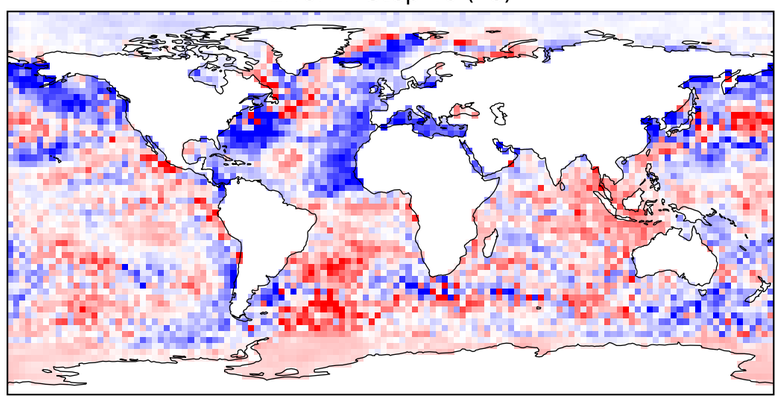
Actual — Step 46 (°C)



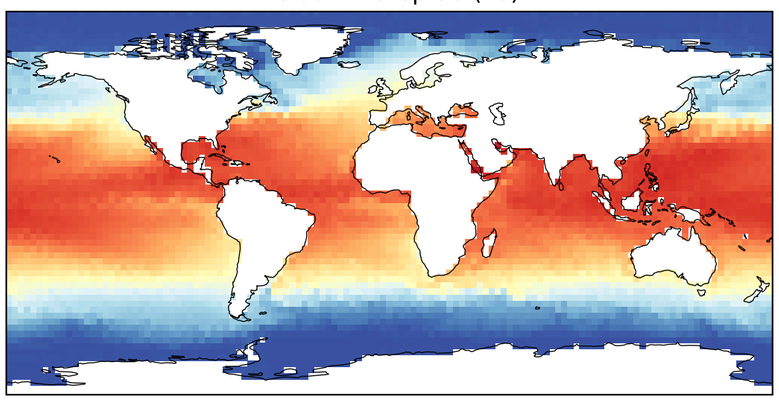
SFNO — Step 46 (°C)



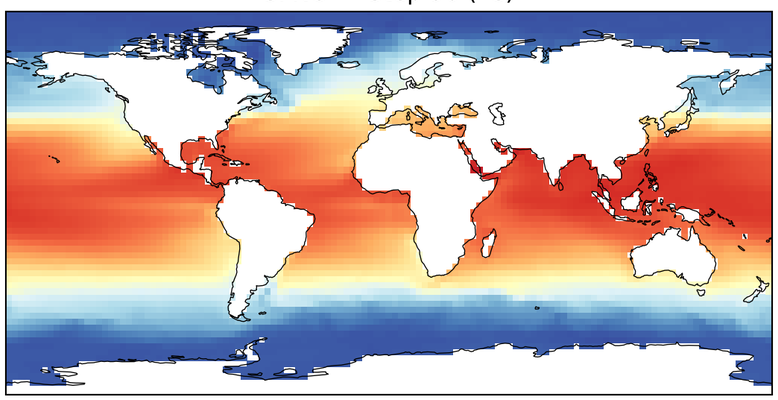
Error — Step 46 (°C)



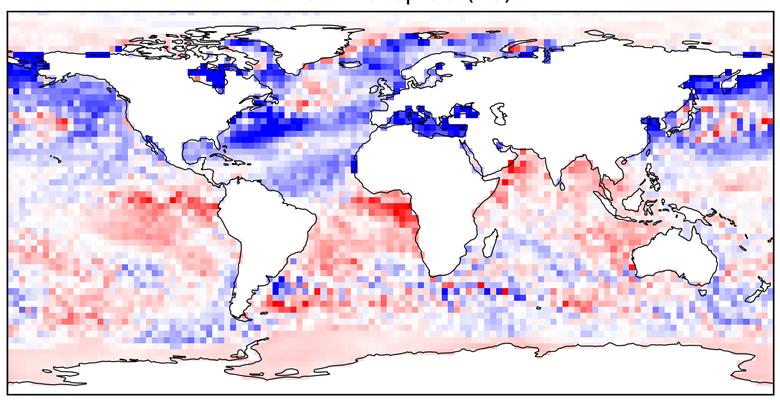
Actual — Step 90 (°C)

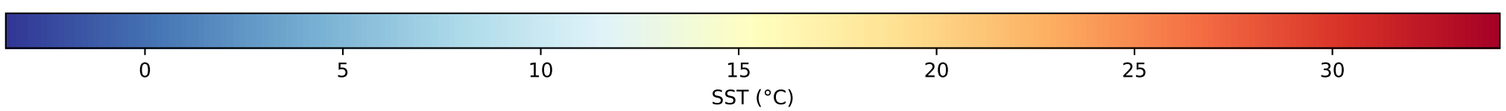


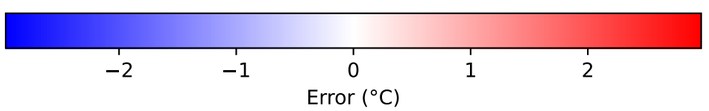
SFNO — Step 90 (°C)



Error — Step 90 (°C)

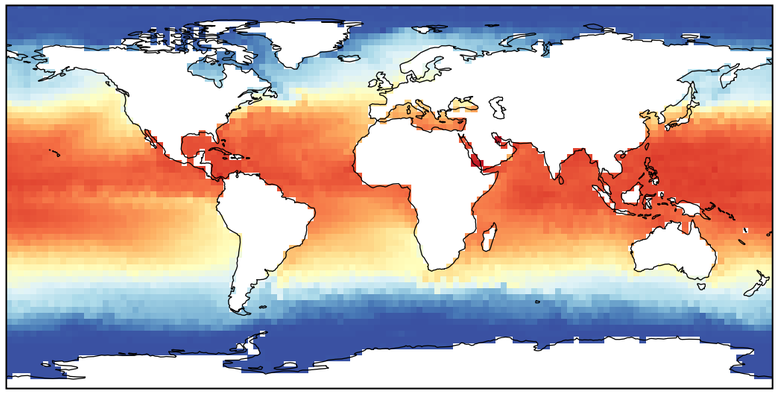




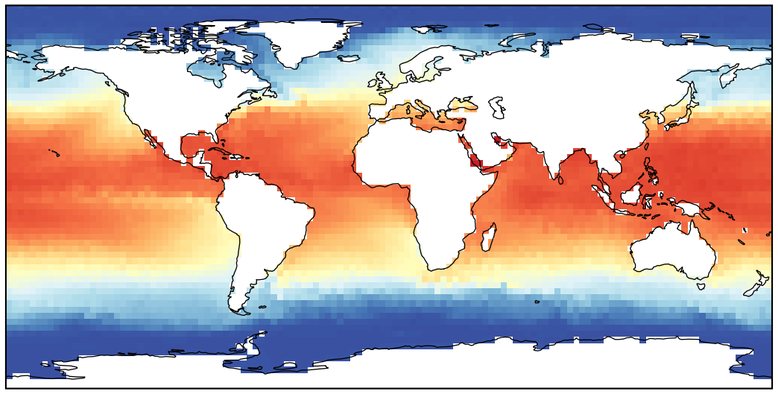


Chain init: 2017-10-02 — Validation (°C)

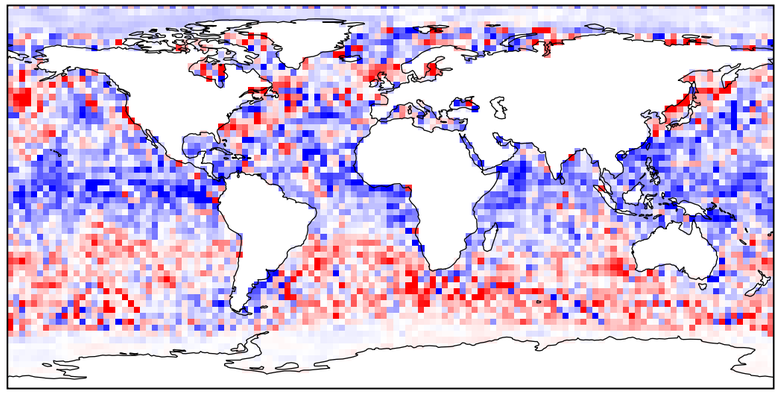
Actual — Step 1 (°C)



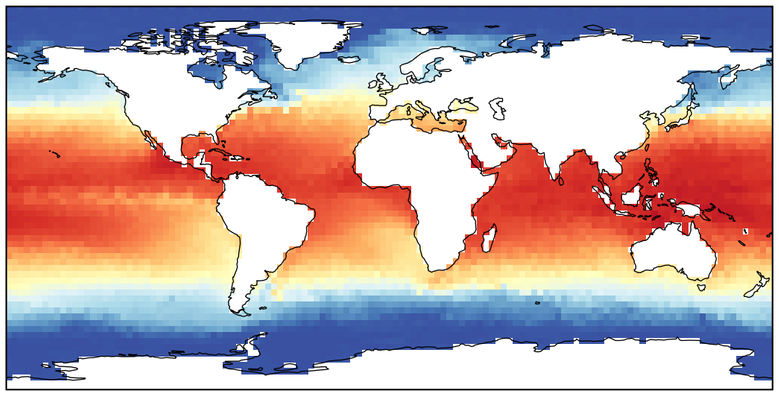
SFNO — Step 1 (°C)



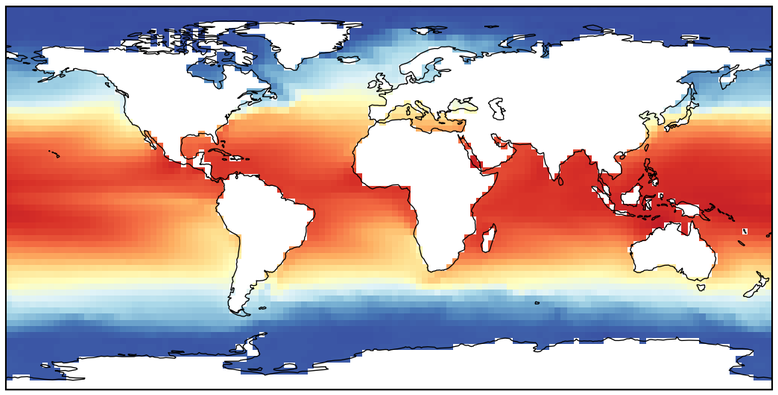
Error — Step 1 (°C)



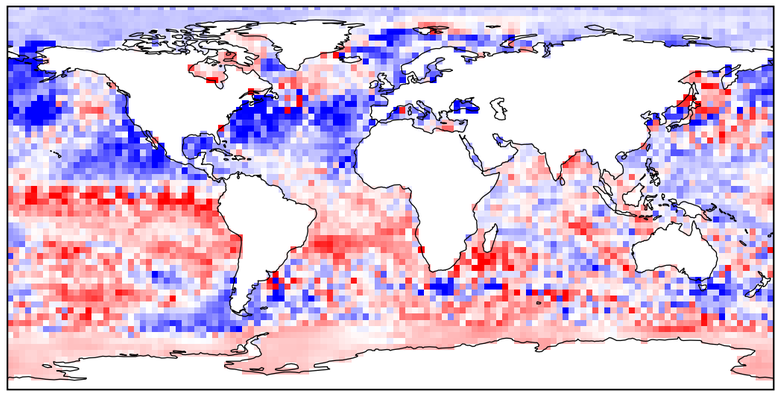
Actual — Step 46 (°C)



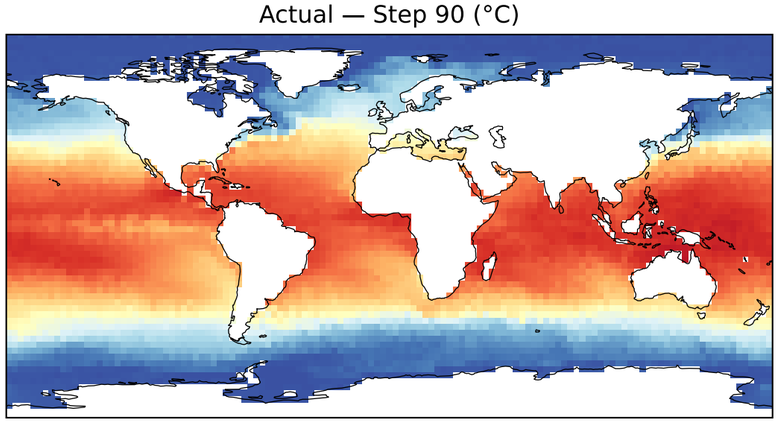
SFNO — Step 46 (°C)



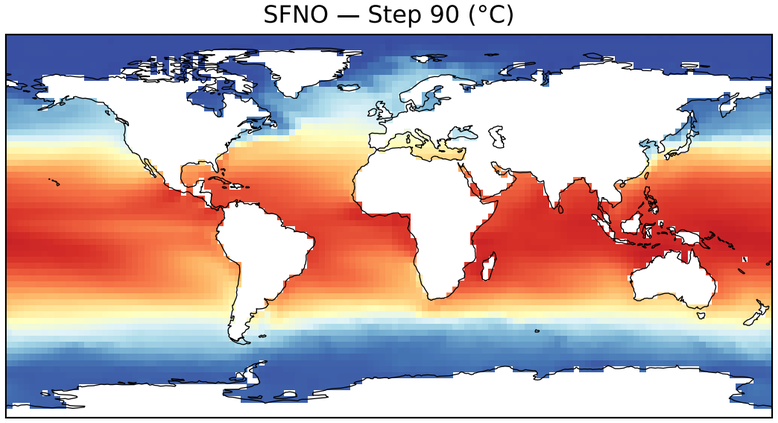
Error — Step 46 (°C)



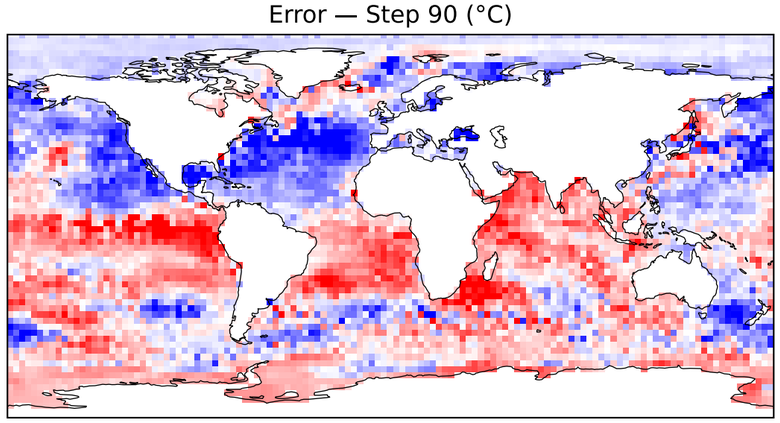
Actual — Step 90 (°C)

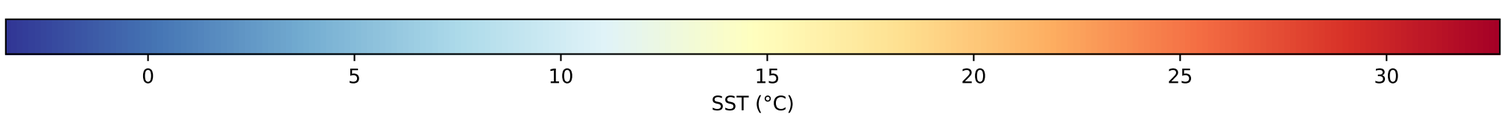


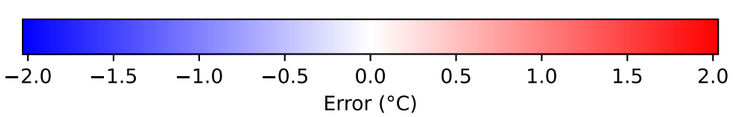
SFNO — Step 90 (°C)



Error — Step 90 (°C)

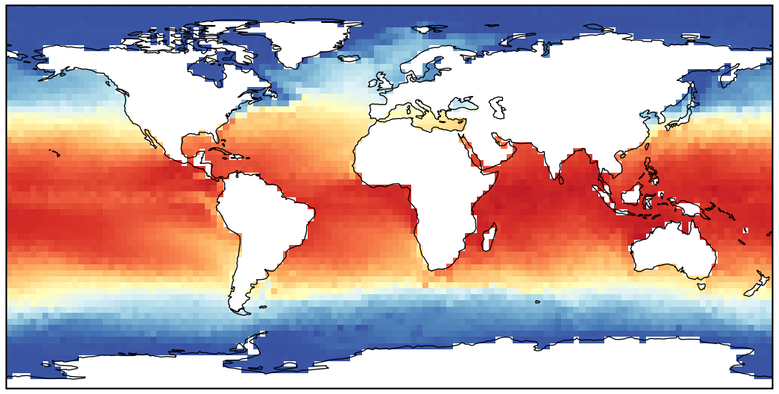




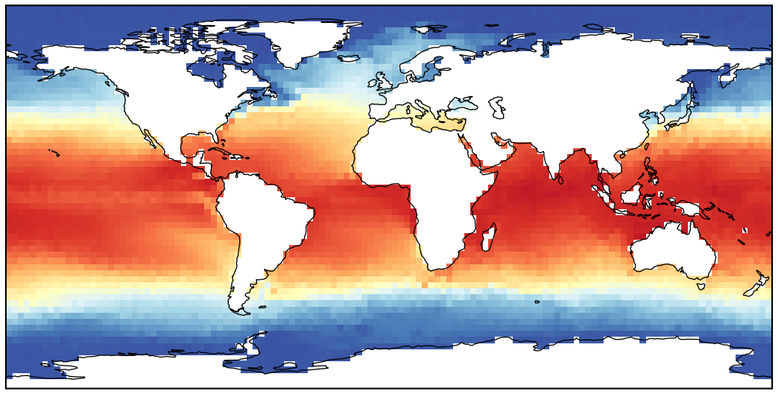


Chain init: 2018-03-31 — Validation (°C)

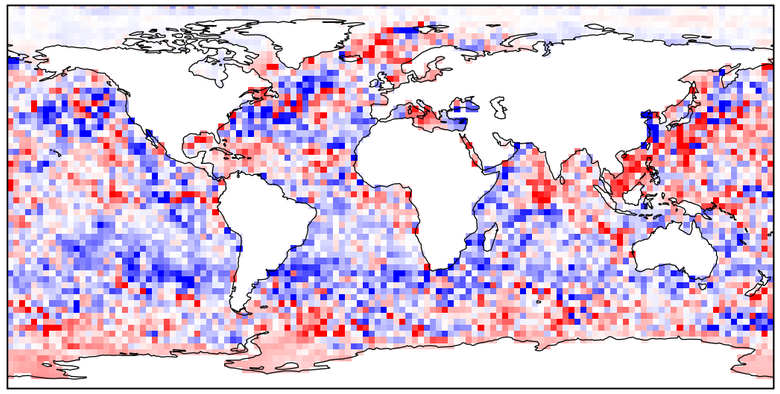
Actual — Step 1 (°C)



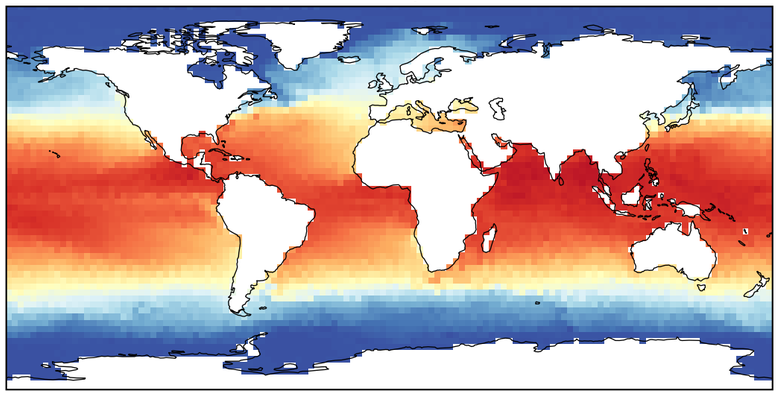
SFNO — Step 1 (°C)



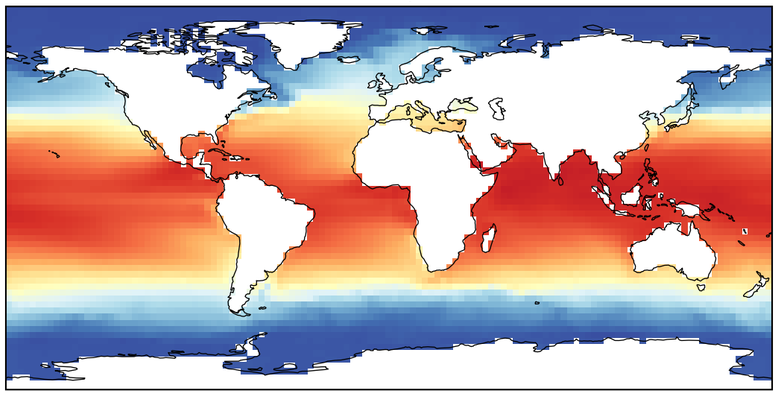
Error — Step 1 (°C)



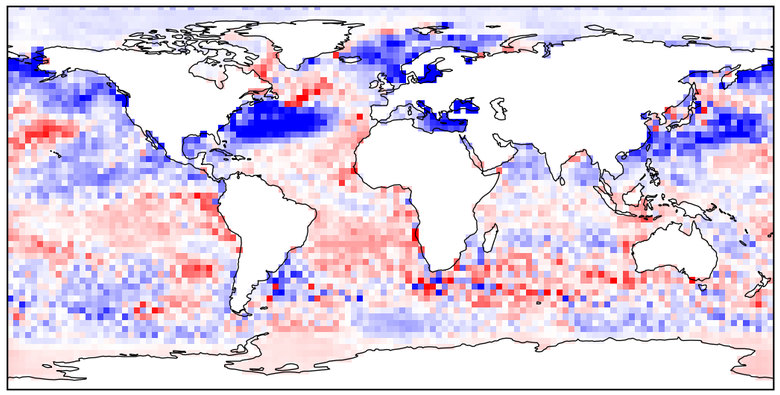
Actual — Step 46 (°C)



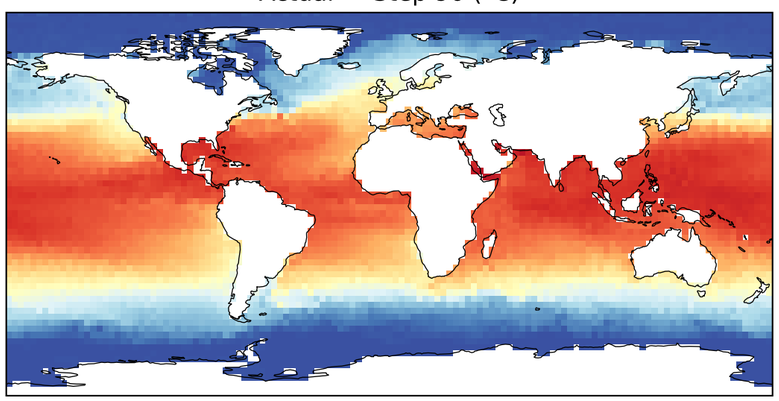
SFNO — Step 46 (°C)



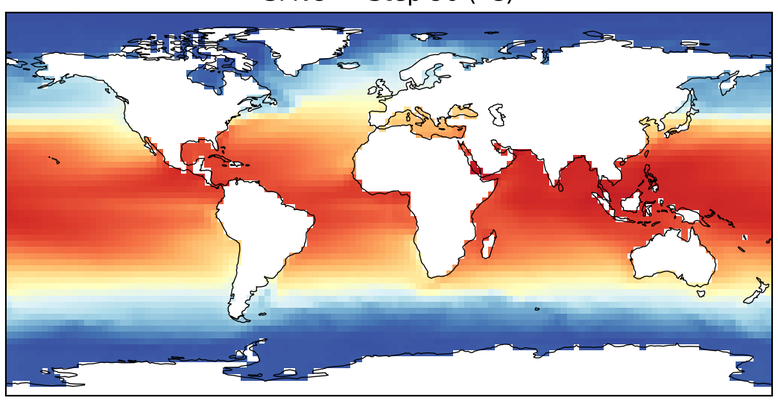
Error — Step 46 (°C)



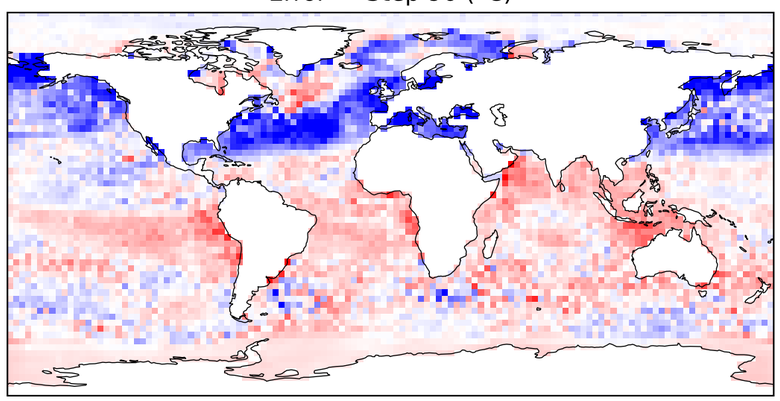
Actual — Step 90 (°C)

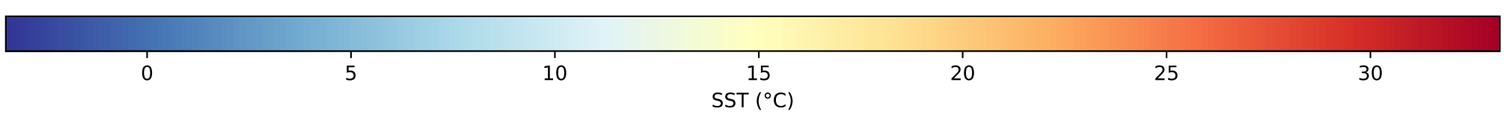


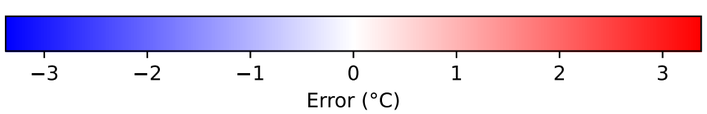
SFNO — Step 90 (°C)



Error — Step 90 (°C)

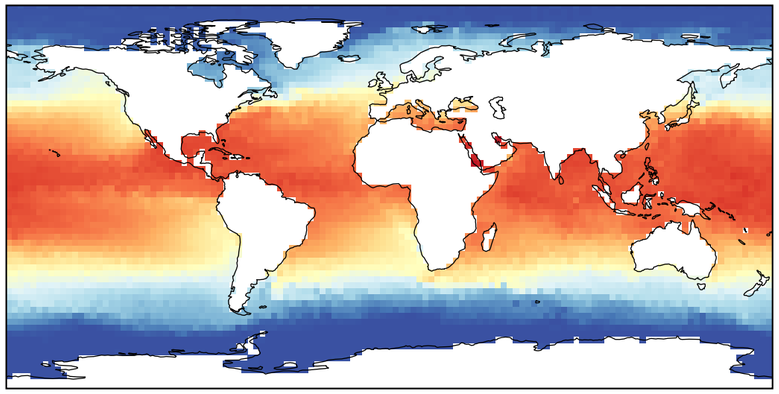




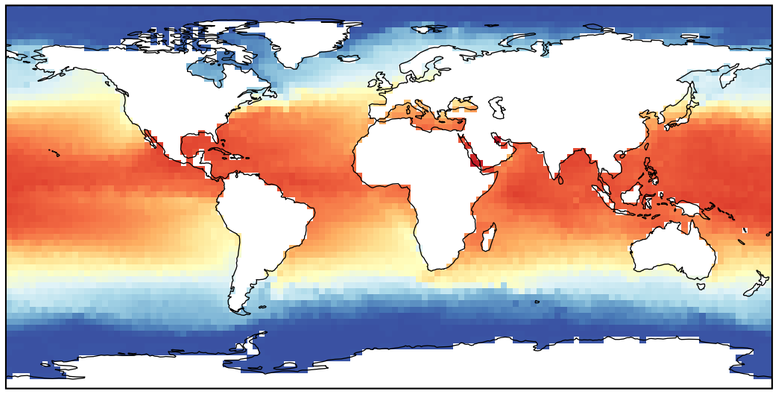


Chain init: 2018-09-27 — Validation (°C)

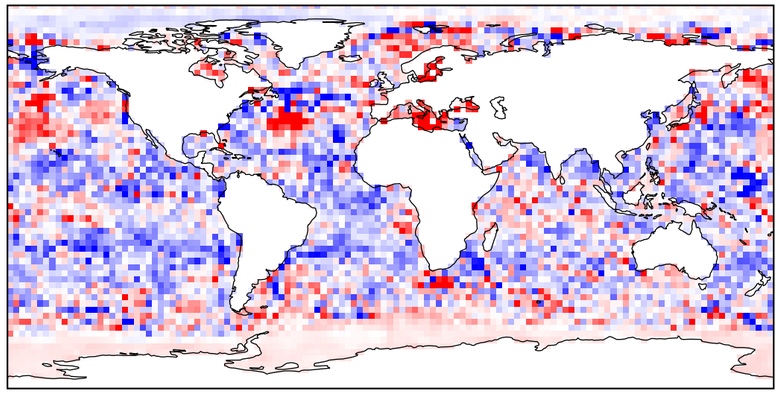
Actual — Step 1 (°C)



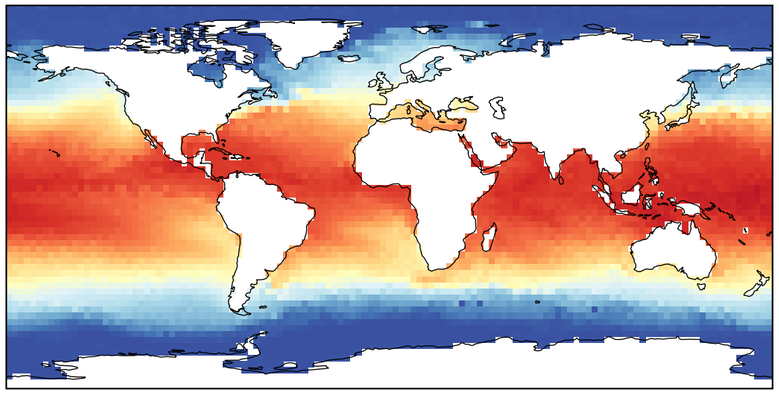
SFNO — Step 1 (°C)



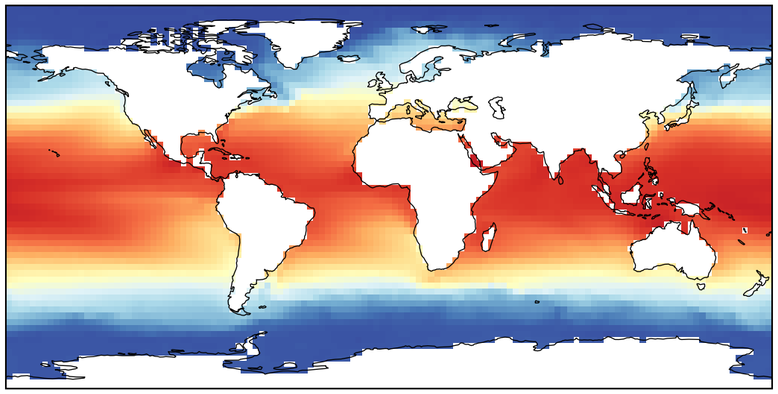
Error — Step 1 (°C)



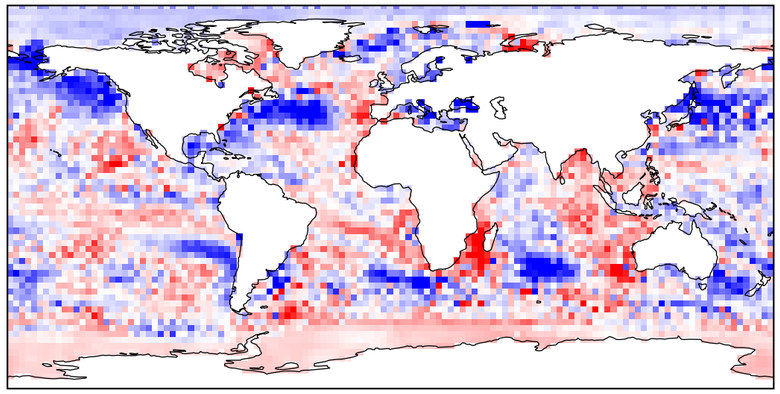
Actual — Step 46 (°C)



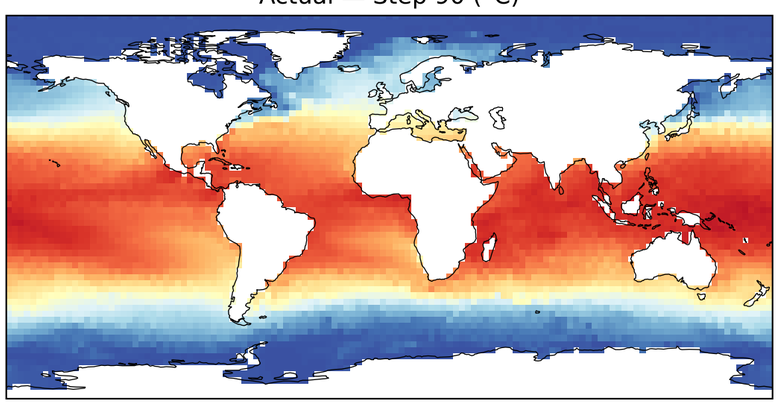
SFNO — Step 46 (°C)



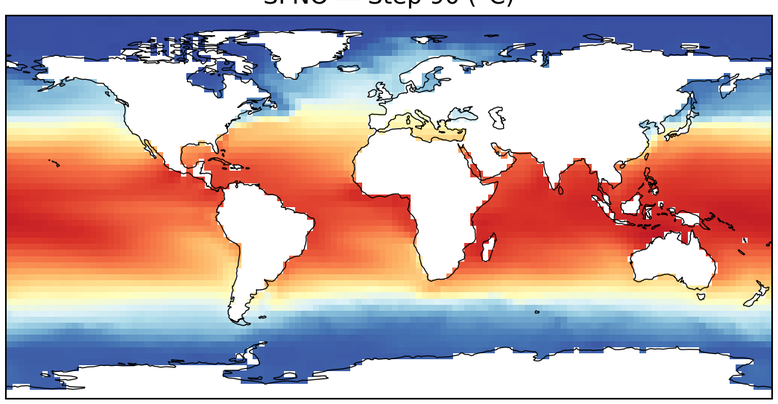
Error — Step 46 (°C)



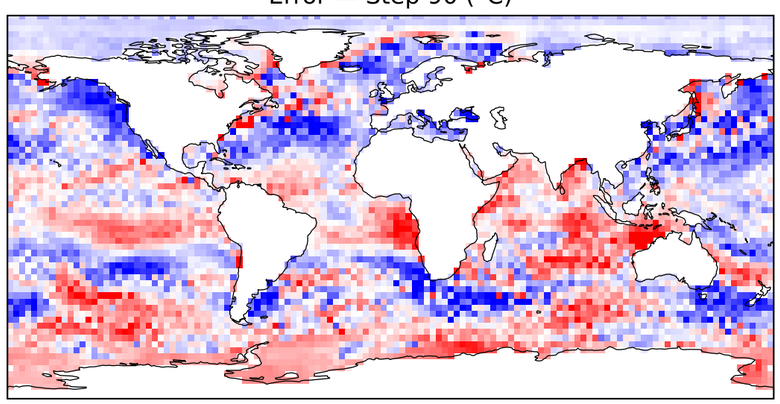
Actual — Step 90 (°C)

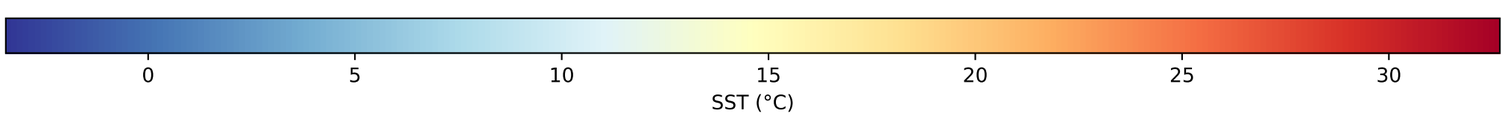


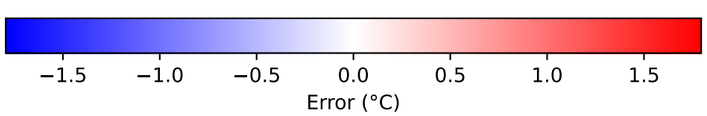
SFNO — Step 90 (°C)



Error — Step 90 (°C)

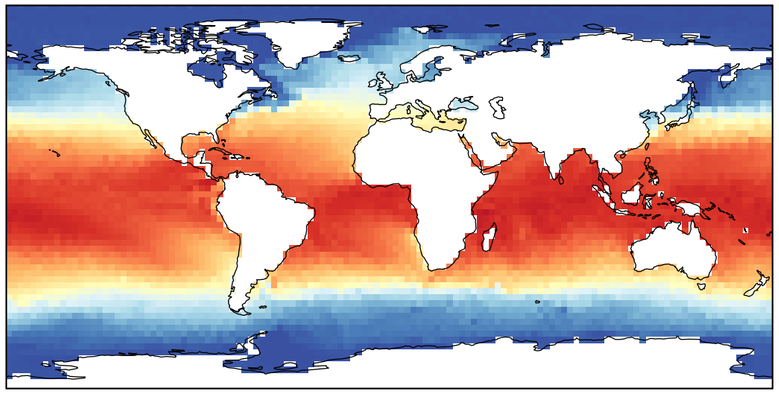




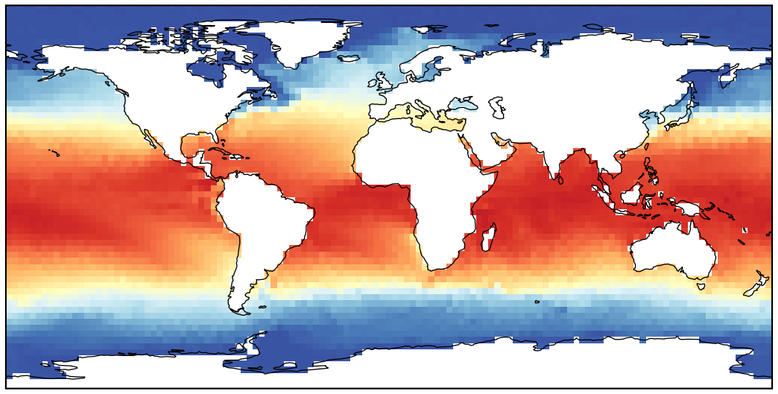


Chain init: 2019-03-26 — Validation (°C)

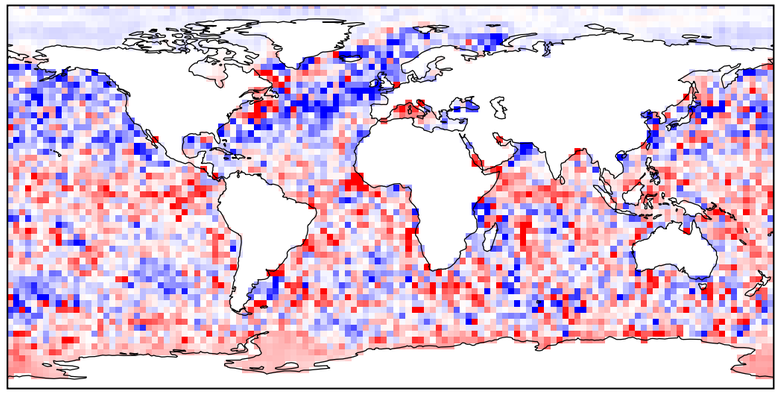
Actual — Step 1 (°C)



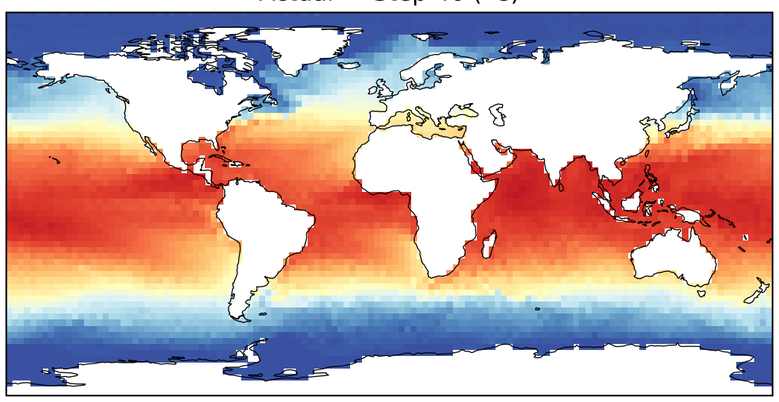
SFNO — Step 1 (°C)



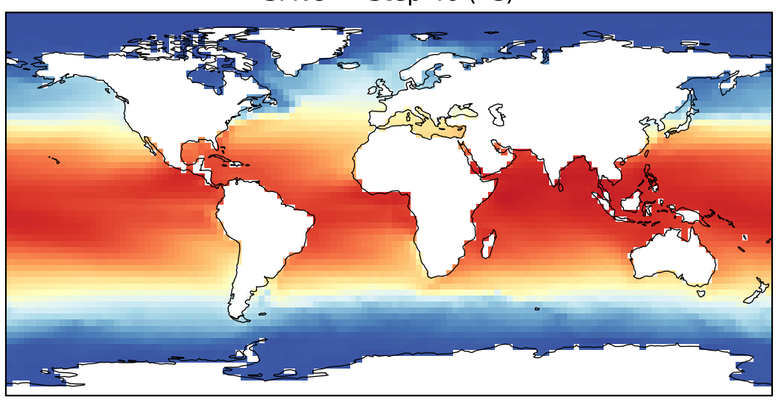
Error — Step 1 (°C)



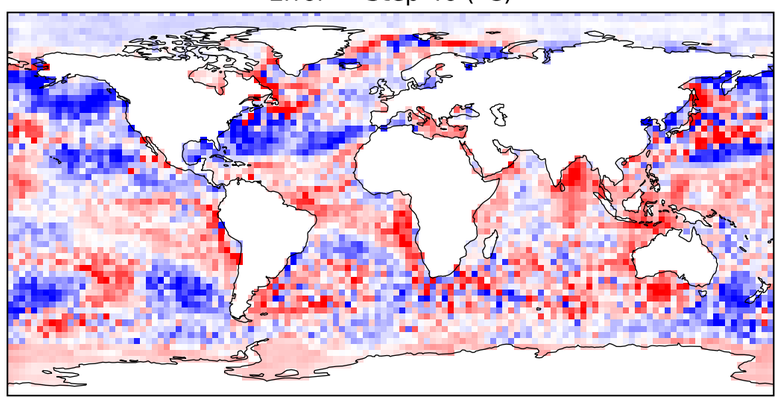
Actual — Step 46 (°C)



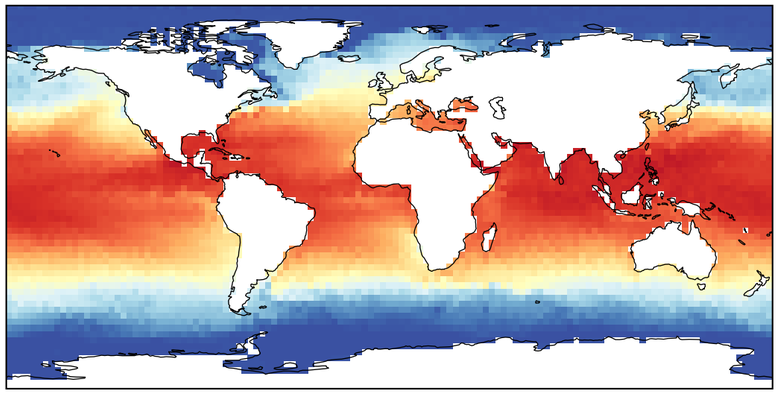
SFNO — Step 46 (°C)



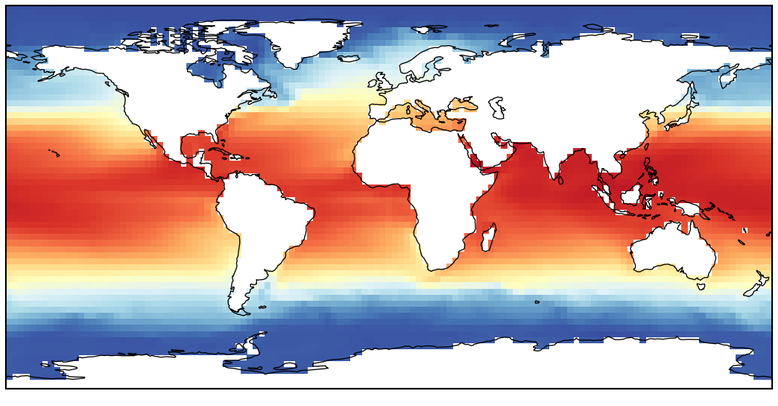
Error — Step 46 (°C)



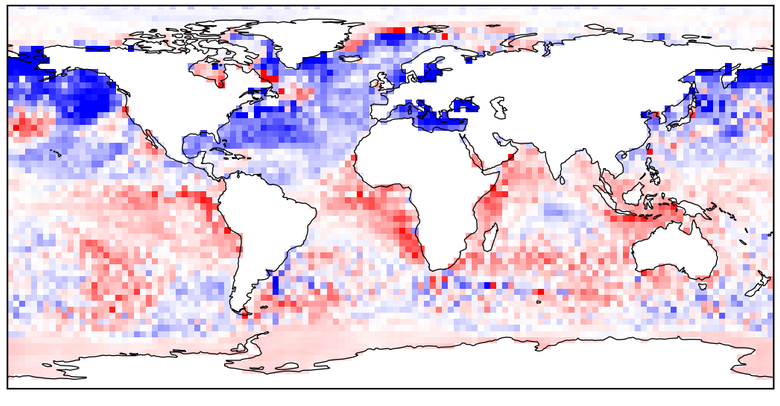
Actual — Step 90 (°C)

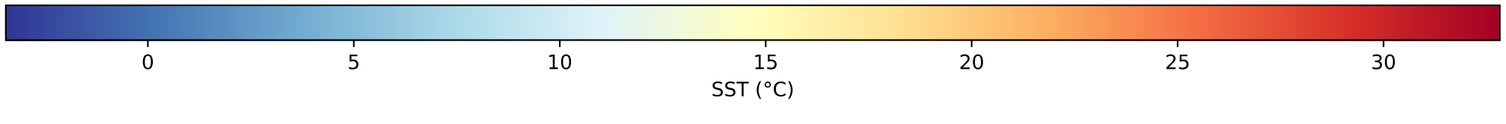


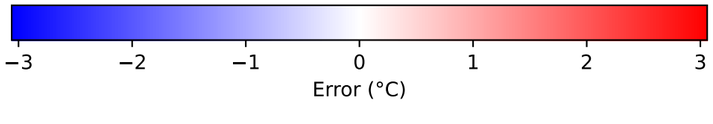
SFNO — Step 90 (°C)



Error — Step 90 (°C)

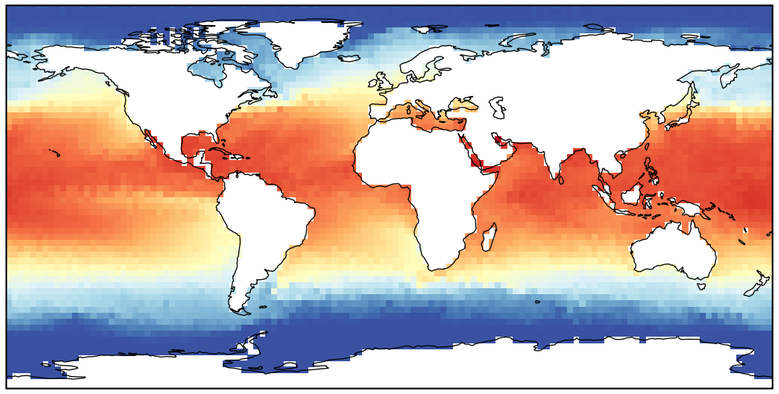




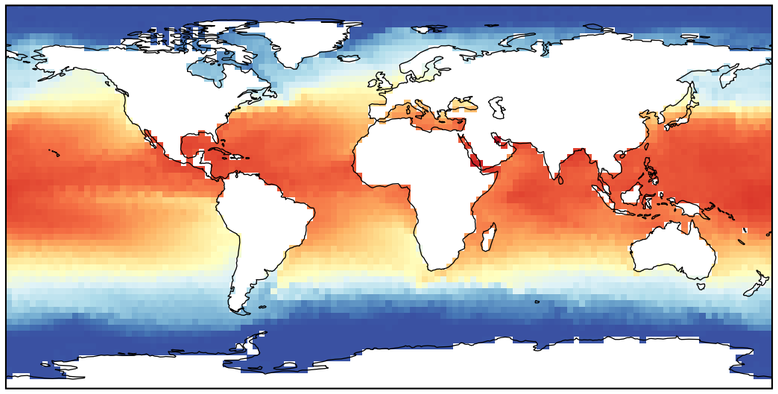


Chain init: 2019-09-22 — Validation (°C)

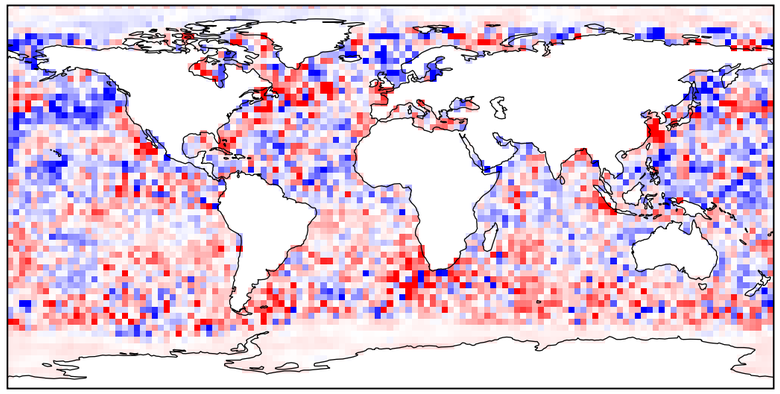
Actual — Step 1 (°C)



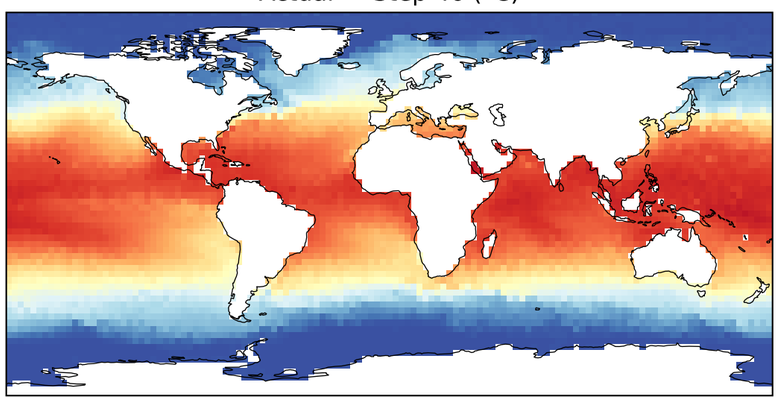
SFNO — Step 1 (°C)



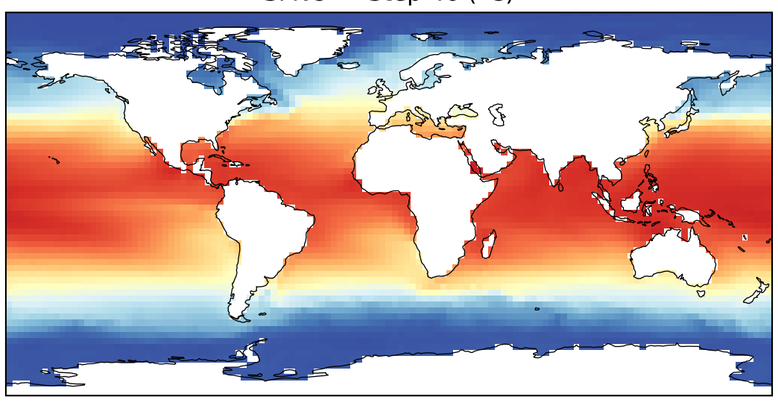
Error — Step 1 (°C)



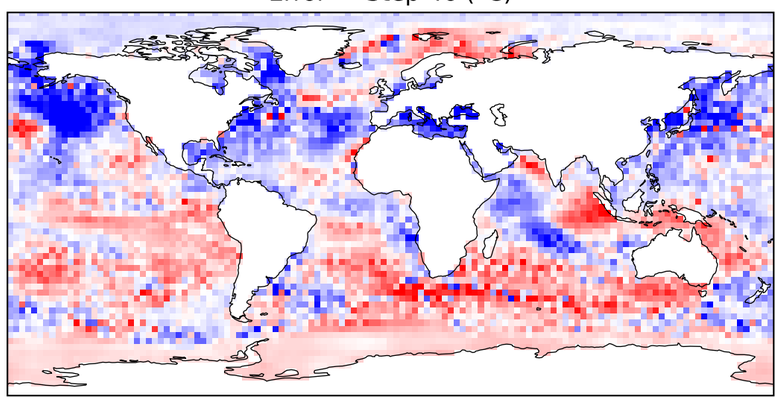
Actual — Step 46 (°C)



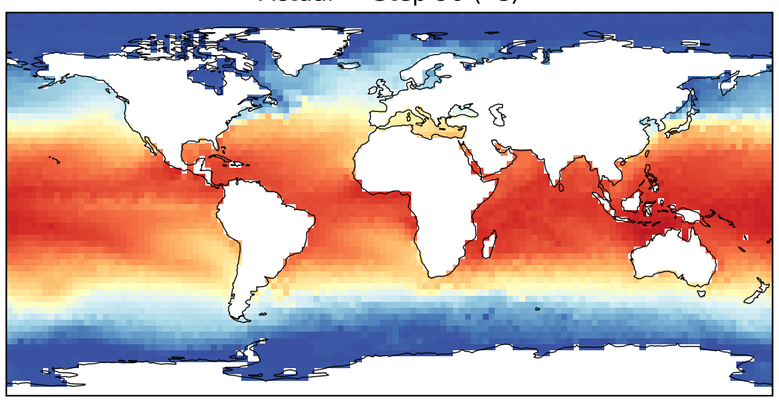
SFNO — Step 46 (°C)



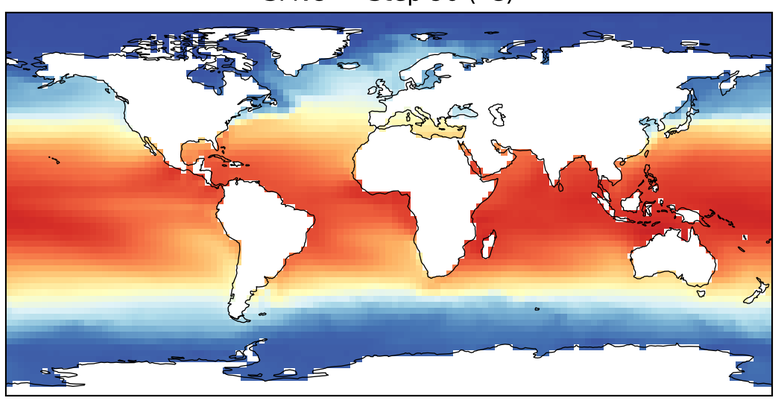
Error — Step 46 (°C)



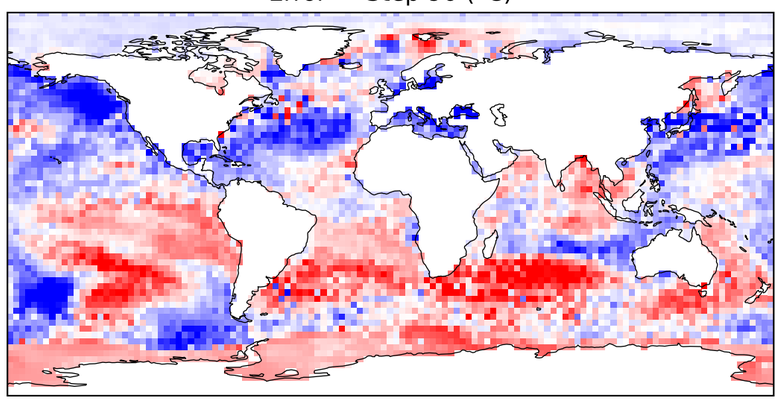
Actual — Step 90 (°C)

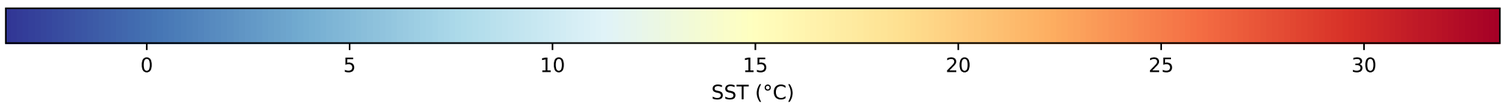


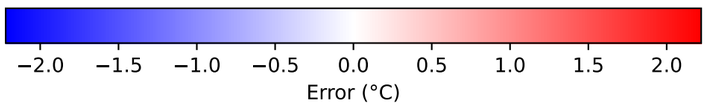
SFNO — Step 90 (°C)



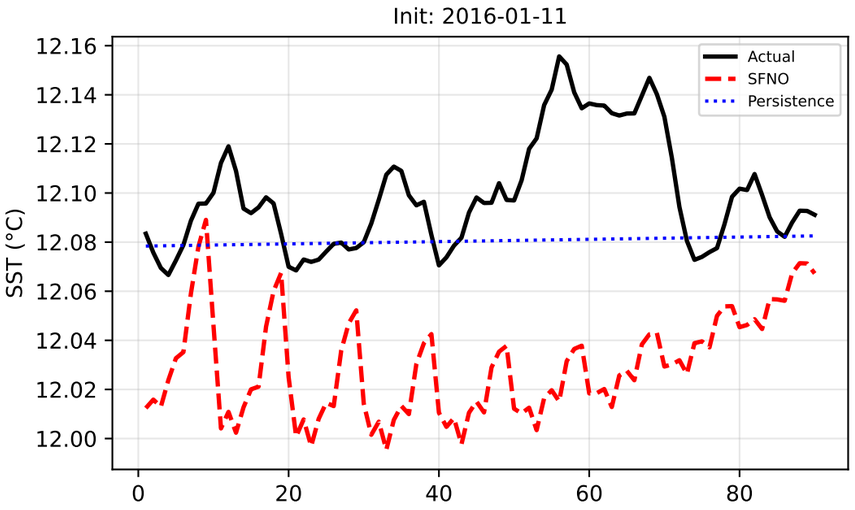
Error — Step 90 (°C)

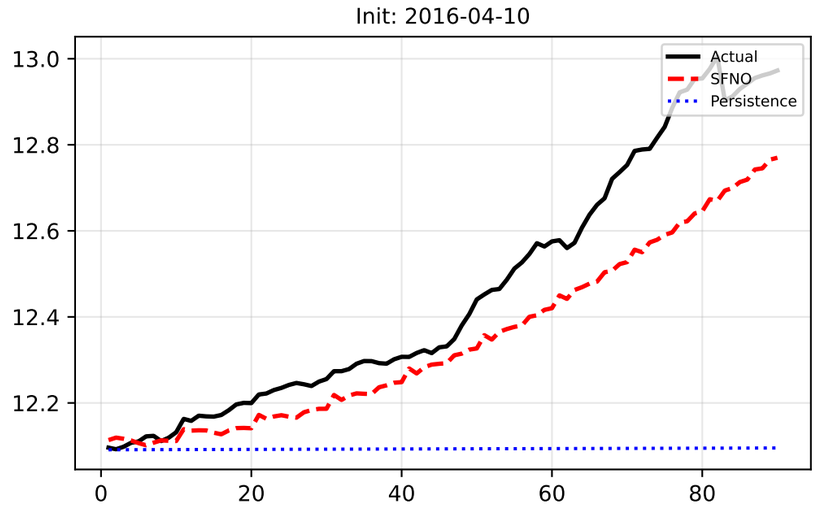


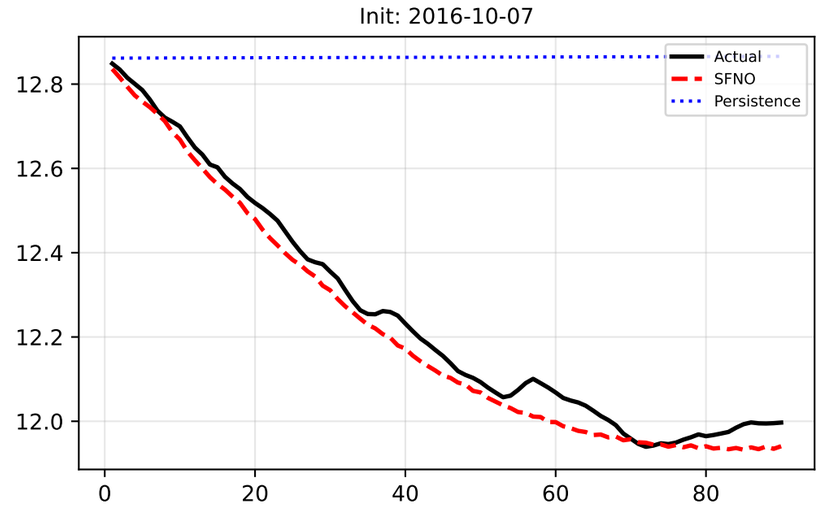


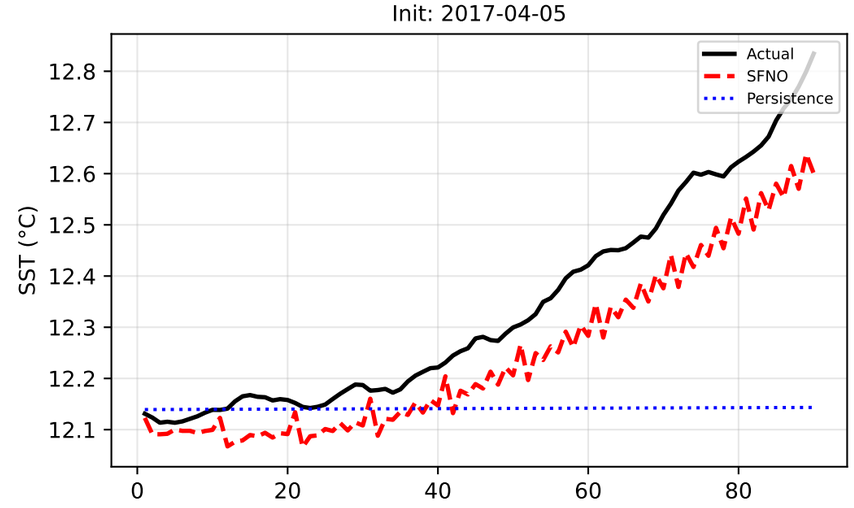


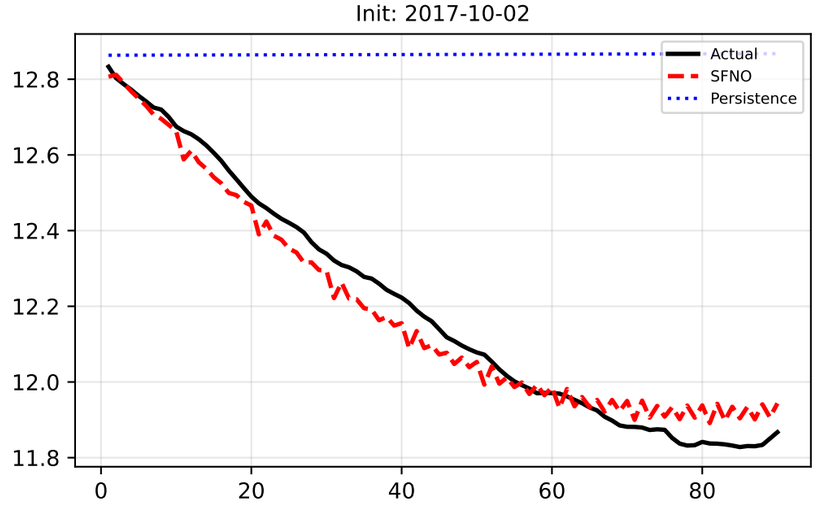
Global-Mean SST — Validation (°C)

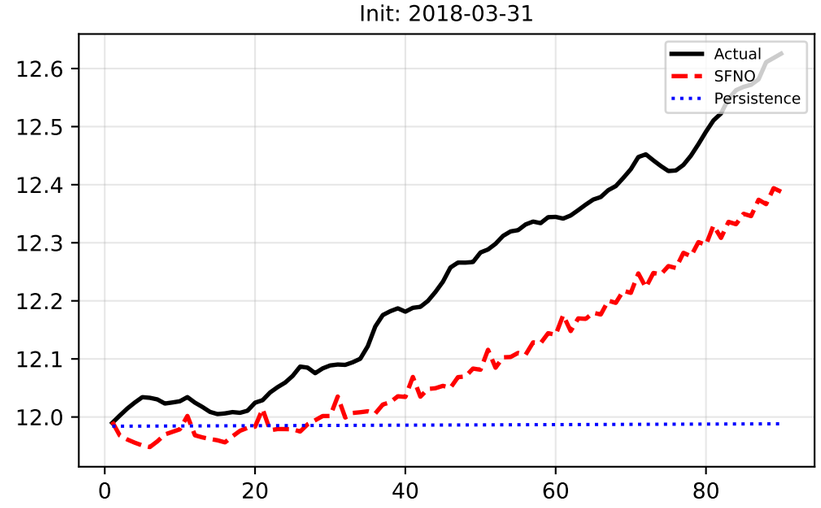


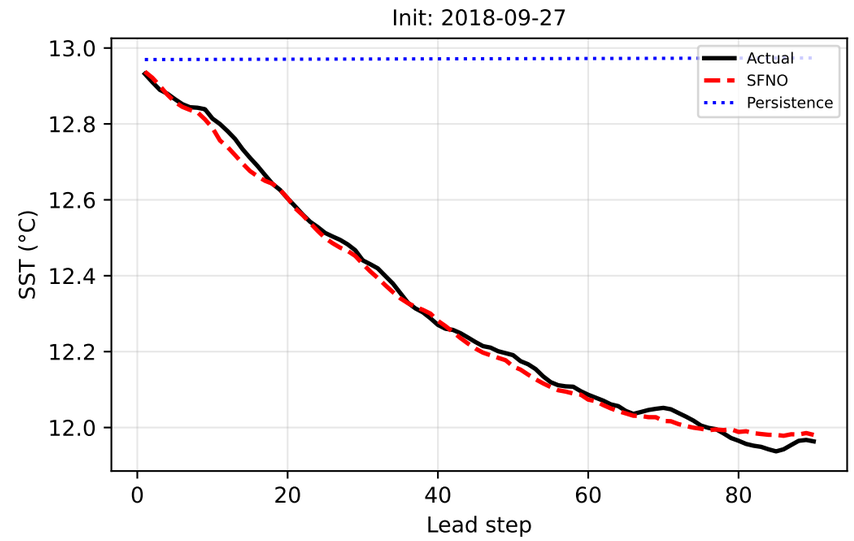


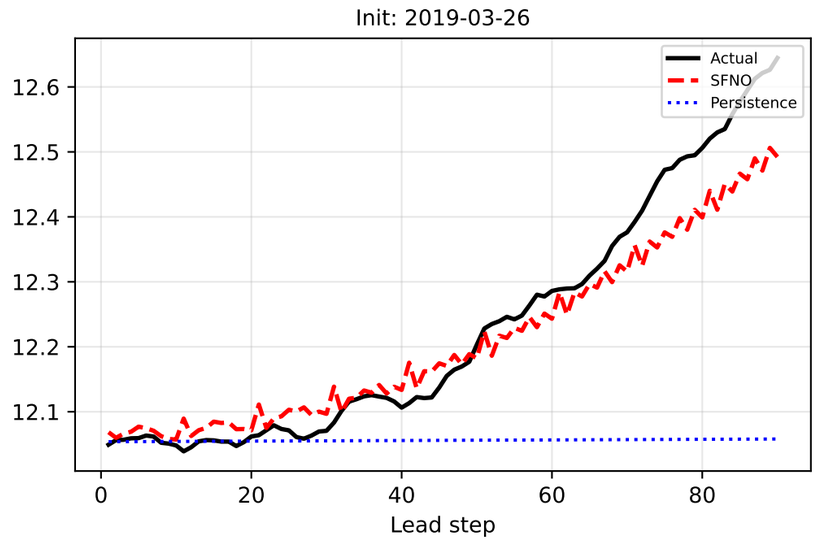


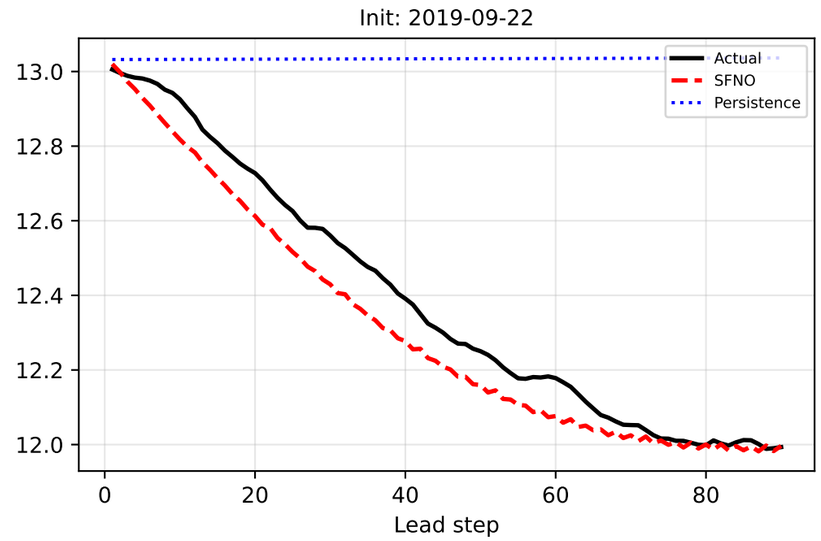






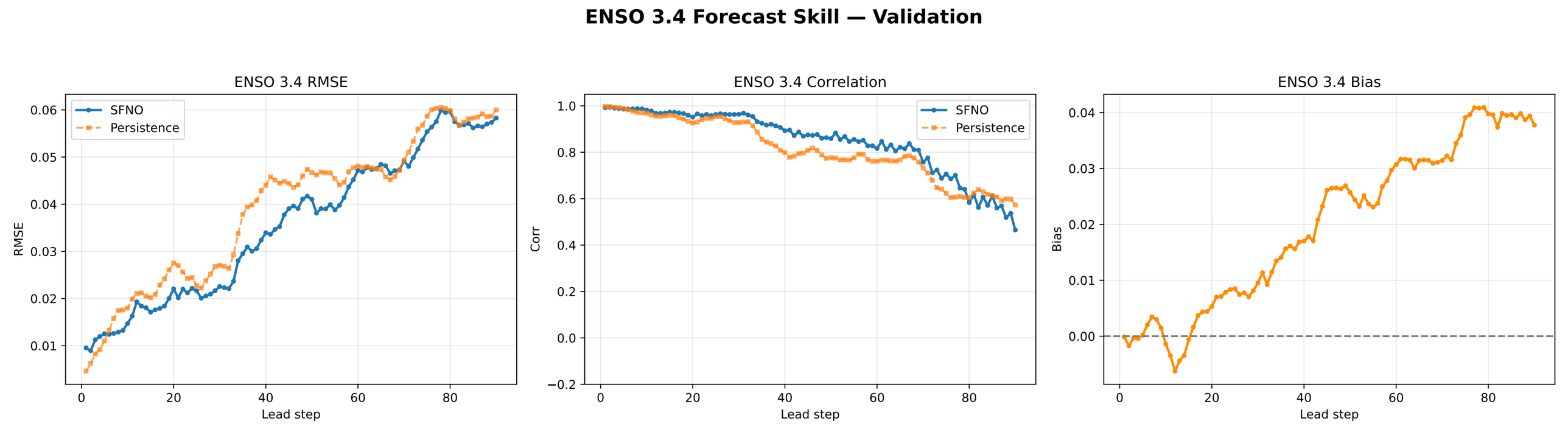




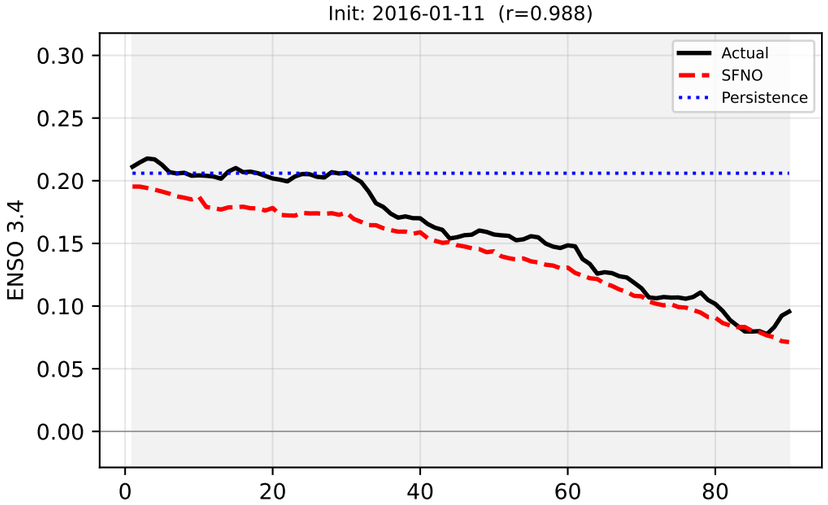


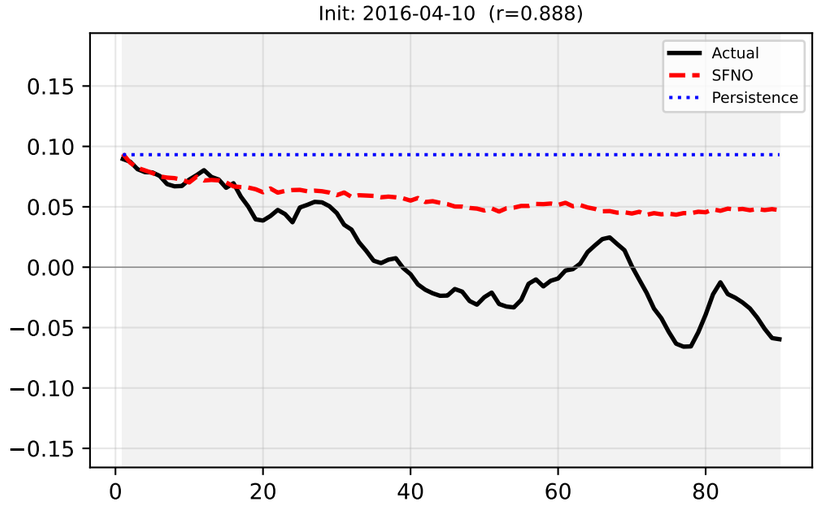
ENSO 3.4 Metrics — Validation

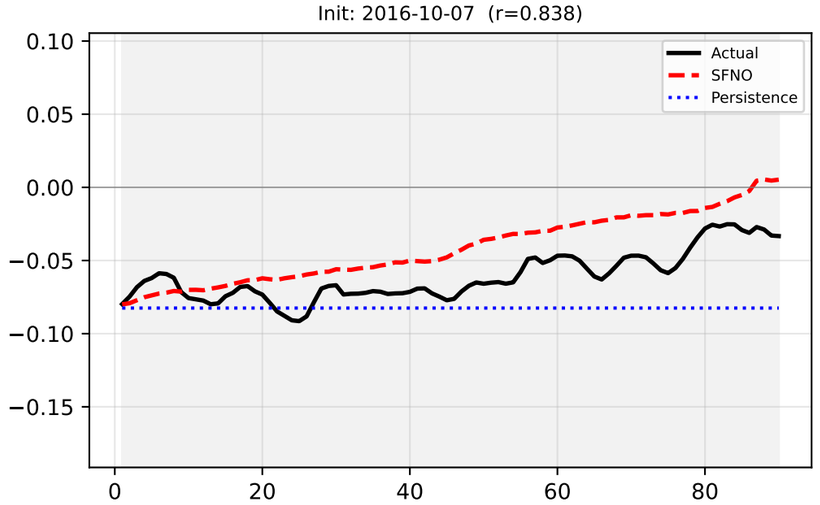
Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0095	0.0046	0.9919	0.9981	-0.0001
2	0.0090	0.0063	0.9939	0.9964	-0.0017
3	0.0113	0.0083	0.9899	0.9938	-0.0004
4	0.0120	0.0092	0.9883	0.9920	-0.0004
5	0.0125	0.0110	0.9857	0.9881	0.0002
6	0.0124	0.0133	0.9860	0.9833	0.0020
7	0.0126	0.0158	0.9868	0.9771	0.0035
8	0.0129	0.0175	0.9877	0.9714	0.0030
9	0.0132	0.0176	0.9870	0.9705	0.0015
10	0.0147	0.0180	0.9821	0.9684	-0.0014
11	0.0163	0.0199	0.9782	0.9615	-0.0035
12	0.0193	0.0211	0.9678	0.9562	-0.0062
13	0.0184	0.0212	0.9681	0.9548	-0.0044
14	0.0181	0.0205	0.9693	0.9576	-0.0035
15	0.0171	0.0202	0.9730	0.9592	-0.0006
16	0.0176	0.0209	0.9723	0.9576	0.0017
17	0.0179	0.0228	0.9701	0.9494	0.0037
18	0.0184	0.0241	0.9667	0.9431	0.0044
19	0.0200	0.0261	0.9595	0.9328	0.0044
20	0.0220	0.0275	0.9514	0.9257	0.0053
21	0.0201	0.0270	0.9652	0.9303	0.0070
22	0.0220	0.0256	0.9577	0.9410	0.0071
23	0.0212	0.0243	0.9634	0.9466	0.0079
24	0.0222	0.0245	0.9579	0.9458	0.0083
25	0.0216	0.0227	0.9628	0.9531	0.0085
26	0.0201	0.0222	0.9660	0.9533	0.0075
27	0.0206	0.0238	0.9639	0.9448	0.0078
28	0.0209	0.0252	0.9628	0.9366	0.0071
29	0.0217	0.0268	0.9629	0.9284	0.0081
30	0.0226	0.0271	0.9636	0.9280	0.0095
31	0.0223	0.0268	0.9684	0.9302	0.0114
32	0.0221	0.0264	0.9606	0.9312	0.0092
33	0.0237	0.0293	0.9546	0.9144	0.0114
34	0.0280	0.0338	0.9316	0.8869	0.0135
35	0.0295	0.0378	0.9246	0.8557	0.0141
36	0.0309	0.0394	0.9170	0.8439	0.0156
37	0.0300	0.0399	0.9209	0.8370	0.0161
38	0.0306	0.0408	0.9137	0.8272	0.0156
39	0.0324	0.0429	0.9069	0.8091	0.0169
40	0.0340	0.0440	0.8928	0.7981	0.0170
41	0.0336	0.0458	0.8962	0.7782	0.0178
42	0.0347	0.0451	0.8716	0.7835	0.0171
43	0.0353	0.0445	0.8873	0.7945	0.0208
44	0.0378	0.0449	0.8692	0.7965	0.0232
45	0.0391	0.0444	0.8751	0.8085	0.0261
46	0.0396	0.0436	0.8723	0.8178	0.0265
47	0.0391	0.0441	0.8769	0.8079	0.0265
48	0.0411	0.0460	0.8604	0.7888	0.0264
49	0.0417	0.0474	0.8629	0.7738	0.0269
50	0.0410	0.0467	0.8587	0.7763	0.0257
51	0.0381	0.0462	0.8832	0.7751	0.0244
52	0.0391	0.0468	0.8554	0.7672	0.0232
53	0.0390	0.0467	0.8678	0.7680	0.0251
54	0.0399	0.0466	0.8462	0.7665	0.0237
55	0.0388	0.0454	0.8555	0.7763	0.0231
56	0.0398	0.0441	0.8457	0.7919	0.0238
57	0.0414	0.0446	0.8516	0.7902	0.0268
58	0.0437	0.0469	0.8276	0.7685	0.0278
59	0.0452	0.0477	0.8280	0.7616	0.0297
60	0.0471	0.0481	0.8165	0.7614	0.0307
61	0.0468	0.0478	0.8466	0.7659	0.0317
62	0.0478	0.0480	0.8123	0.7649	0.0317
63	0.0474	0.0477	0.8318	0.7628	0.0315
64	0.0475	0.0474	0.8053	0.7625	0.0300
65	0.0485	0.0473	0.8218	0.7666	0.0314
66	0.0482	0.0458	0.8149	0.7823	0.0315
67	0.0465	0.0452	0.8376	0.7845	0.0315
68	0.0471	0.0459	0.8113	0.7758	0.0309
69	0.0473	0.0472	0.8093	0.7581	0.0311
70	0.0492	0.0493	0.7572	0.7330	0.0314
71	0.0480	0.0510	0.7759	0.7100	0.0322
72	0.0499	0.0533	0.7112	0.6791	0.0316
73	0.0517	0.0559	0.7236	0.6473	0.0345
74	0.0536	0.0567	0.6876	0.6412	0.0360
75	0.0554	0.0587	0.7060	0.6229	0.0391
76	0.0564	0.0600	0.6854	0.6055	0.0396
77	0.0575	0.0603	0.7006	0.6065	0.0409
78	0.0599	0.0605	0.6456	0.6097	0.0408
79	0.0594	0.0603	0.6409	0.6050	0.0409
80	0.0596	0.0599	0.5832	0.6048	0.0398
81	0.0575	0.0581	0.6173	0.6243	0.0396
82	0.0568	0.0566	0.5619	0.6391	0.0374
83	0.0568	0.0574	0.6069	0.6298	0.0399
84	0.0571	0.0581	0.5710	0.6190	0.0395
85	0.0562	0.0583	0.6116	0.6136	0.0397
86	0.0566	0.0585	0.5596	0.6072	0.0391
87	0.0564	0.0591	0.5689	0.5917	0.0398
88	0.0570	0.0586	0.5189	0.5994	0.0387
89	0.0573	0.0587	0.5368	0.5977	0.0394
90	0.0583	0.0600	0.4649	0.5725	0.0377

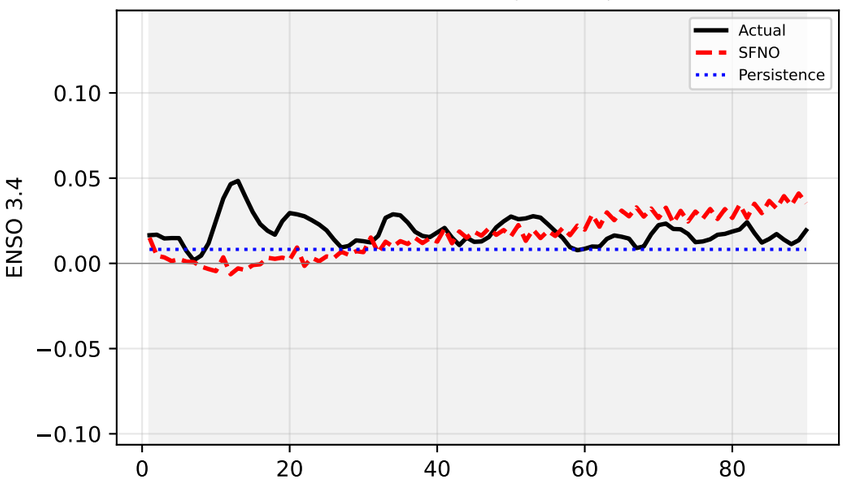

ENSO 3.4 Forecast Skill — Validation
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
SFNO
Persistence
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

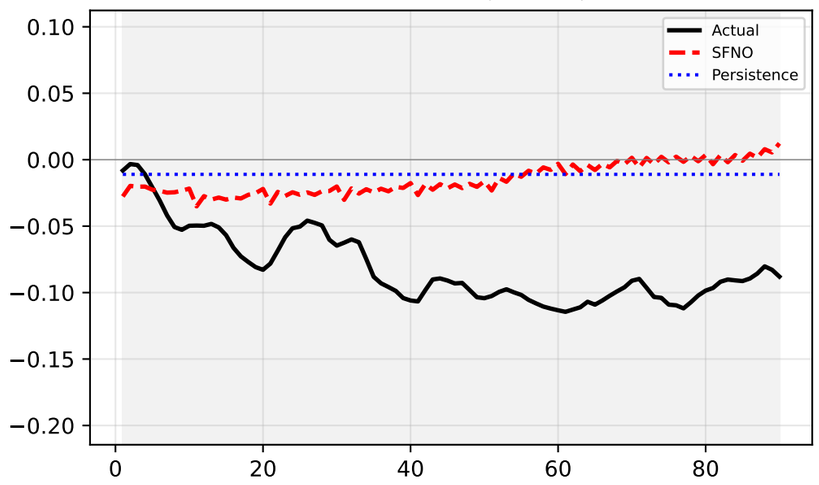
ENSO 3.4 Index — Validation

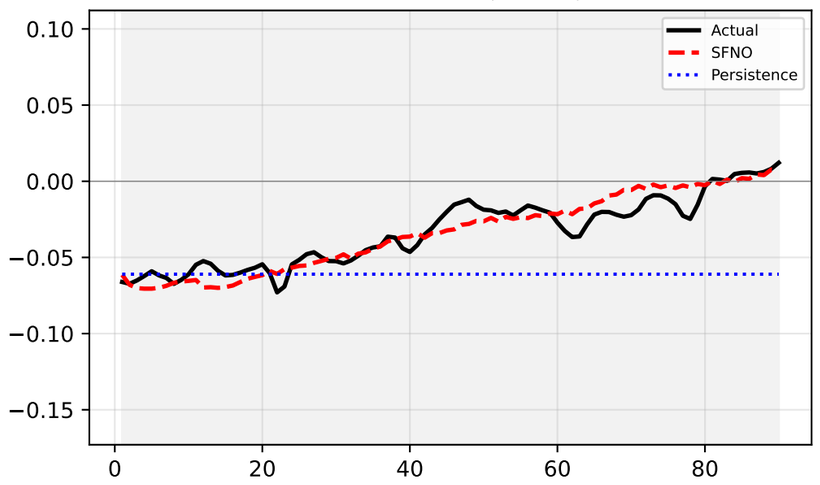


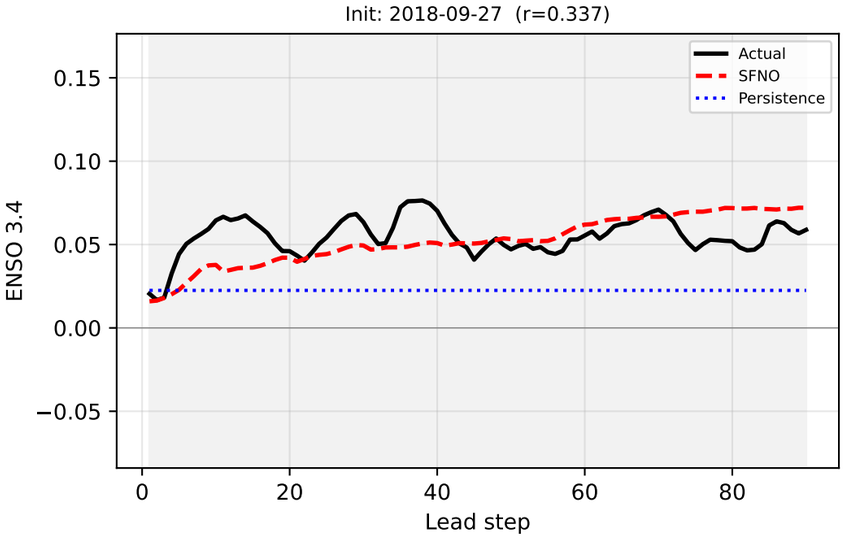


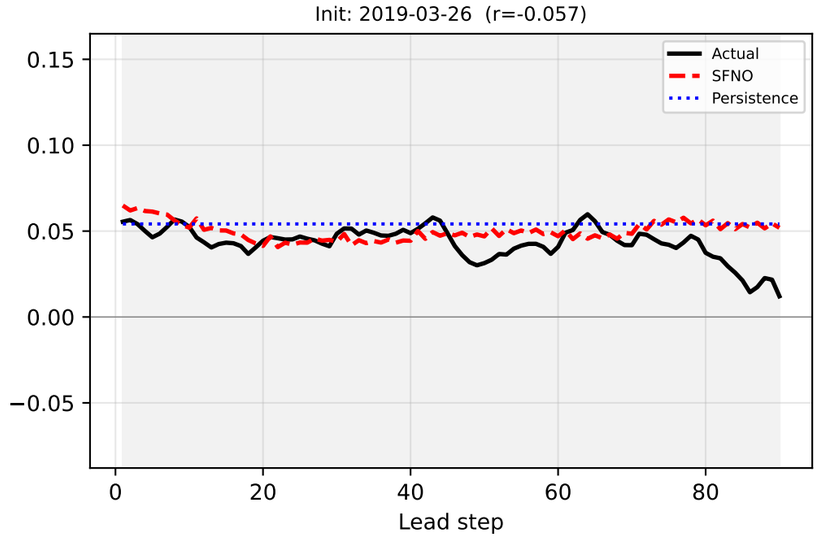


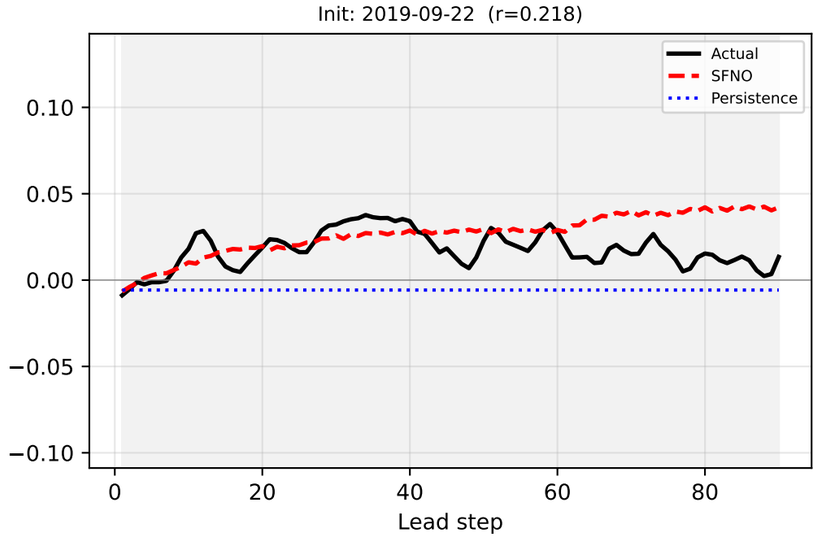




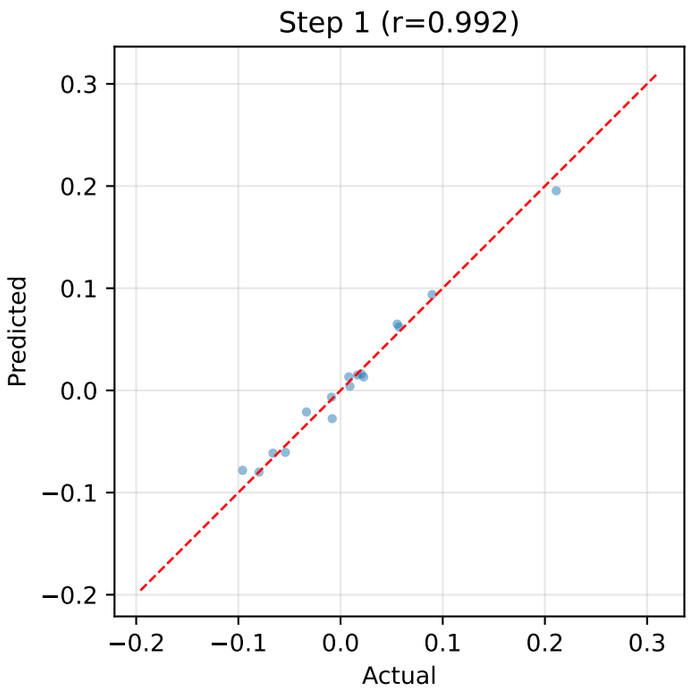


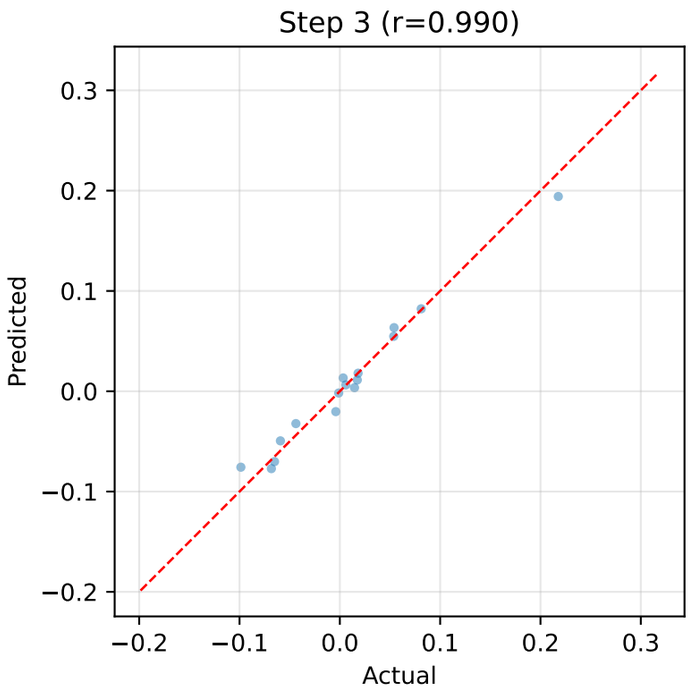


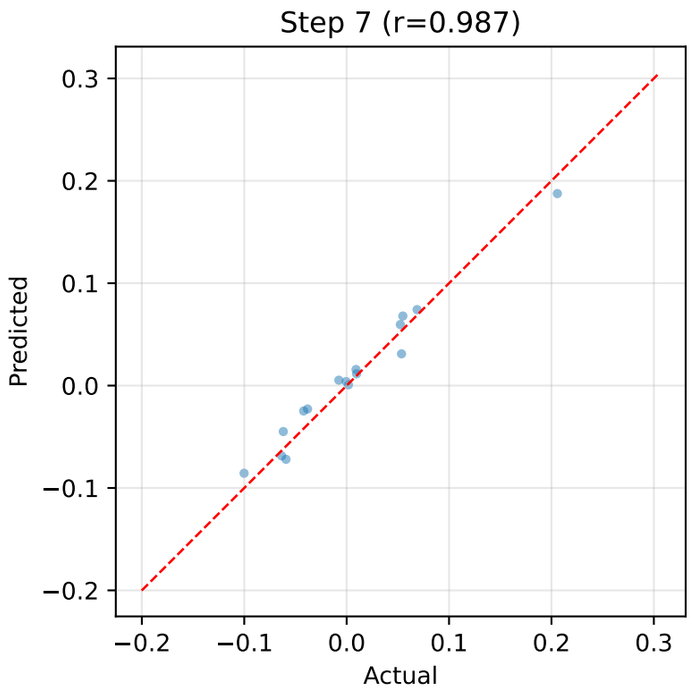


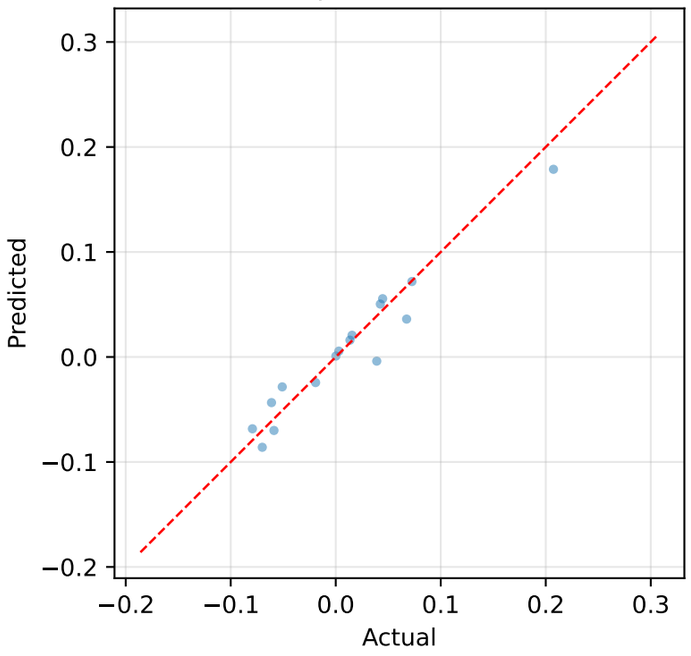


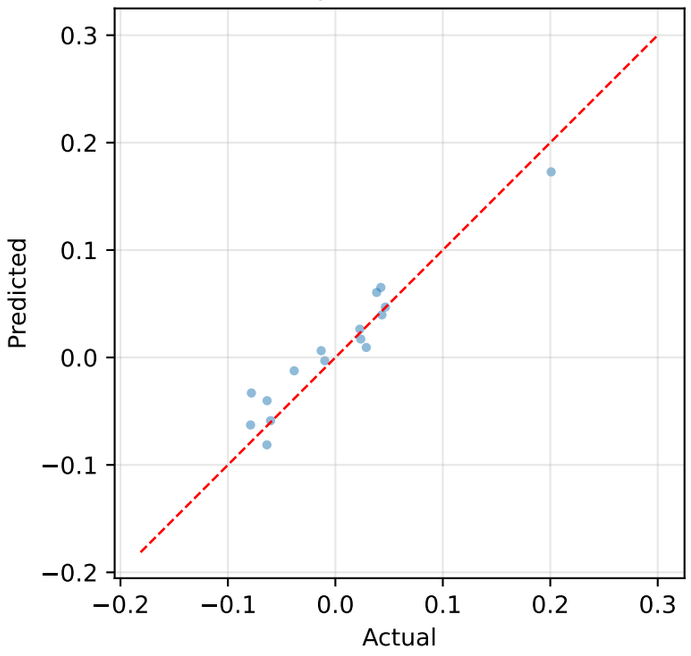
ENSO 3.4 Scatter — Validation

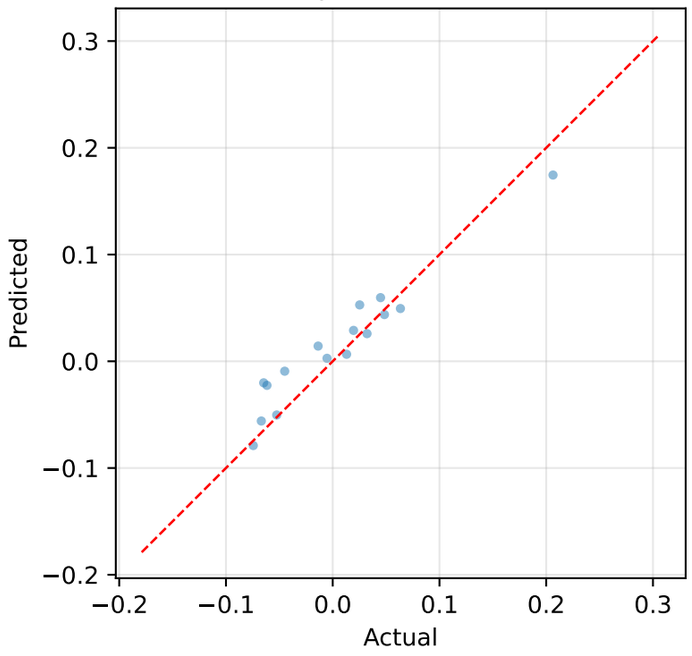


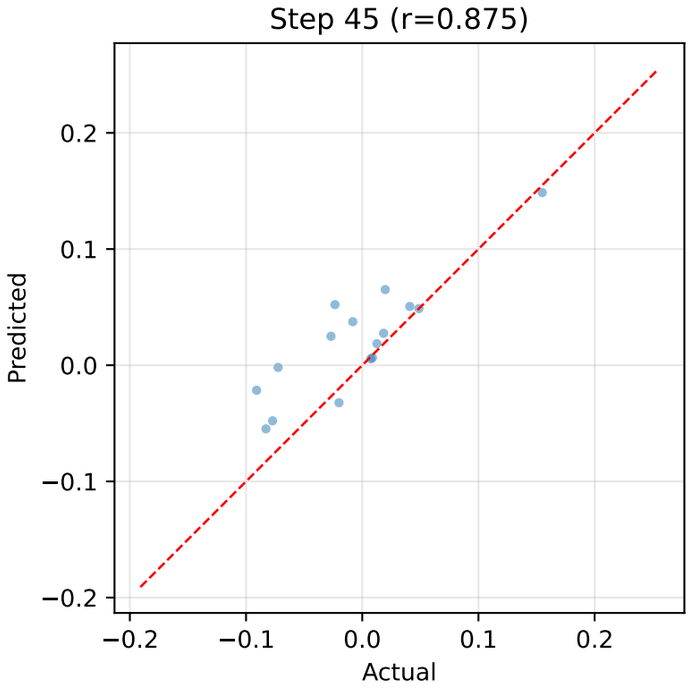


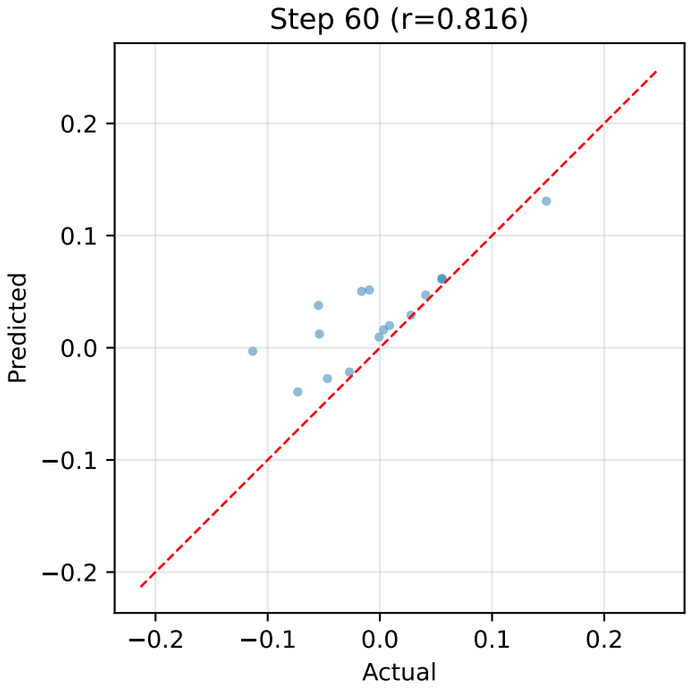


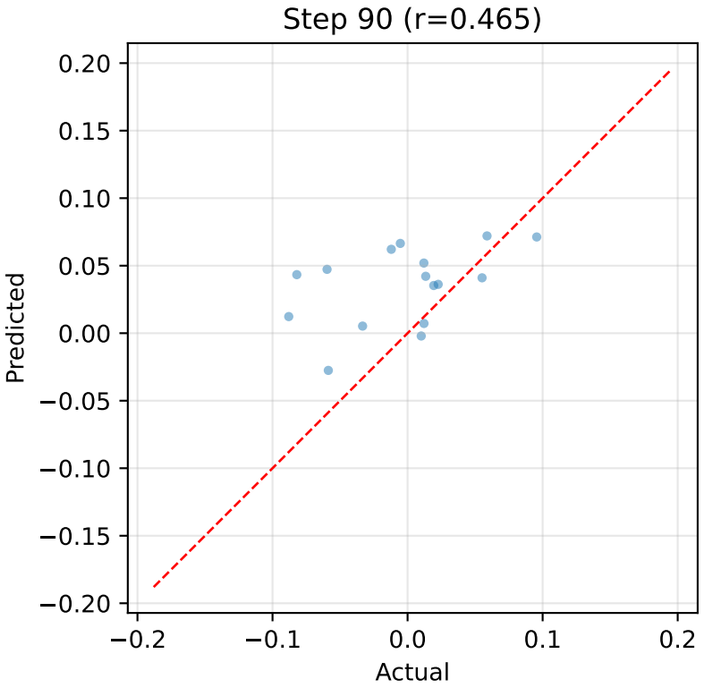








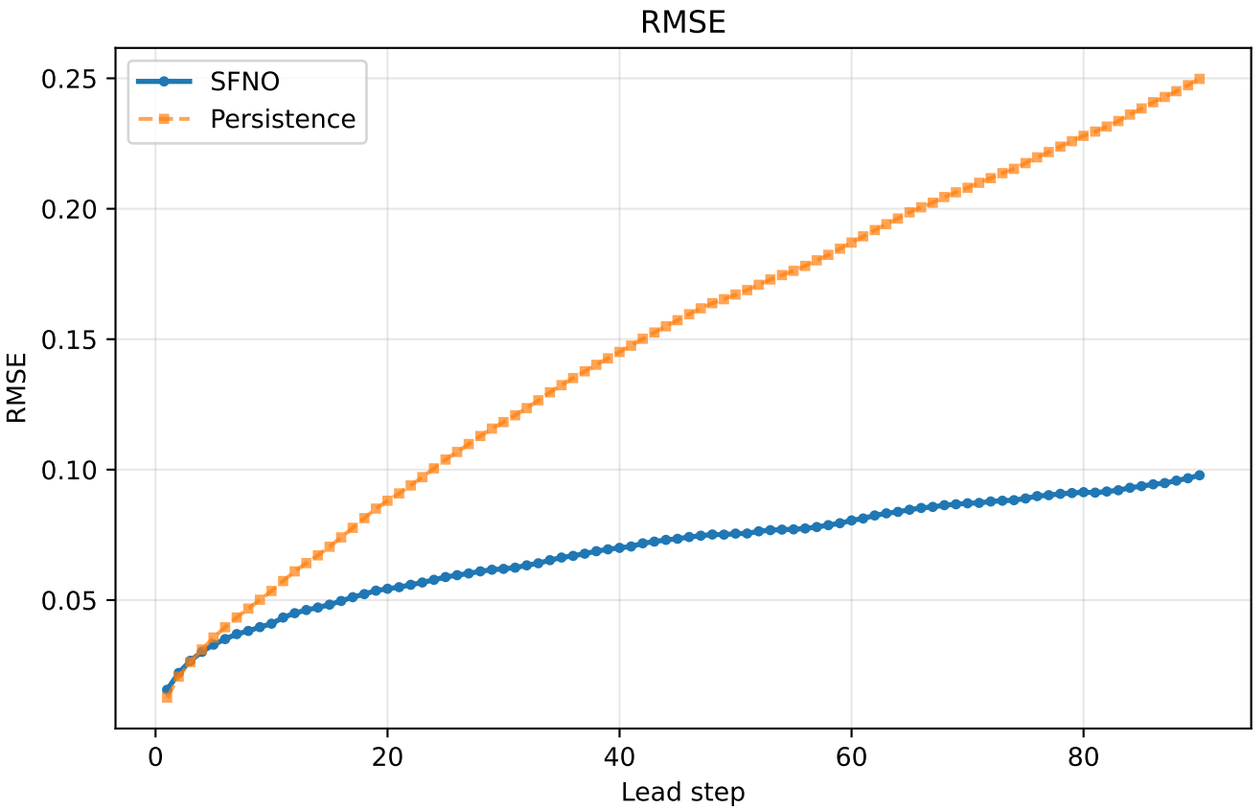


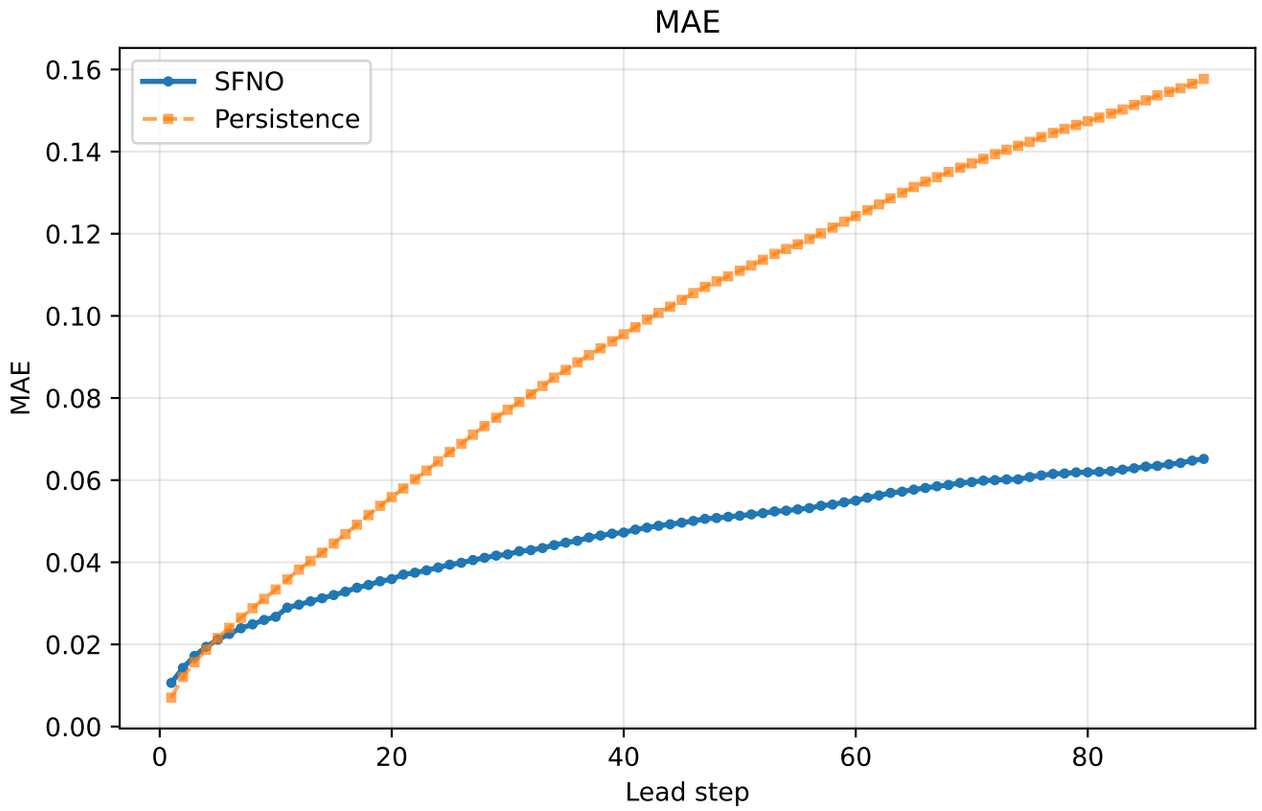


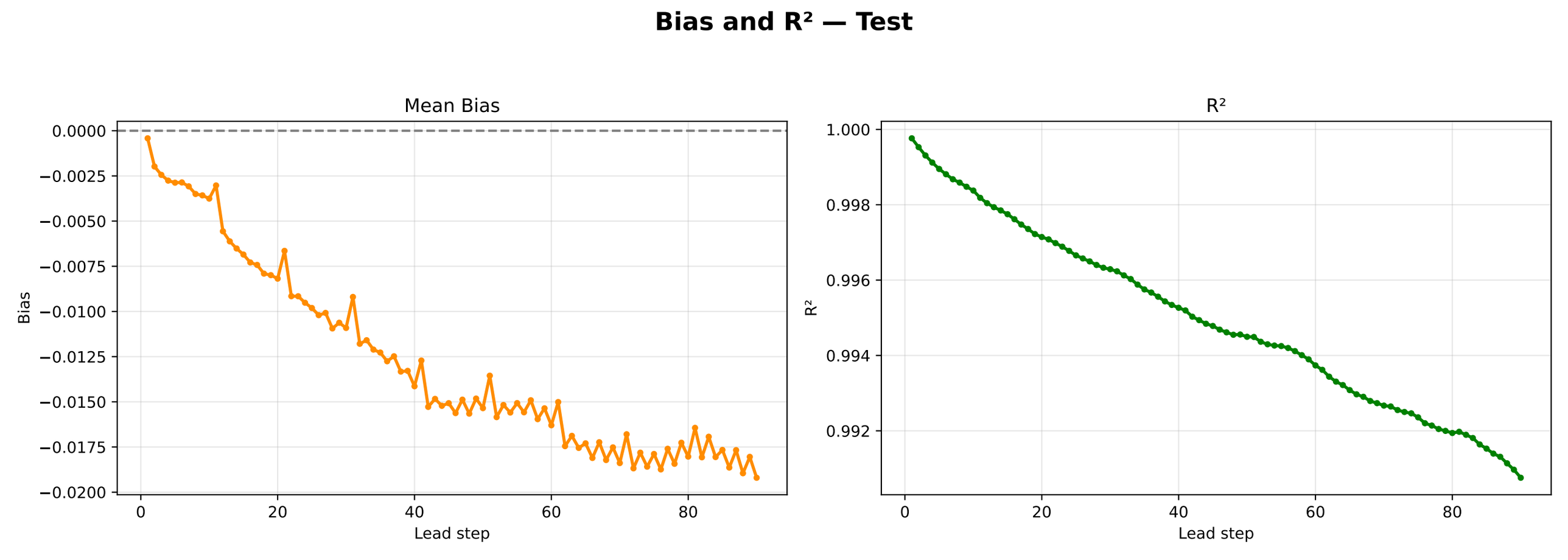
Metrics by Lead Step — Test

Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0156	0.0126	-23.7%	0.0106	-0.0004	0.9730	0.9796	0.9998
2	0.0221	0.0206	-6.9%	0.0143	-0.0020	0.9422	0.9462	0.9995
3	0.0267	0.0262	-2.0%	0.0172	-0.0024	0.9114	0.9132	0.9993
4	0.0301	0.0311	2.9%	0.0194	-0.0028	0.8835	0.8780	0.9991
5	0.0329	0.0357	7.7%	0.0212	-0.0029	0.8576	0.8411	0.9990
6	0.0351	0.0396	11.5%	0.0226	-0.0029	0.8346	0.8072	0.9988
7	0.0370	0.0433	14.7%	0.0239	-0.0031	0.8126	0.7728	0.9987
8	0.0382	0.0467	18.3%	0.0249	-0.0035	0.7942	0.7389	0.9986
9	0.0396	0.0501	20.9%	0.0260	-0.0036	0.7764	0.7063	0.9985
10	0.0409	0.0535	23.4%	0.0267	-0.0038	0.7602	0.6743	0.9984
11	0.0433	0.0573	24.4%	0.0289	-0.0030	0.7355	0.6410	0.9982
12	0.0450	0.0610	26.3%	0.0297	-0.0056	0.7130	0.6068	0.9980
13	0.0462	0.0642	28.0%	0.0305	-0.0061	0.6940	0.5769	0.9979
14	0.0471	0.0672	29.8%	0.0312	-0.0065	0.6784	0.5500	0.9979
15	0.0482	0.0705	31.5%	0.0320	-0.0069	0.6620	0.5219	0.9977
16	0.0497	0.0740	32.9%	0.0328	-0.0073	0.6433	0.4938	0.9976
17	0.0511	0.0777	34.2%	0.0338	-0.0074	0.6241	0.4629	0.9975
18	0.0523	0.0814	35.8%	0.0345	-0.0079	0.6069	0.4320	0.9974
19	0.0536	0.0851	37.0%	0.0354	-0.0080	0.5878	0.4030	0.9972
20	0.0543	0.0881	38.3%	0.0359	-0.0082	0.5728	0.3801	0.9971
21	0.0549	0.0909	39.5%	0.0370	-0.0066	0.5575	0.3592	0.9971
22	0.0559	0.0939	40.5%	0.0375	-0.0092	0.5383	0.3356	0.9970
23	0.0567	0.0971	41.6%	0.0380	-0.0092	0.5244	0.3131	0.9969
24	0.0578	0.1005	42.5%	0.0387	-0.0095	0.5088	0.2889	0.9968
25	0.0588	0.1039	43.4%	0.0394	-0.0098	0.4918	0.2670	0.9967
26	0.0595	0.1068	44.2%	0.0399	-0.0102	0.4745	0.2487	0.9966
27	0.0602	0.1098	45.2%	0.0405	-0.0101	0.4568	0.2287	0.9965
28	0.0610	0.1129	45.9%	0.0411	-0.0109	0.4398	0.2138	0.9964
29	0.0616	0.1157	46.7%	0.0416	-0.0106	0.4242	0.2032	0.9963
30	0.0620	0.1182	47.6%	0.0419	-0.0109	0.4094	0.1947	0.9963
31	0.0624	0.1208	48.3%	0.0427	-0.0092	0.3917	0.1854	0.9962
32	0.0633	0.1236	48.8%	0.0430	-0.0118	0.3714	0.1758	0.9961
33	0.0641	0.1266	49.3%	0.0435	-0.0116	0.3548	0.1635	0.9960
34	0.0653	0.1296	49.6%	0.0442	-0.0121	0.3356	0.1532	0.9959
35	0.0663	0.1324	49.9%	0.0448	-0.0123	0.3164	0.1445	0.9958
36	0.0669	0.1351	50.5%	0.0452	-0.0128	0.2985	0.1347	0.9957
37	0.0678	0.1377	50.8%	0.0460	-0.0125	0.2798	0.1282	0.9956
38	0.0687	0.1402	51.0%	0.0465	-0.0133	0.2622	0.1243	0.9954
39	0.0695	0.1427	51.3%	0.0470	-0.0133	0.2503	0.1209	0.9953
40	0.0700	0.1451	51.7%	0.0472	-0.0141	0.2383	0.1145	0.9953
41	0.0705	0.1475	52.2%	0.0480	-0.0127	0.2223	0.1073	0.9952
42	0.0718	0.1502	52.2%	0.0484	-0.0153	0.2040	0.1002	0.9950
43	0.0724	0.1526	52.5%	0.0489	-0.0148	0.1953	0.0960	0.9949
44	0.0731	0.1549	52.8%	0.0493	-0.0152	0.1831	0.0912	0.9948
45	0.0735	0.1573	53.3%	0.0496	-0.0151	0.1722	0.0831	0.9948
46	0.0742	0.1596	53.5%	0.0501	-0.0156	0.1593	0.0757	0.9947
47	0.0747	0.1618	53.8%	0.0506	-0.0149	0.1516	0.0700	0.9946
48	0.0751	0.1638	54.1%	0.0508	-0.0157	0.1420	0.0685	0.9945
49	0.0751	0.1653	54.6%	0.0511	-0.0148	0.1362	0.0718	0.9946
50	0.0755	0.1671	54.8%	0.0513	-0.0153	0.1282	0.0747	0.9945
51	0.0755	0.1688	55.3%	0.0517	-0.0136	0.1231	0.0768	0.9945
52	0.0764	0.1709	55.3%	0.0520	-0.0158	0.1104	0.0753	0.9944
53	0.0768	0.1729	55.6%	0.0524	-0.0152	0.1029	0.0730	0.9943
54	0.0770	0.1746	55.9%	0.0526	-0.0156	0.0972	0.0736	0.9943
55	0.0771	0.1762	56.2%	0.0529	-0.0151	0.0937	0.0758	0.9942
56	0.0774	0.1781	56.5%	0.0532	-0.0156	0.0868	0.0736	0.9942
57	0.0780	0.1802	56.7%	0.0537	-0.0149	0.0798	0.0694	0.9941
58	0.0787	0.1823	56.8%	0.0541	-0.0160	0.0730	0.0654	0.9940
59	0.0795	0.1847	57.0%	0.0546	-0.0154	0.0660	0.0584	0.9939
60	0.0805	0.1870	56.9%	0.0550	-0.0163	0.0599	0.0535	0.9937
61	0.0813	0.1894	57.1%	0.0557	-0.0150	0.0527	0.0470	0.9936
62	0.0824	0.1918	57.0%	0.0563	-0.0175	0.0398	0.0385	0.9934
63	0.0833	0.1941	57.1%	0.0569	-0.0169	0.0286	0.0316	0.9933
64	0.0838	0.1963	57.3%	0.0572	-0.0176	0.0218	0.0248	0.9932
65	0.0846	0.1986	57.4%	0.0577	-0.0173	0.0169	0.0192	0.9931
66	0.0853	0.2006	57.5%	0.0581	-0.0181	0.0097	0.0171	0.9930
67	0.0857	0.2023	57.6%	0.0585	-0.0172	0.0072	0.0183	0.9929
68	0.0864	0.2045	57.8%	0.0588	-0.0182	0.0034	0.0135	0.9928
69	0.0867	0.2063	58.0%	0.0593	-0.0175	0.0012	0.0125	0.9927
70	0.0871	0.2080	58.1%	0.0595	-0.0184	-0.0019	0.0122	0.9927
71	0.0872	0.2099	58.4%	0.0599	-0.0168	-0.0032	0.0110	0.9926
72	0.0878	0.2117	58.5%	0.0600	-0.0187	-0.0046	0.0120	0.9926
73	0.0881	0.2136	58.8%	0.0602	-0.0178	-0.0021	0.0109	0.9925
74	0.0883	0.2153	59.0%	0.0602	-0.0186	-0.0015	0.0133	0.9925
75	0.0889	0.2175	59.1%	0.0608	-0.0179	-0.0034	0.0086	0.9924
76	0.0899	0.2197	59.1%	0.0612	-0.0187	-0.0092	0.0050	0.9922
77	0.0902	0.2217	59.3%	0.0615	-0.0176	-0.0093	0.0049	0.9921
78	0.0907	0.2238	59.5%	0.0616	-0.0184	-0.0100	0.0037	0.9920
79	0.0910	0.2259	59.7%	0.0619	-0.0173	-0.0092	0.0019	0.9920
80	0.0914	0.2279	59.9%	0.0619	-0.0180	-0.0089	-0.0015	0.9919
81	0.0912	0.2296	60.3%	0.0620	-0.0164	-0.0041	-0.0020	0.9920
82	0.0916	0.2315	60.4%	0.0622	-0.0181	-0.0058	-0.0056	0.9919
83	0.0921	0.2336	60.6%	0.0626	-0.0169	-0.0052	-0.0076	0.9918
84	0.0931	0.2361	60.6%	0.0629	-0.0181	-0.0069	-0.0090	0.9916
85	0.0937	0.2385	60.7%	0.0633	-0.0177	-0.0041	-0.0119	0.9915
86	0.0944	0.2408	60.8%	0.0635	-0.0186	-0.0048	-0.0147	0.9914
87	0.0948	0.2428	60.9%	0.0639	-0.0177	-0.0037	-0.0142	0.9913
88	0.0958	0.2451	60.9%	0.0642	-0.0190	-0.0062	-0.0133	0.9911
89	0.0967	0.2474	60.9%	0.0647	-0.0180	-0.0075	-0.0151	0.9910
90	0.0978	0.2498	60.8%	0.0652	-0.0192	-0.0108	-0.0180	0.9908

Forecast Skill vs Lead Step — Test

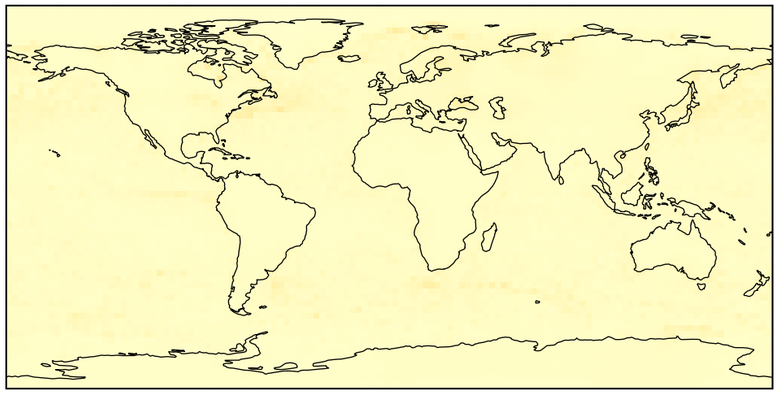




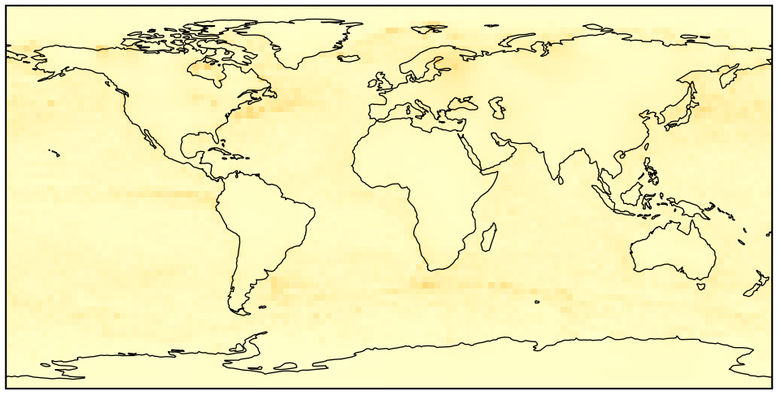

Bias and R² — Test
Mean Bias
Bias
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
−0.0125
−0.0150
−0.0175
−0.0200
0
20
40
60
80
Lead step
R²
R²
1.000
0.998
0.996
0.994
0.992
0
20
40
60
80
Lead step

Spatial RMSE (all grid) — Test

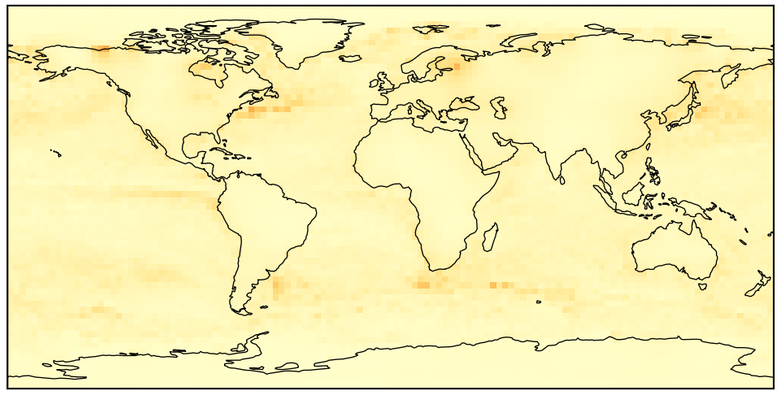
Step 1



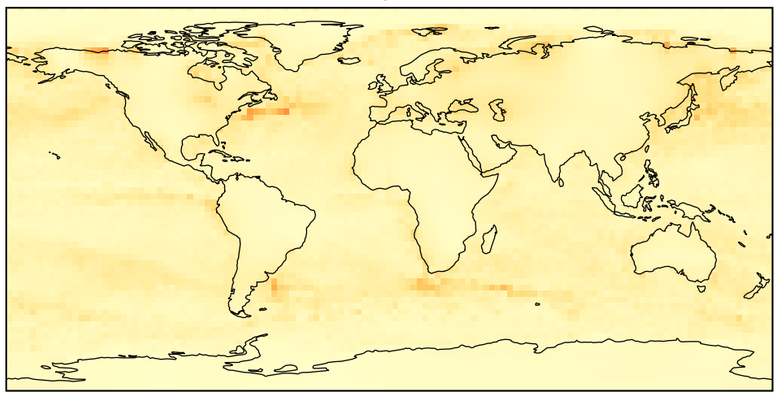
Step 3



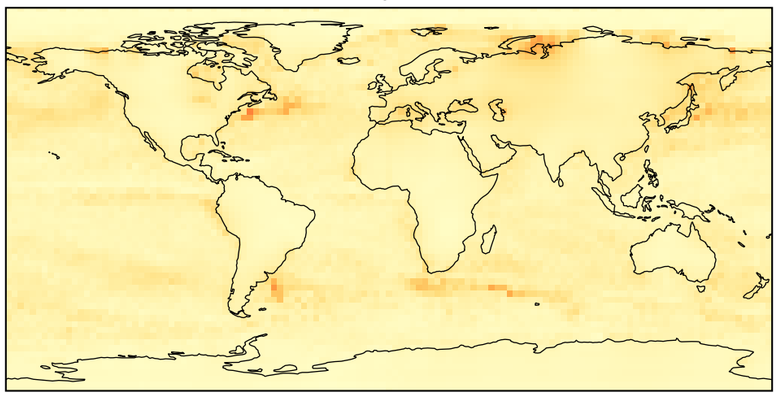
Step 7



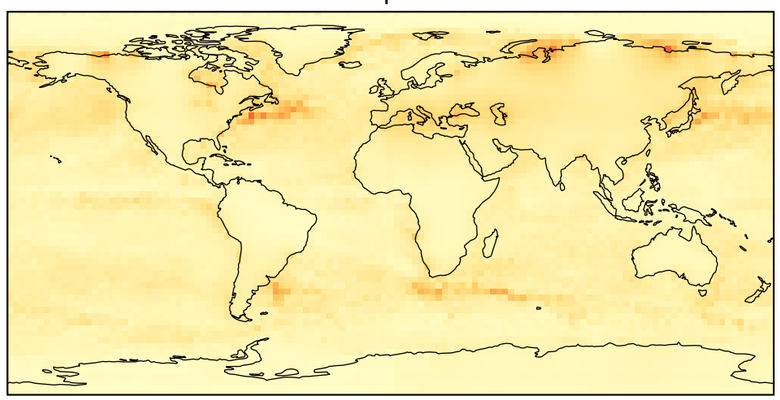
Step 14



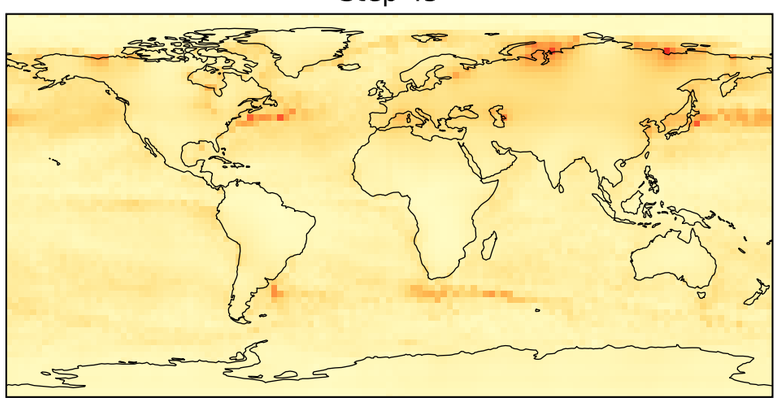
Step 21



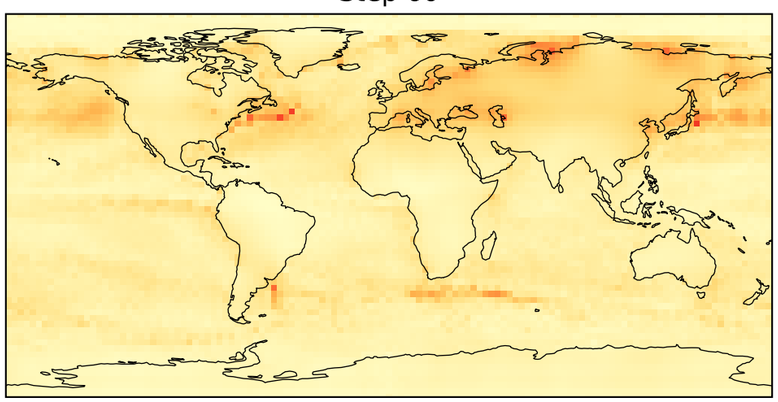
Step 30



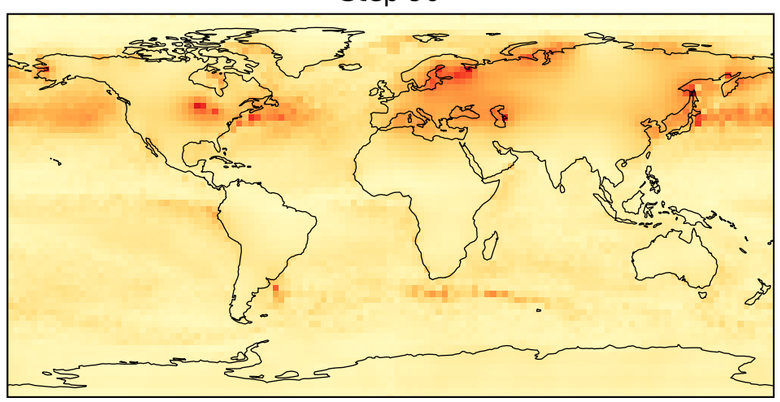
Step 45



Step 60

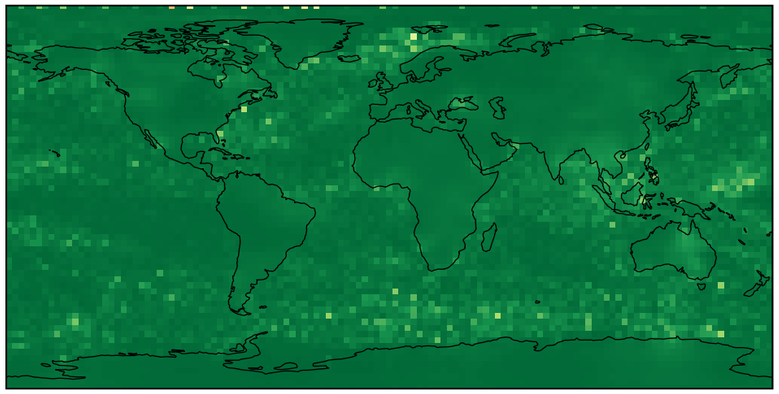


Step 90

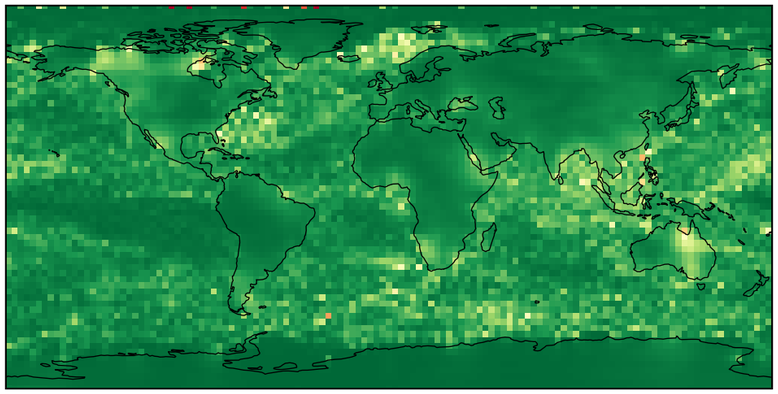


Spatial ACC (all grid) — Test

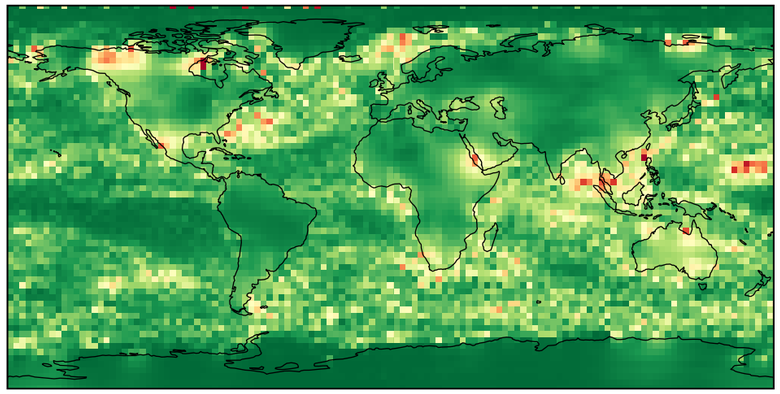
Step 1



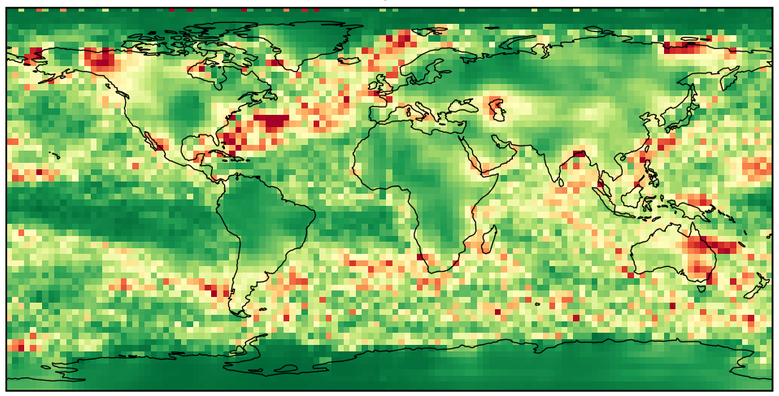
Step 3



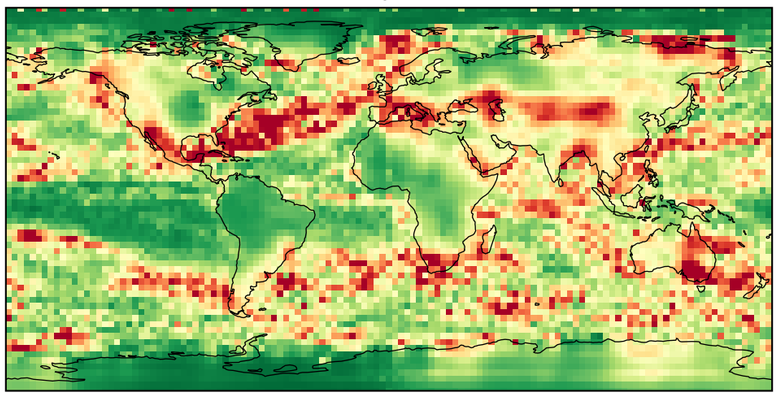
Step 7



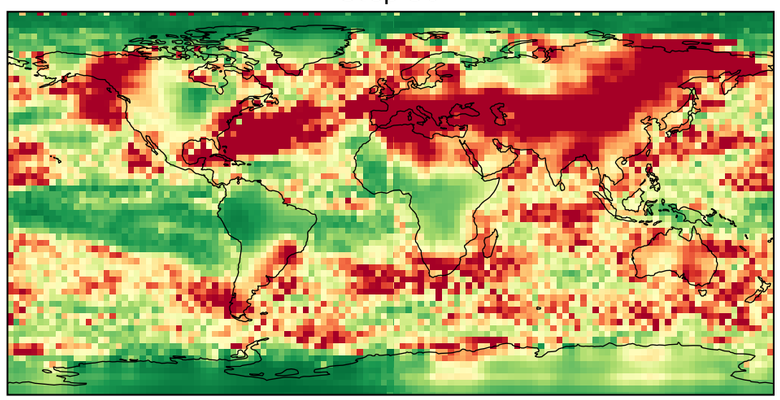
Step 14



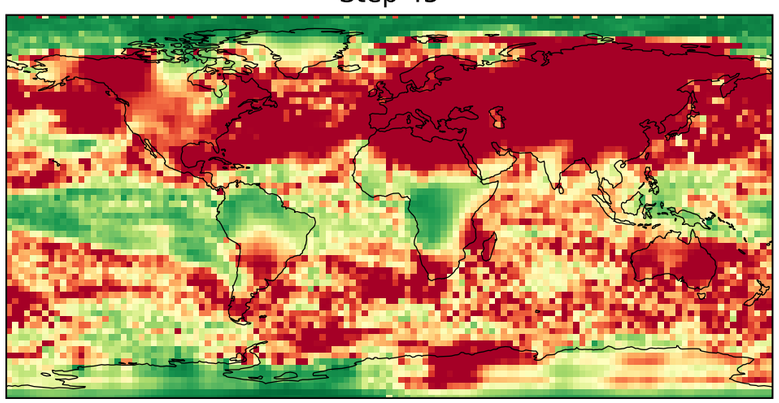
Step 21



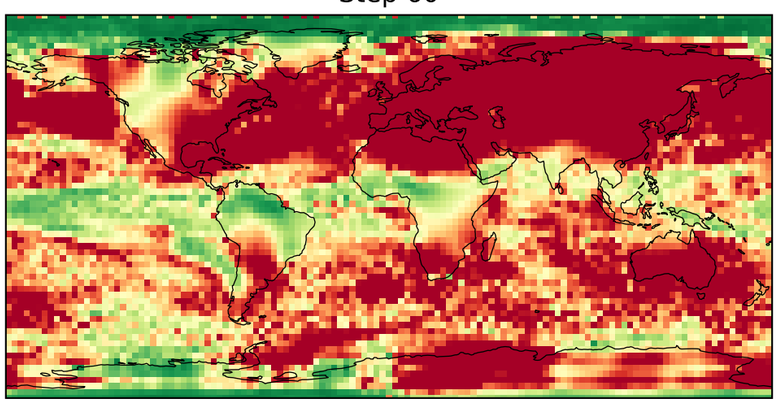
Step 30



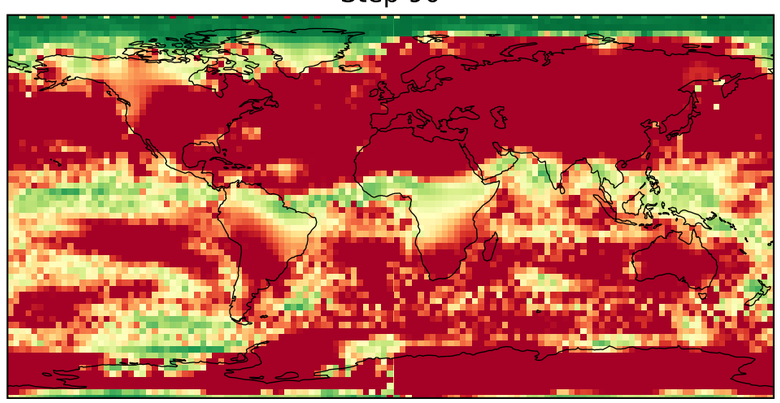
Step 45

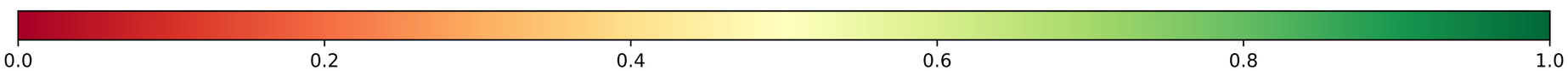


Step 60



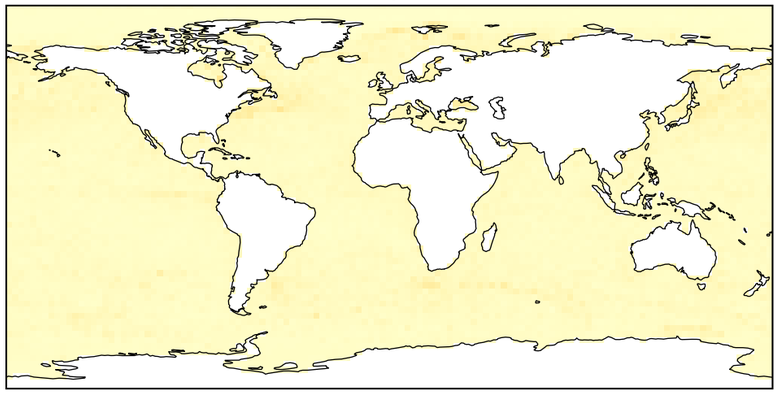
Step 90



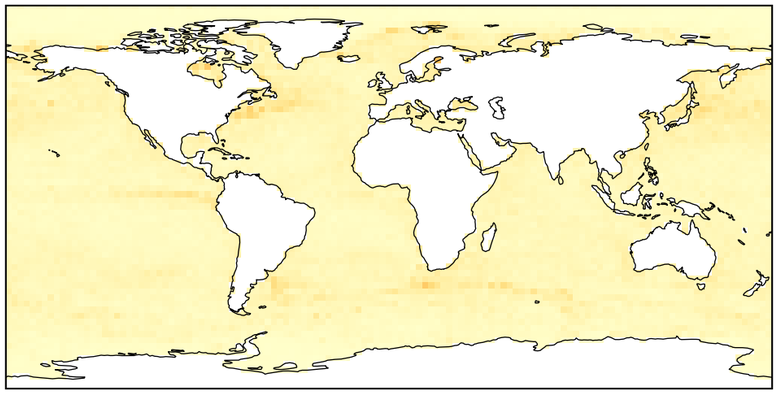


Spatial RMSE (ocean only) — Test

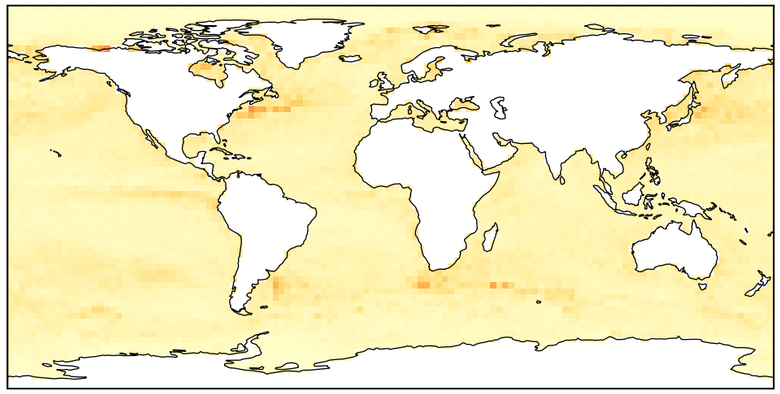
Step 1



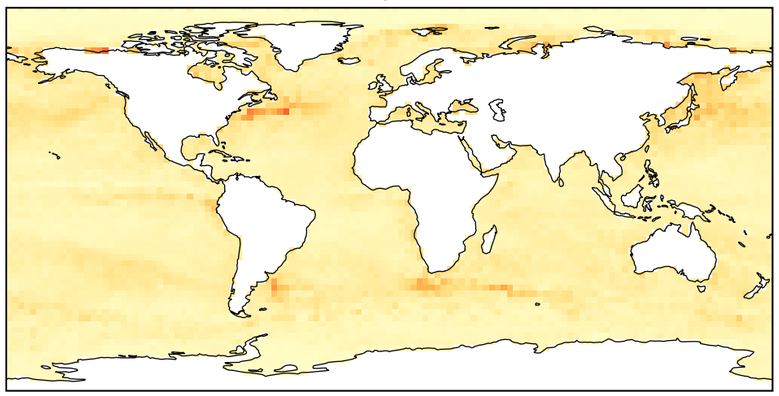
Step 3



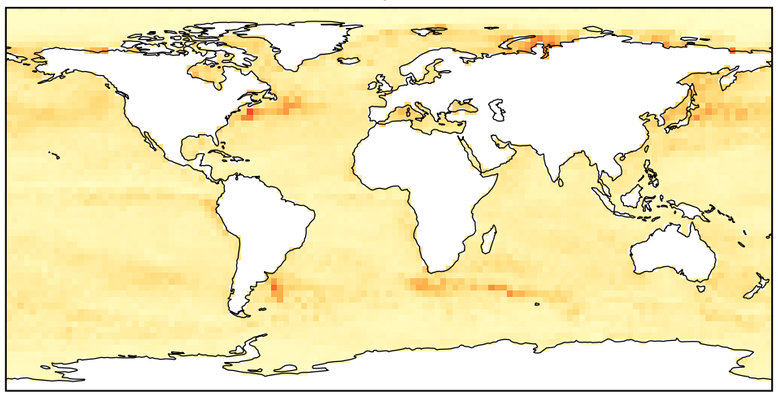
Step 7



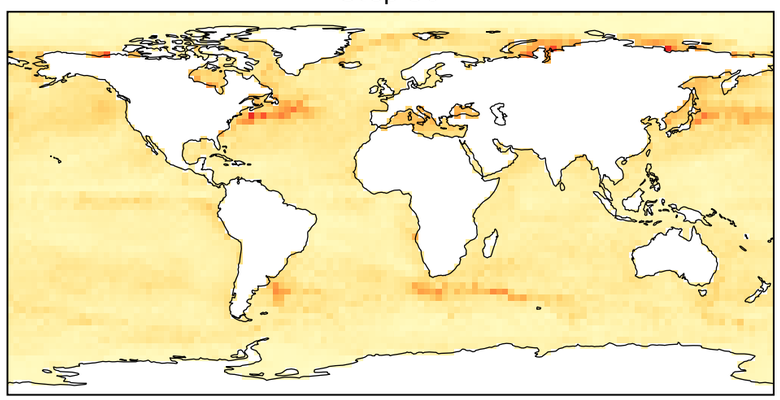
Step 14



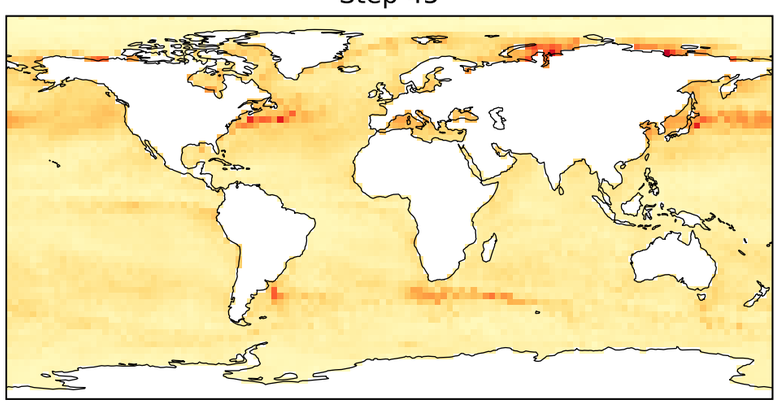
Step 21



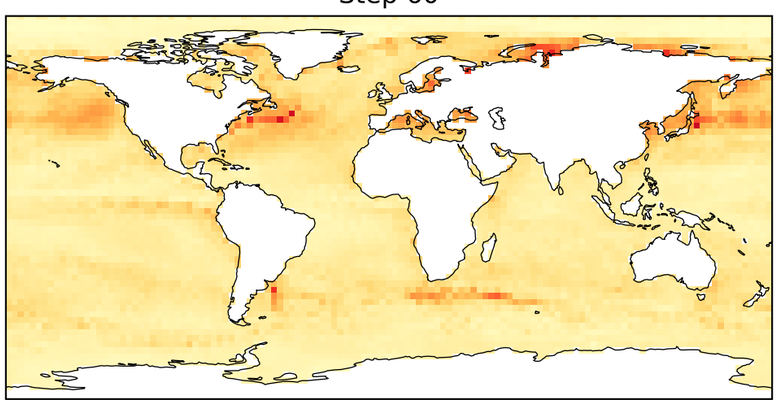
Step 30



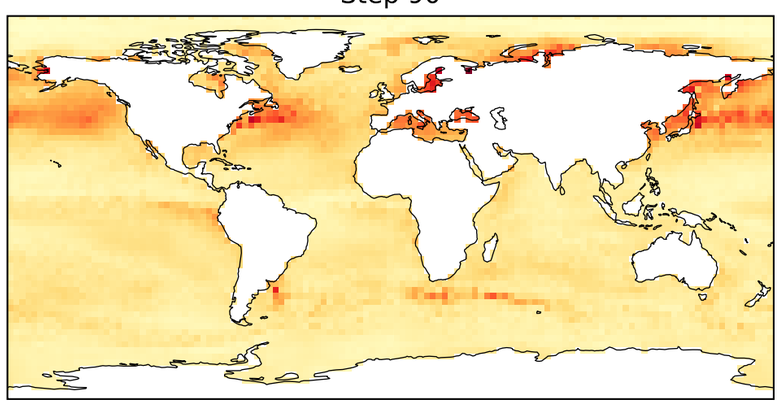
Step 45



Step 60

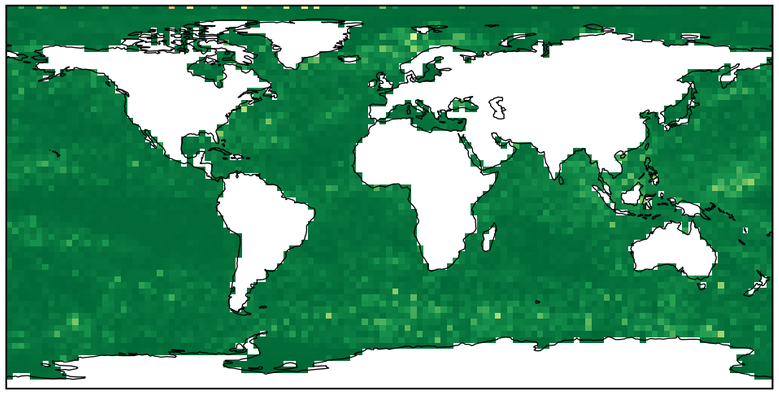


Step 90

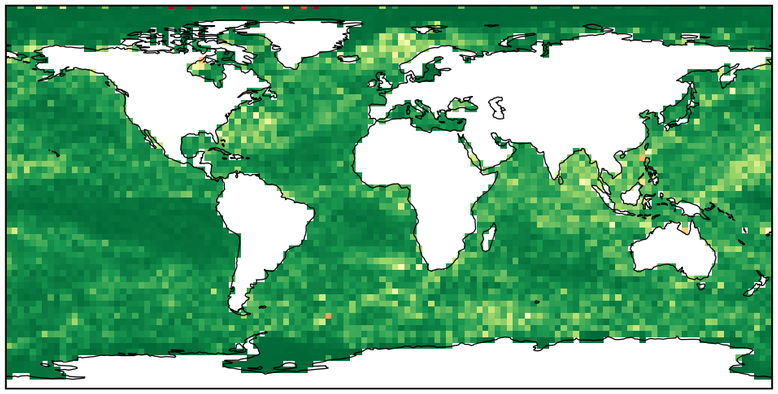


Spatial ACC (ocean only) — Test

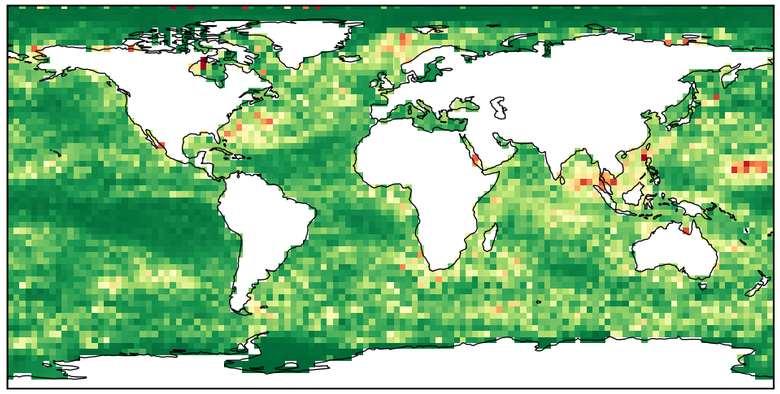
Step 1



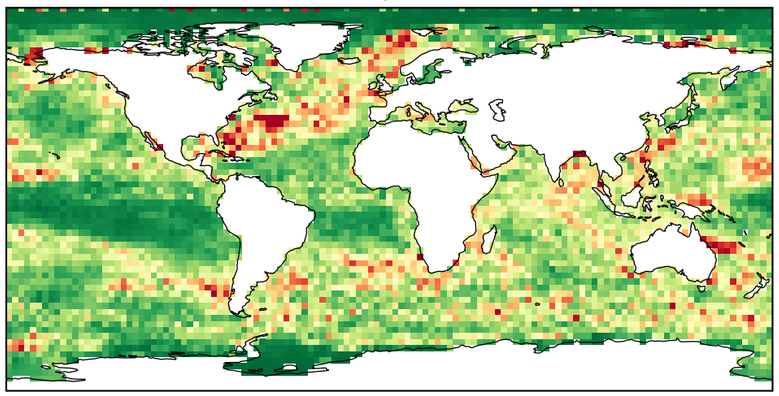
Step 3



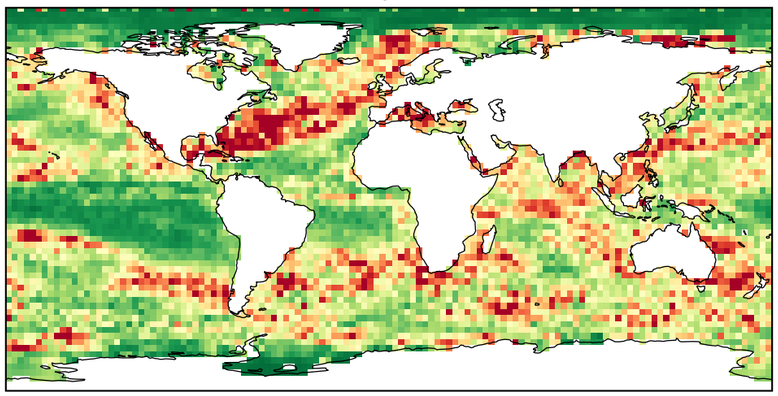
Step 7



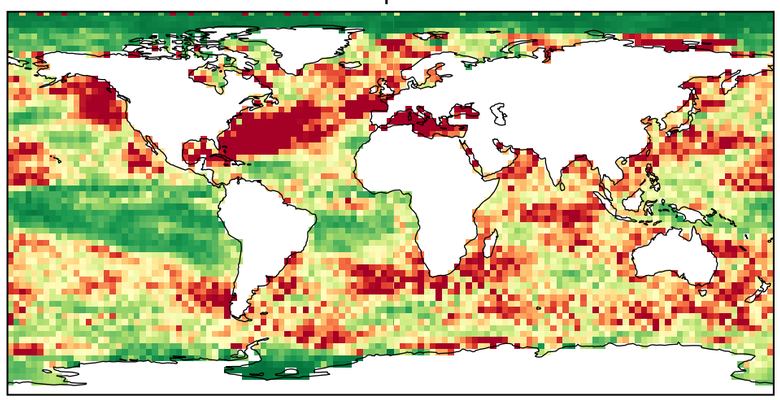
Step 14



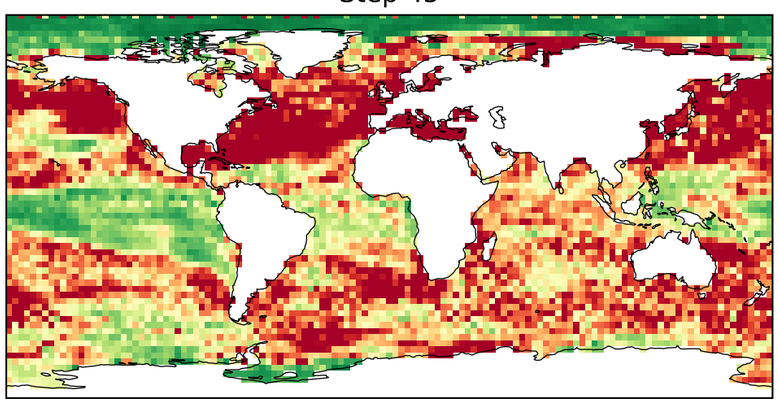
Step 21



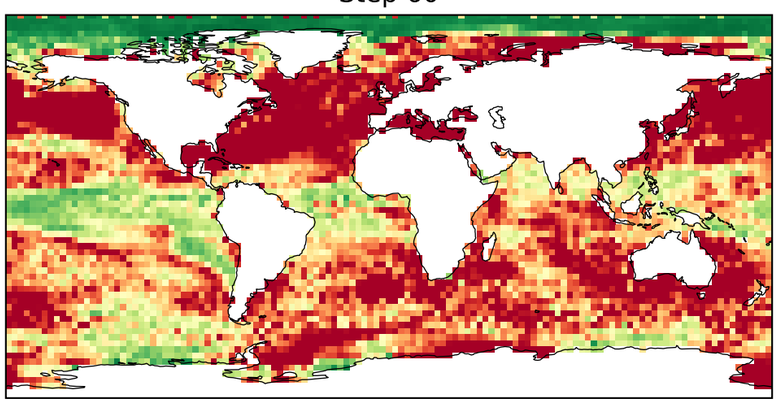
Step 30



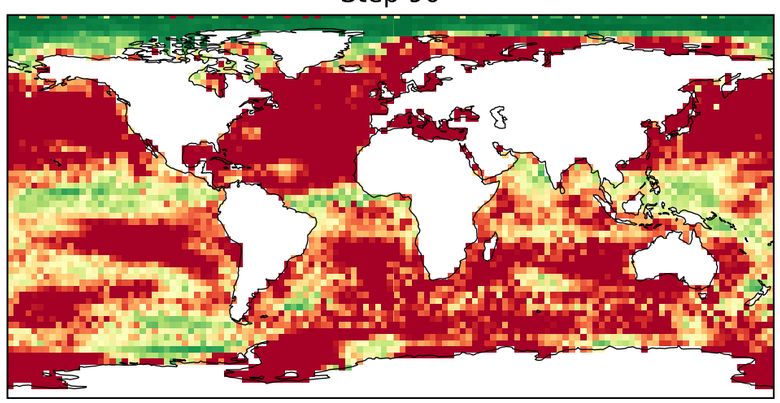
Step 45

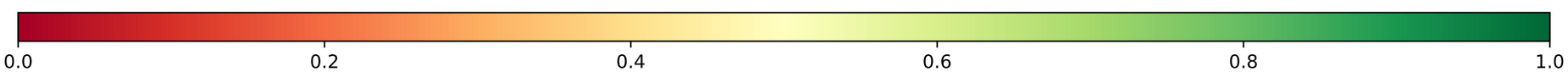


Step 60



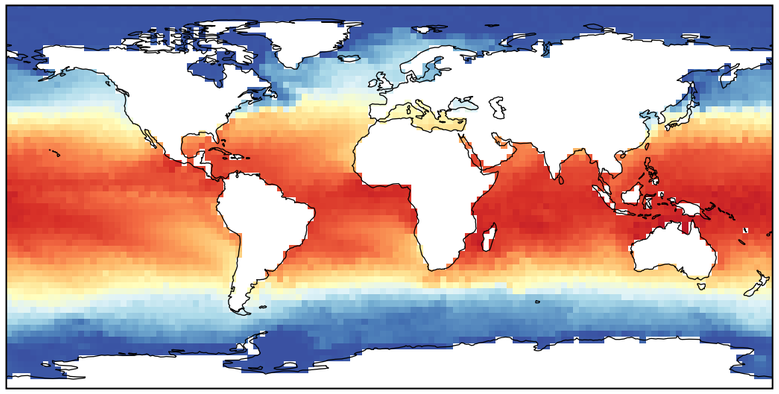
Step 90



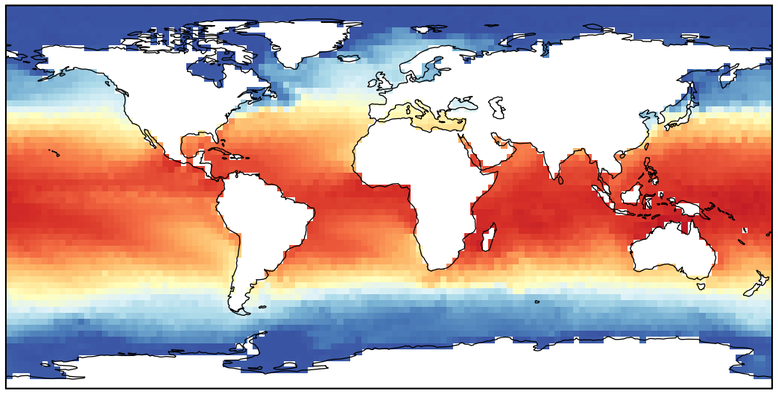


Chain init: 2020-01-11 — Test (°C)

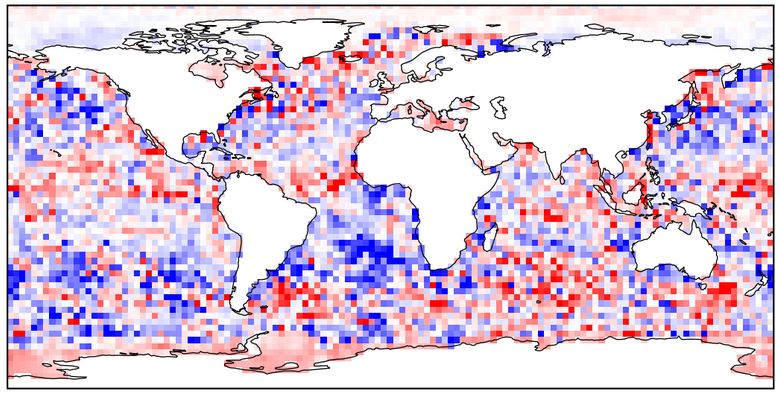
Actual — Step 1 (°C)



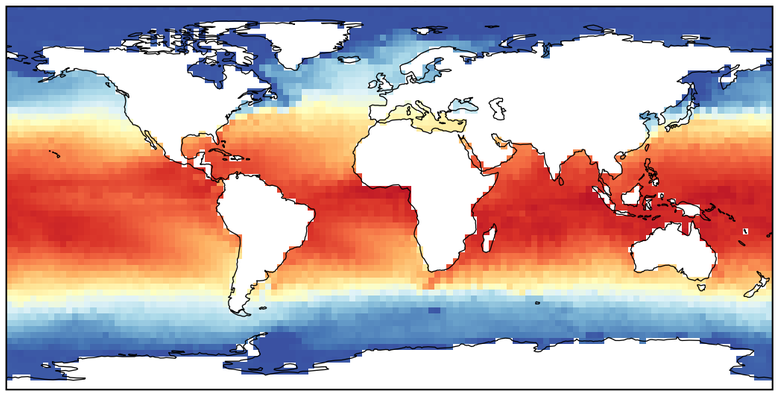
SFNO — Step 1 (°C)



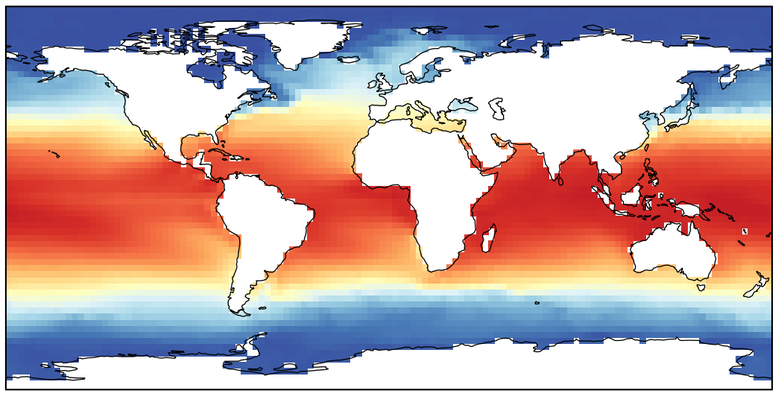
Error — Step 1 (°C)



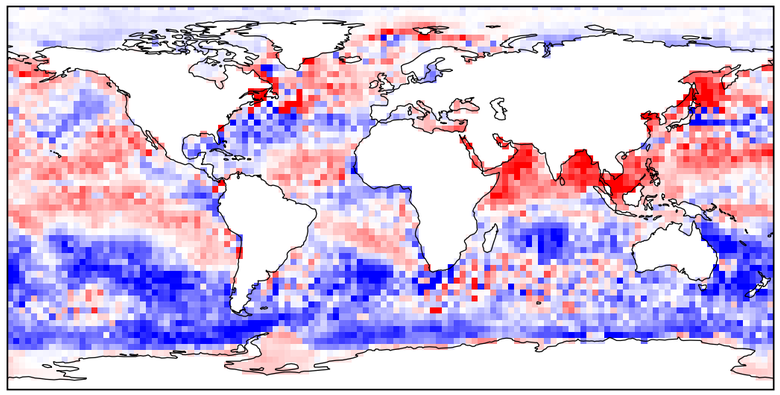
Actual — Step 46 (°C)



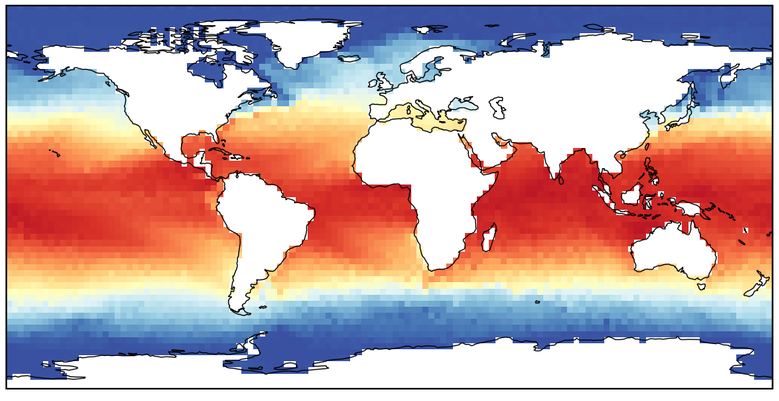
SFNO — Step 46 (°C)



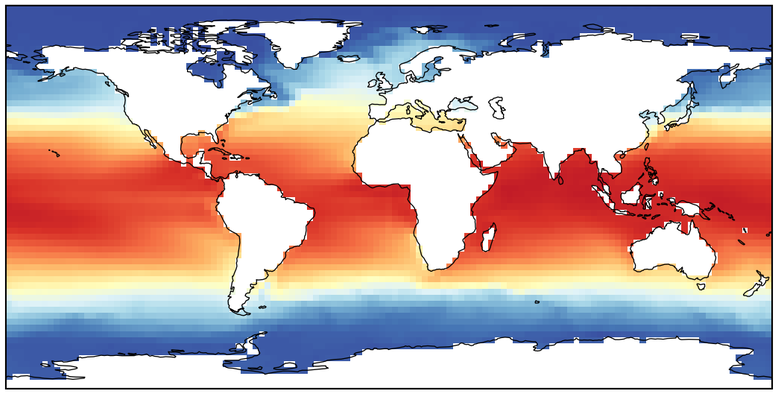
Error — Step 46 (°C)



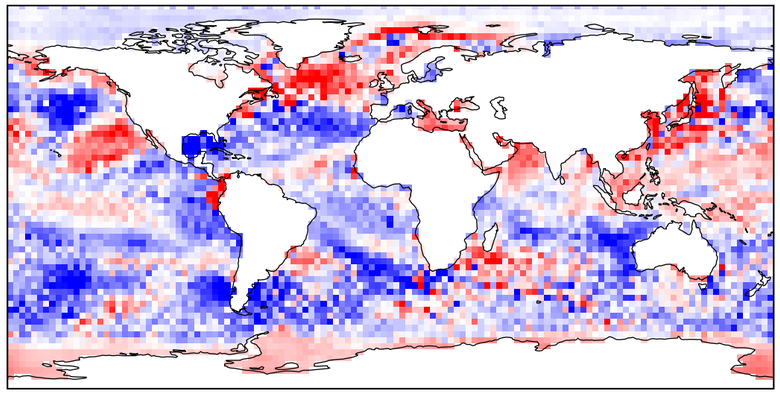
Actual — Step 90 (°C)

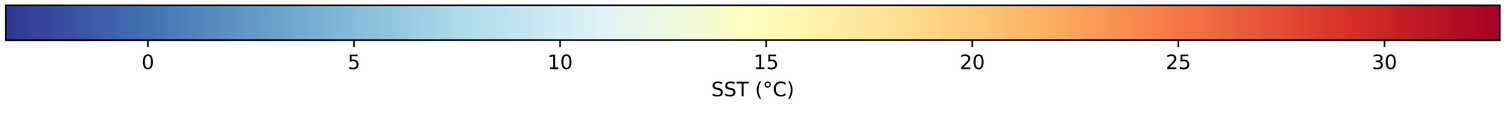


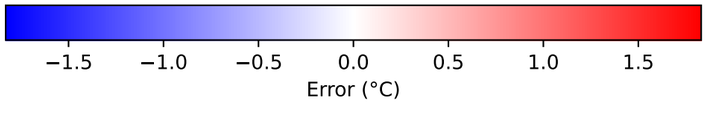
SFNO — Step 90 (°C)



Error — Step 90 (°C)

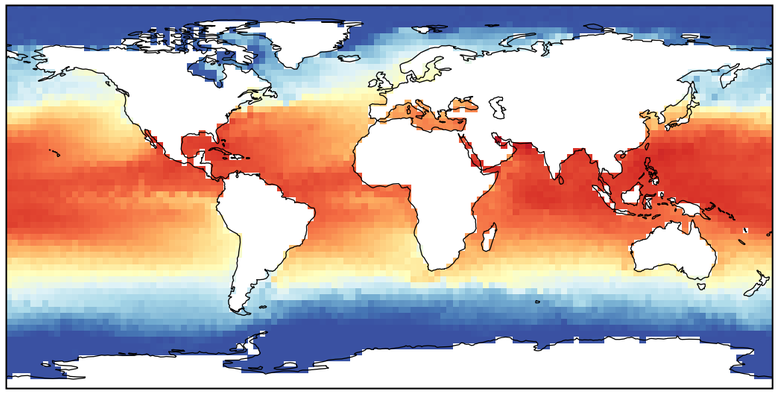




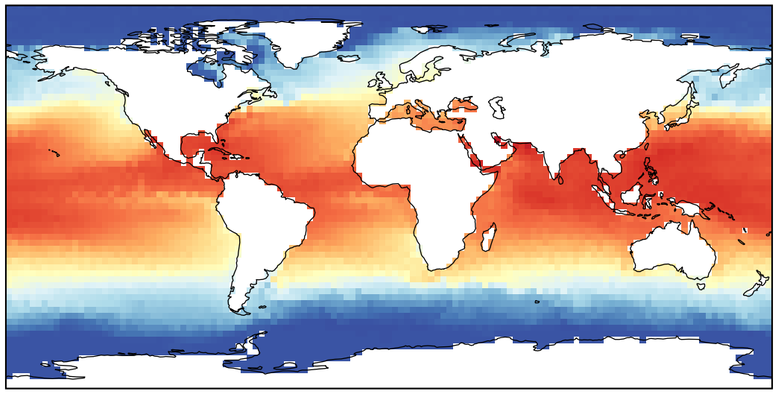


Chain init: 2020-07-09 — Test (°C)

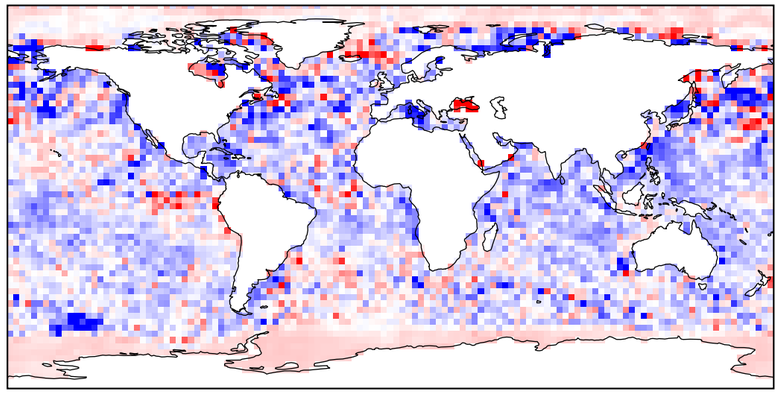
Actual — Step 1 (°C)



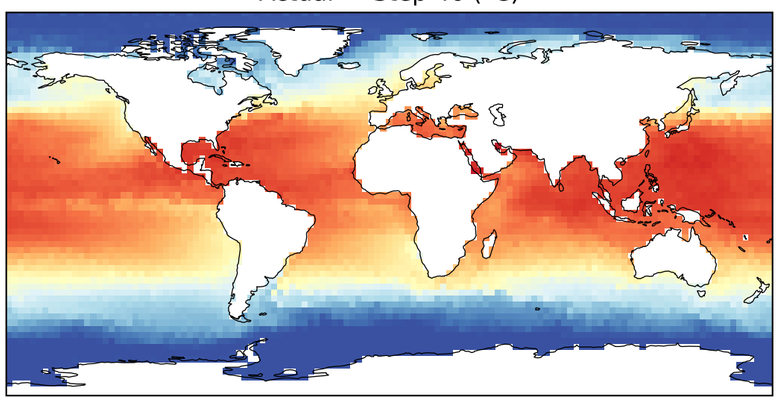
SFNO — Step 1 (°C)



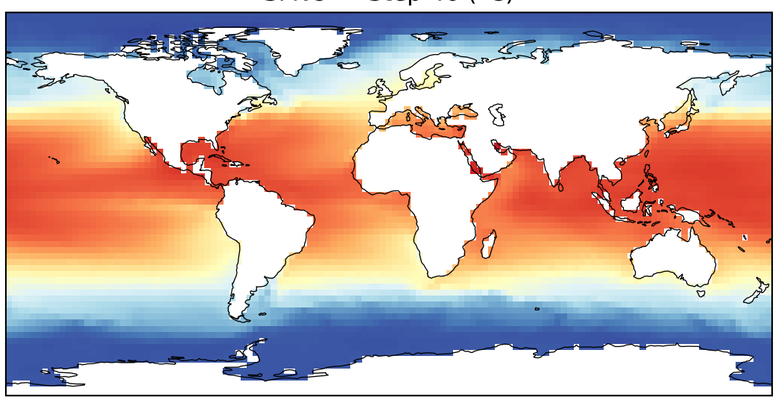
Error — Step 1 (°C)



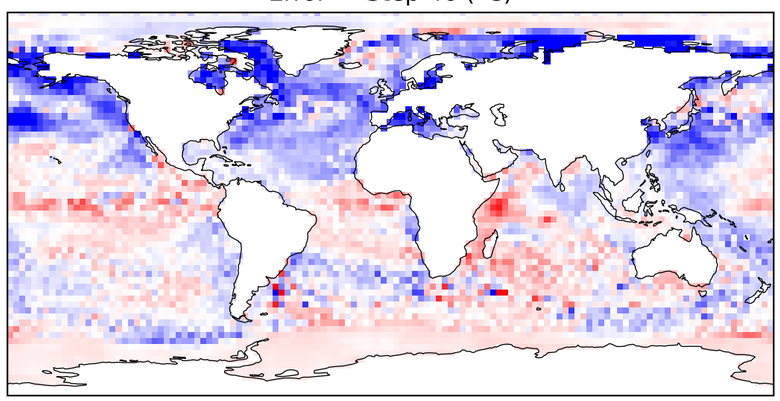
Actual — Step 46 (°C)



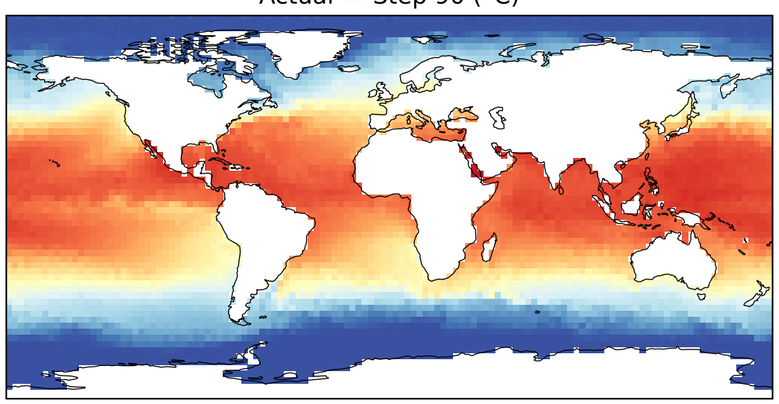
SFNO — Step 46 (°C)



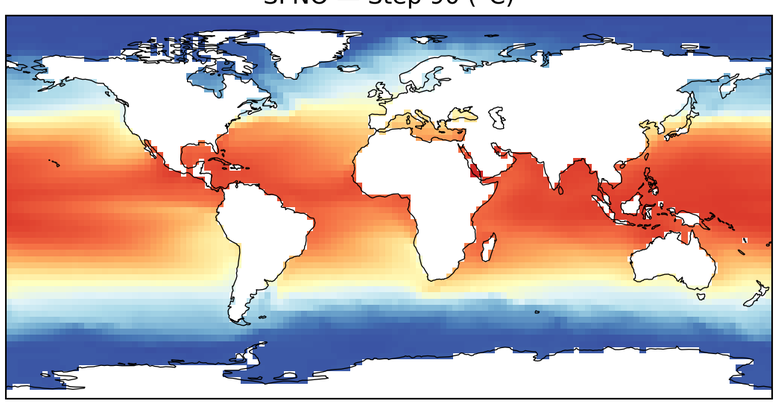
Error — Step 46 (°C)



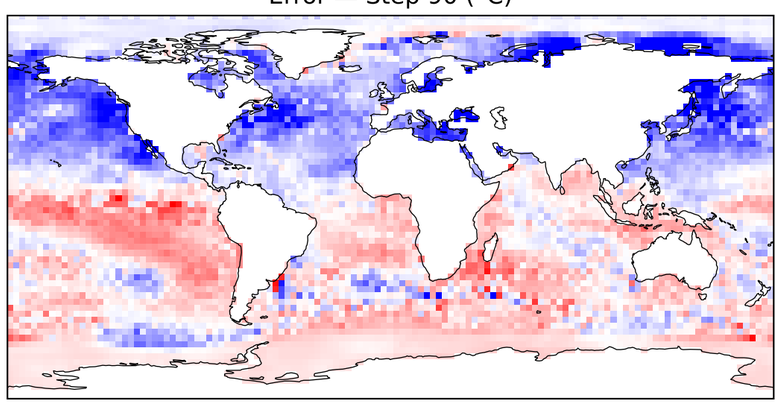
Actual — Step 90 (°C)

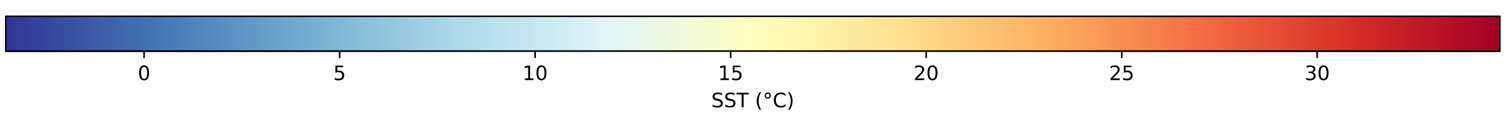


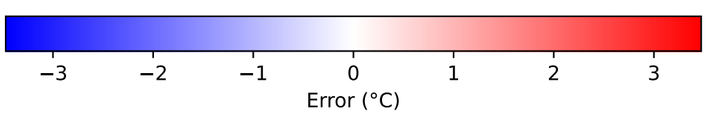
SFNO — Step 90 (°C)



Error — Step 90 (°C)

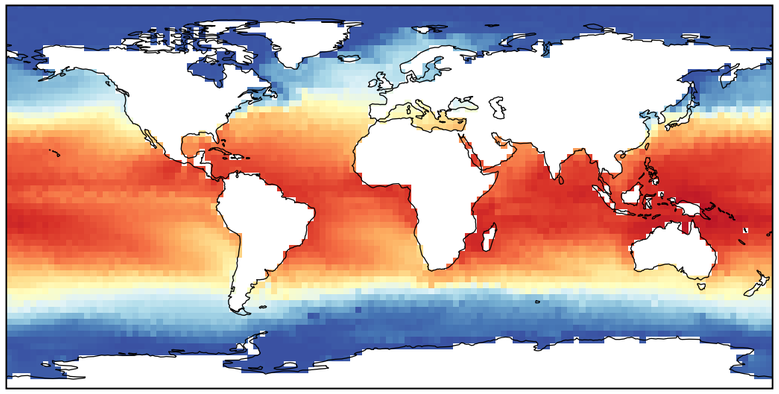




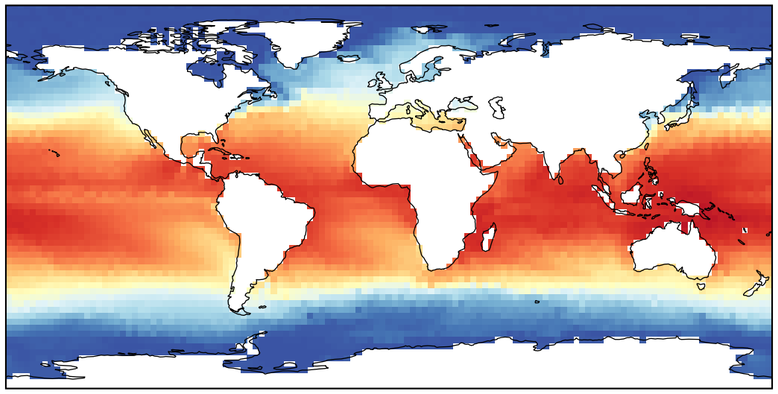


Chain init: 2021-01-06 — Test (°C)

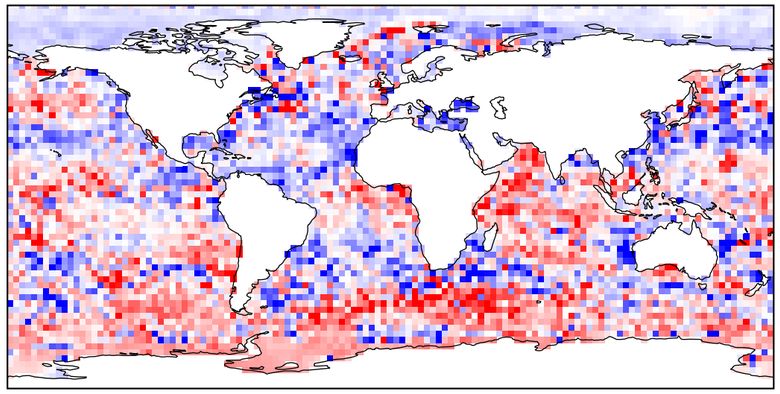
Actual — Step 1 (°C)



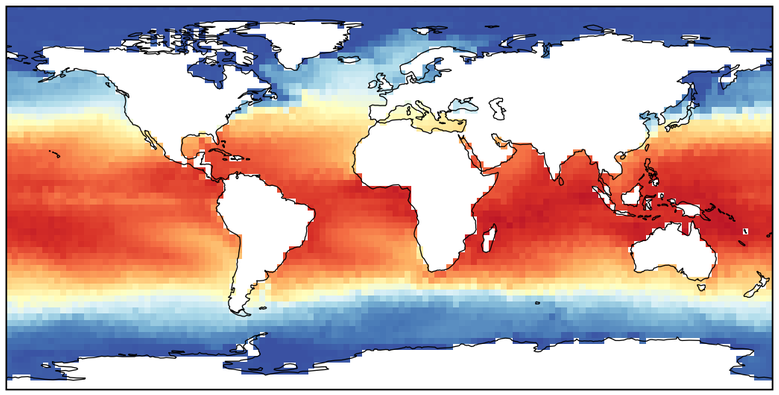
SFNO — Step 1 (°C)



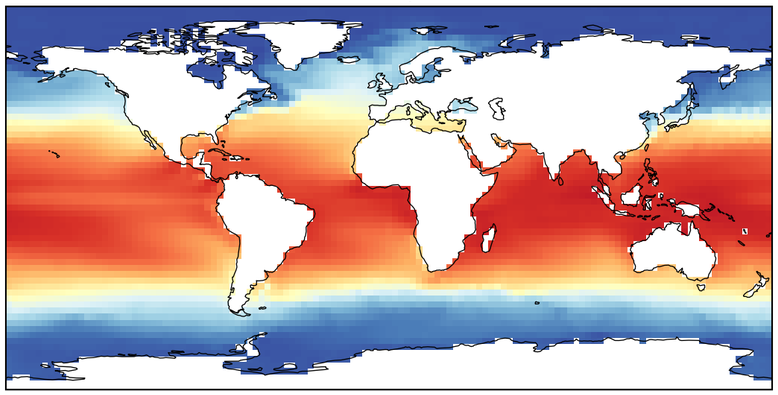
Error — Step 1 (°C)



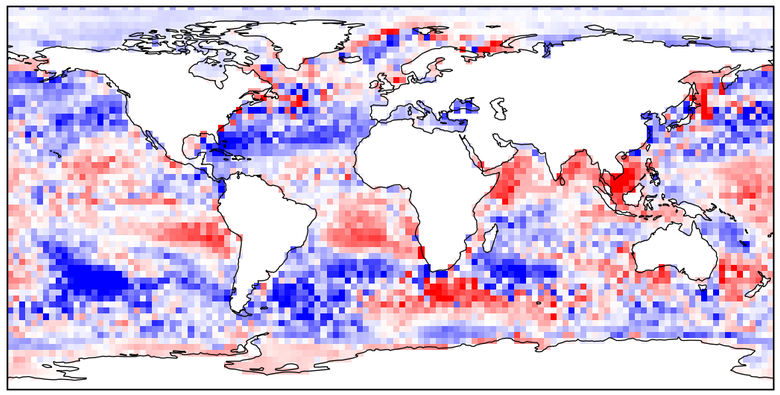
Actual — Step 46 (°C)



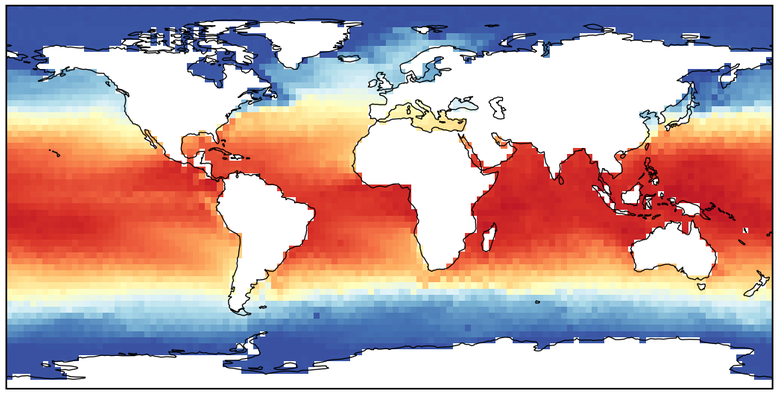
SFNO — Step 46 (°C)



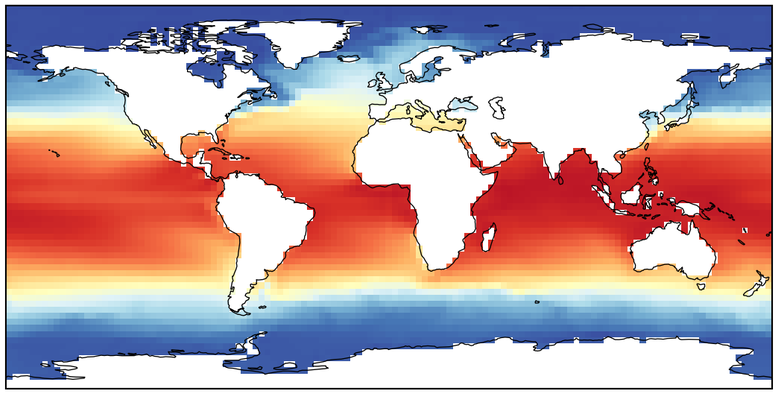
Error — Step 46 (°C)



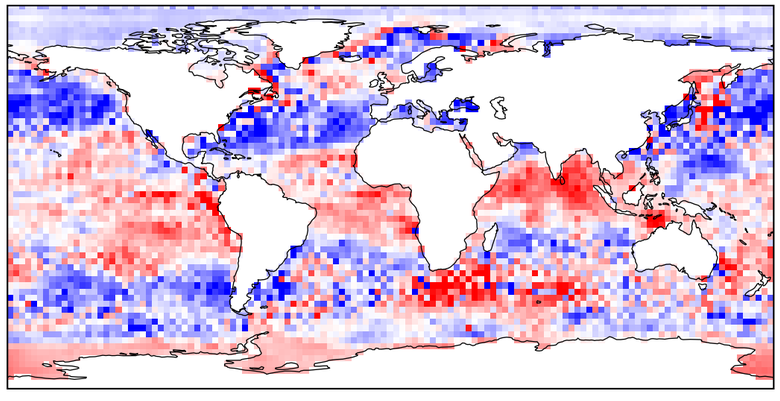
Actual — Step 90 (°C)

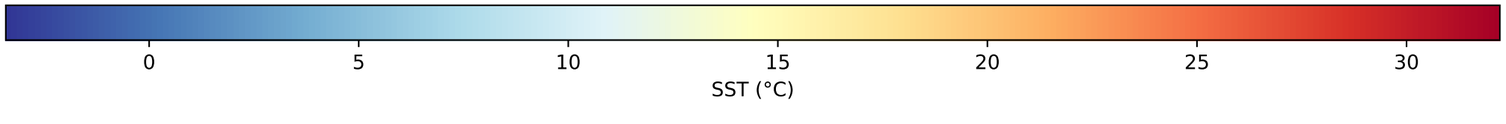


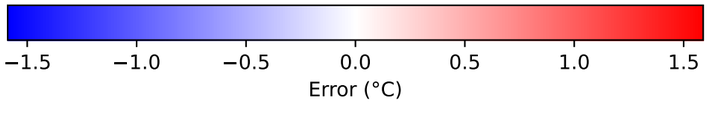
SFNO — Step 90 (°C)



Error — Step 90 (°C)

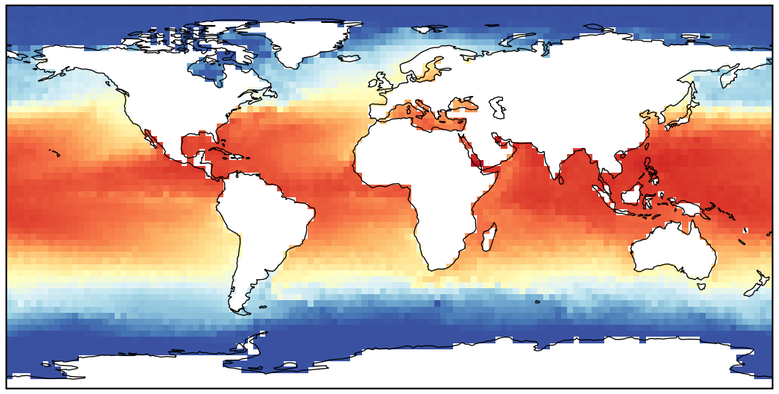




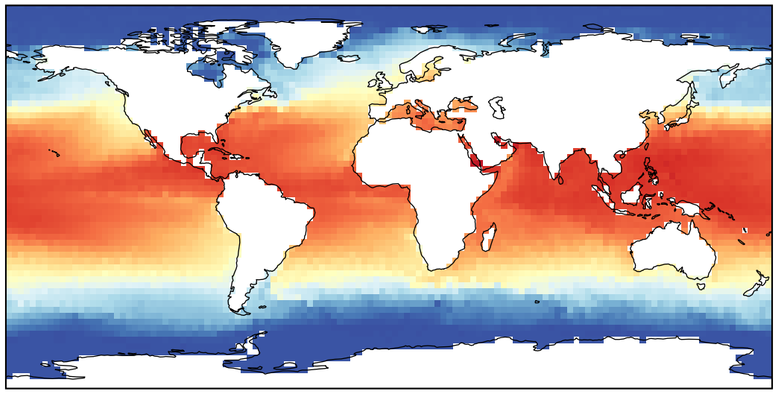


Chain init: 2021-07-05 — Test (°C)

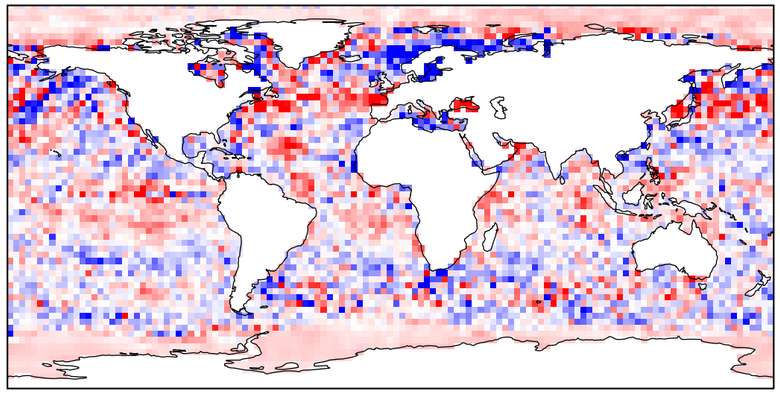
Actual — Step 1 (°C)



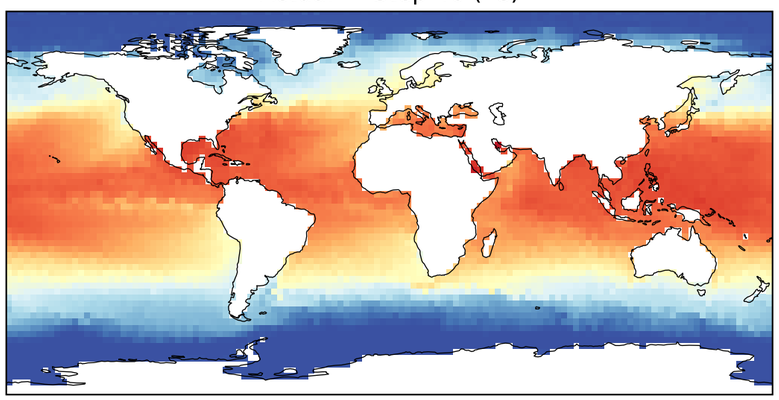
SFNO — Step 1 (°C)



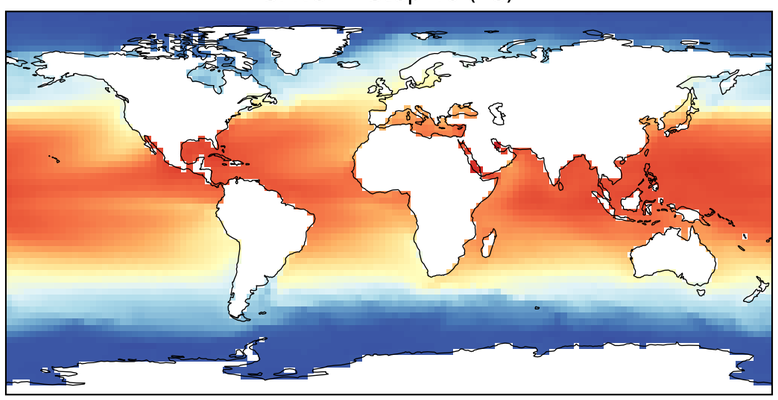
Error — Step 1 (°C)



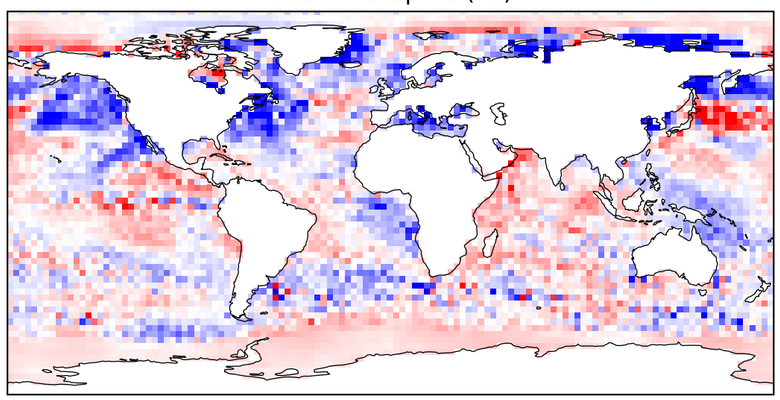
Actual — Step 46 (°C)



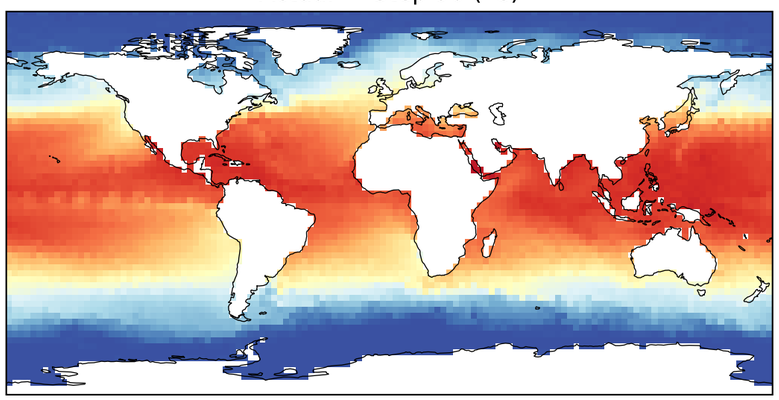
SFNO — Step 46 (°C)



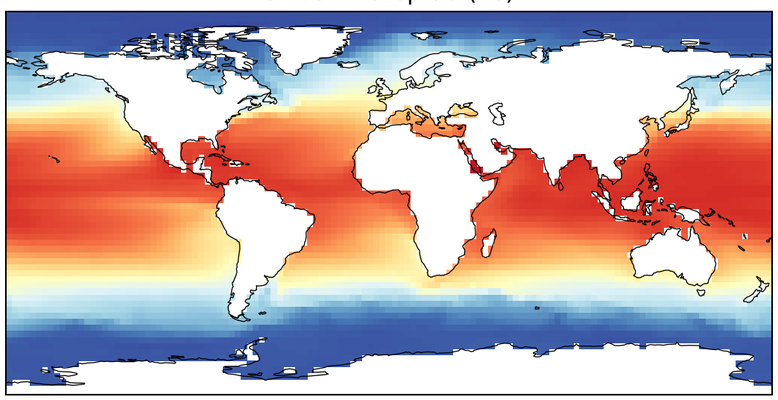
Error — Step 46 (°C)



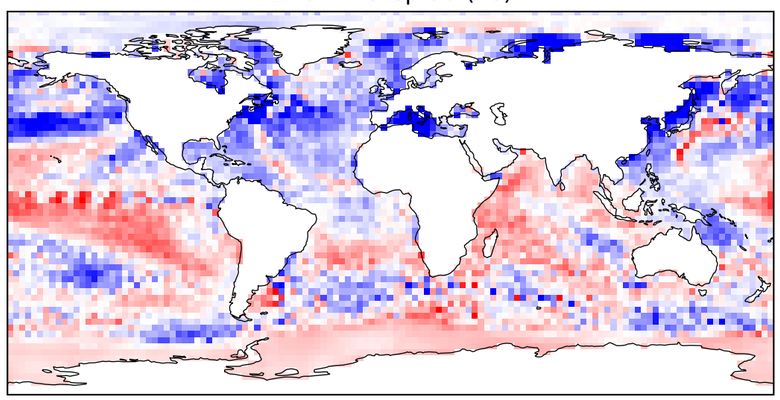
Actual — Step 90 (°C)

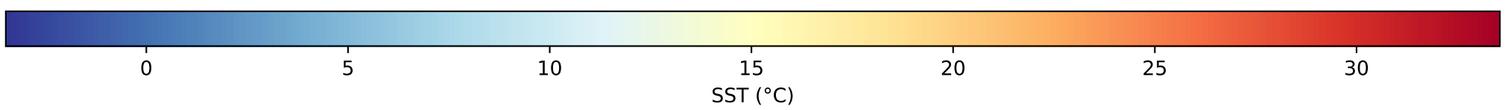


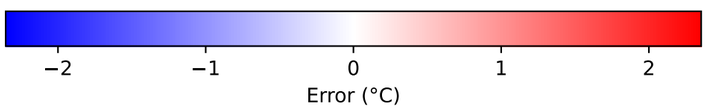
SFNO — Step 90 (°C)



Error — Step 90 (°C)

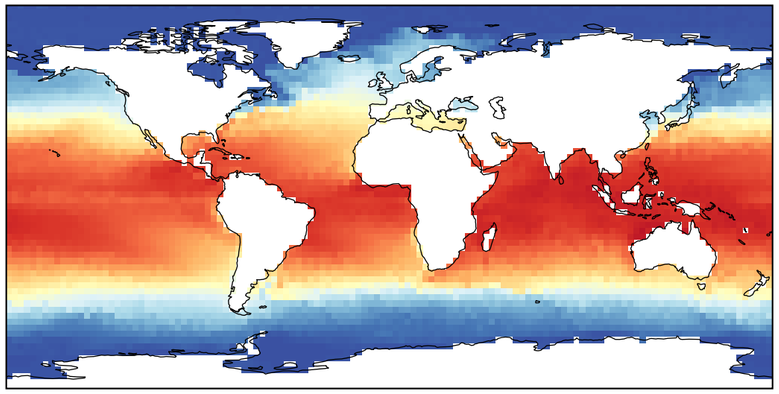




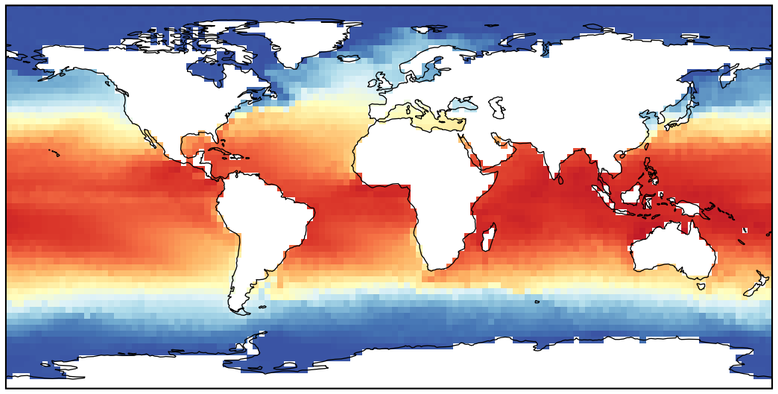


Chain init: 2022-04-01 — Test (°C)

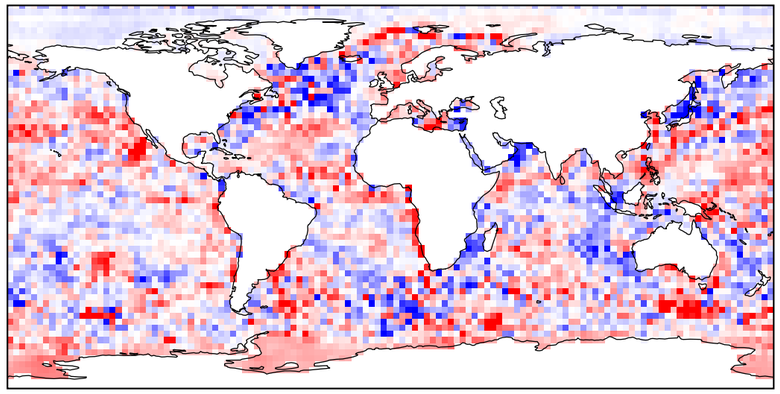
Actual — Step 1 (°C)



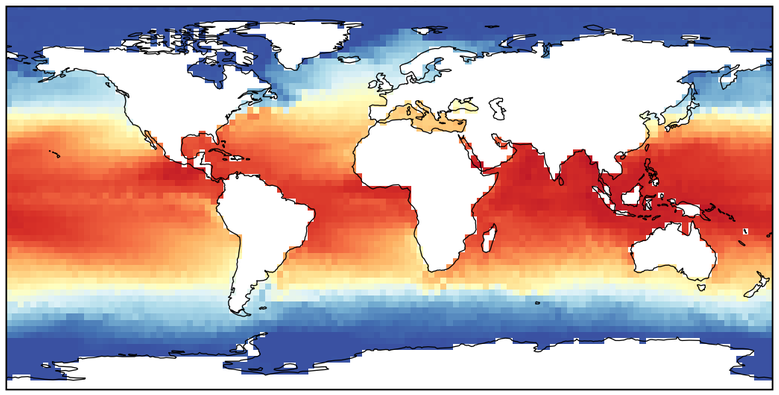
SFNO — Step 1 (°C)



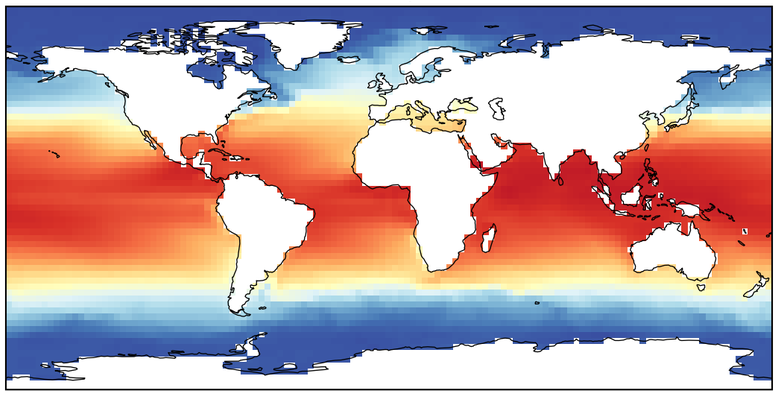
Error — Step 1 (°C)



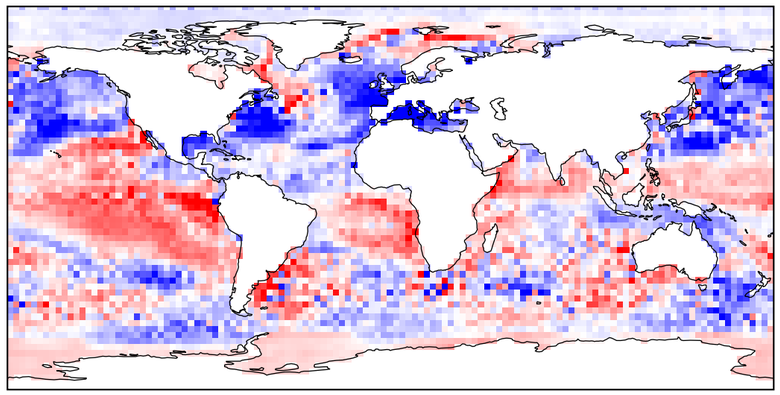
Actual — Step 46 (°C)



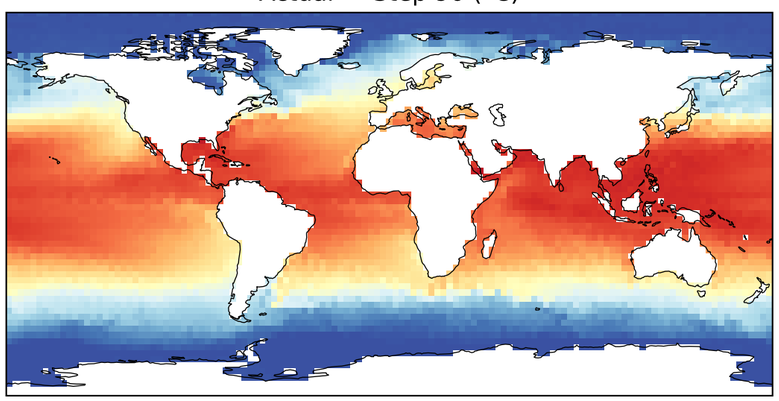
SFNO — Step 46 (°C)



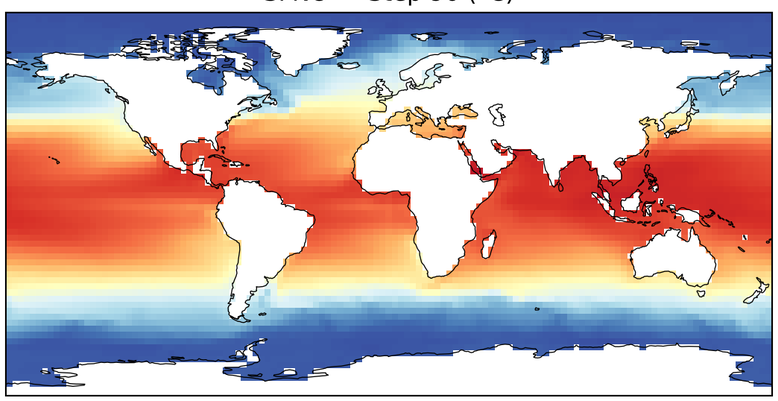
Error — Step 46 (°C)



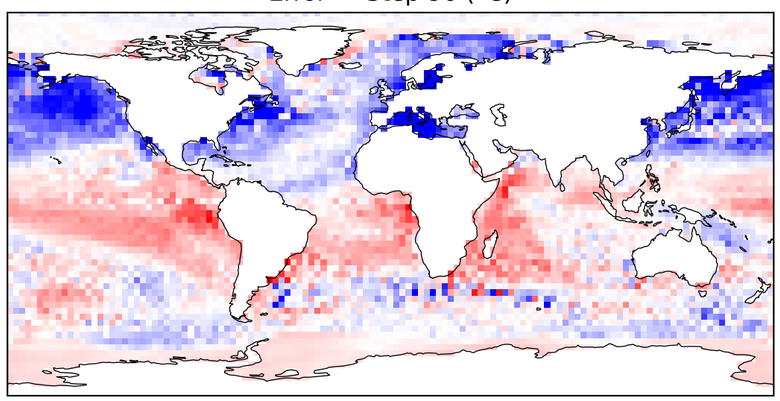
Actual — Step 90 (°C)

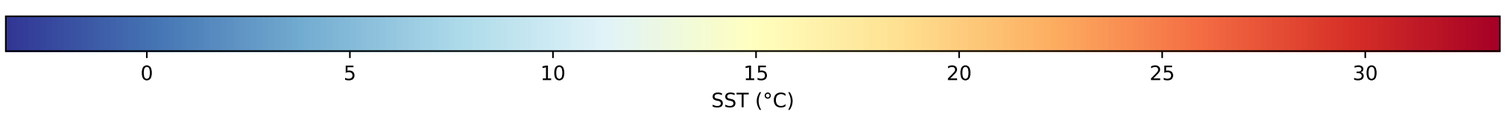


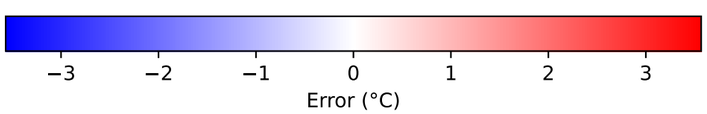
SFNO — Step 90 (°C)



Error — Step 90 (°C)

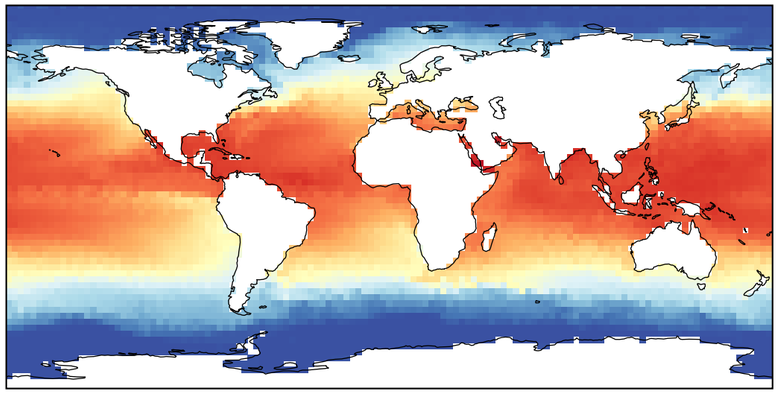




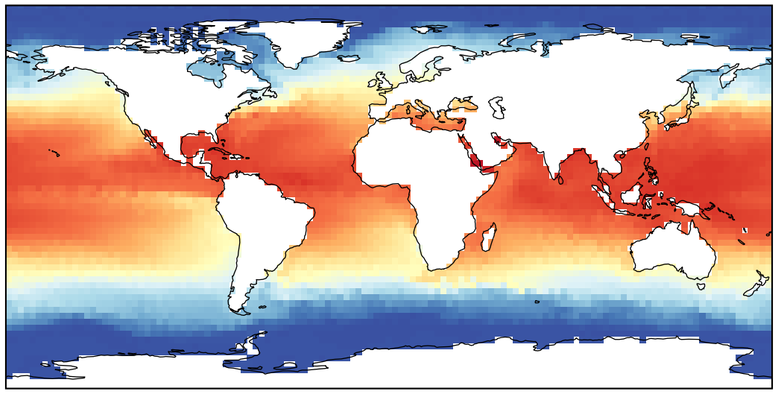


Chain init: 2022-09-28 — Test (°C)

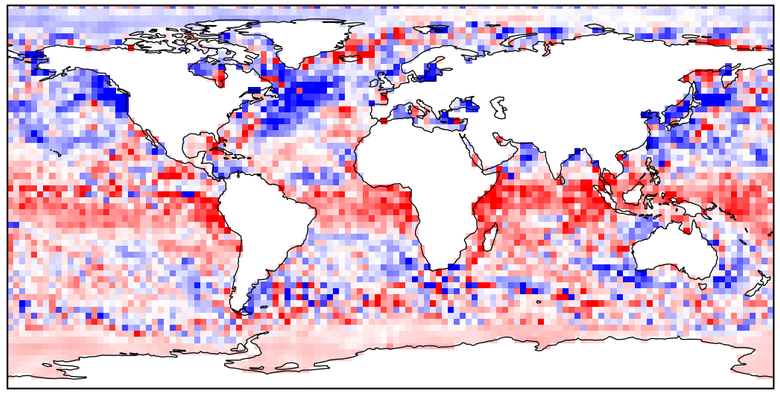
Actual — Step 1 (°C)



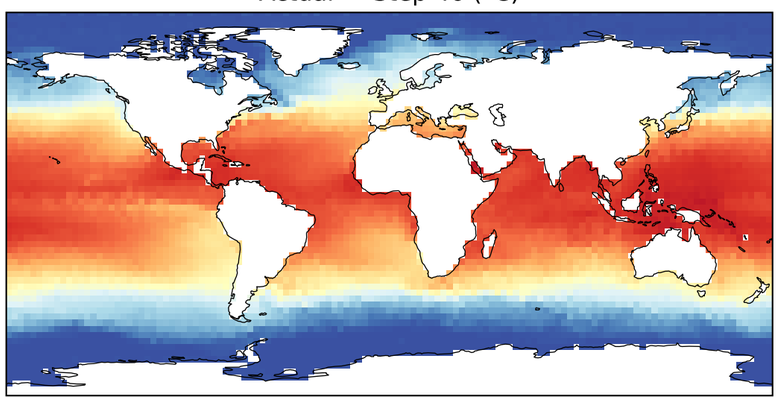
SFNO — Step 1 (°C)



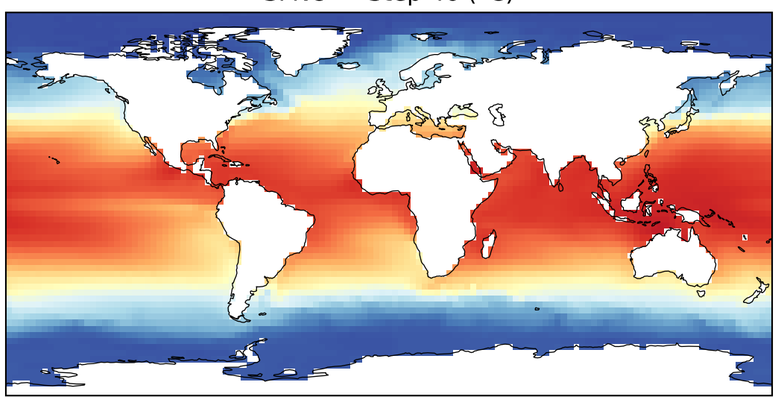
Error — Step 1 (°C)



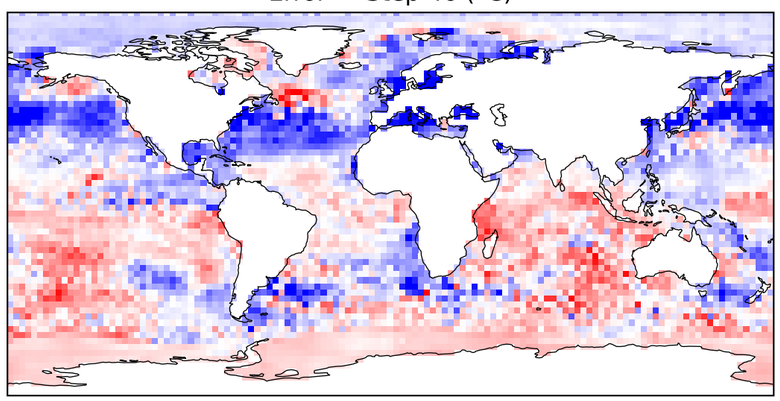
Actual — Step 46 (°C)



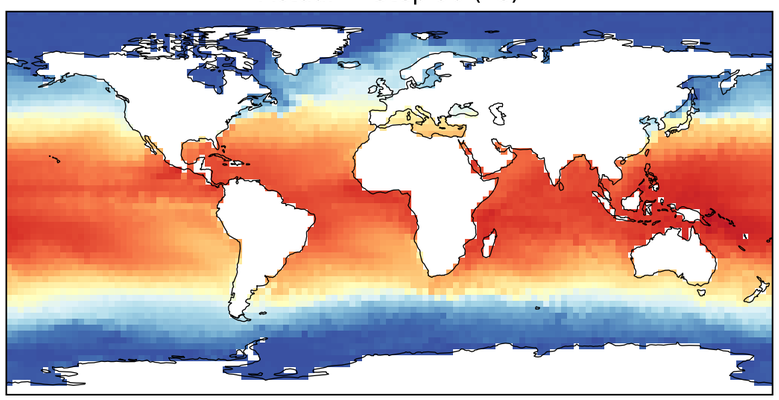
SFNO — Step 46 (°C)



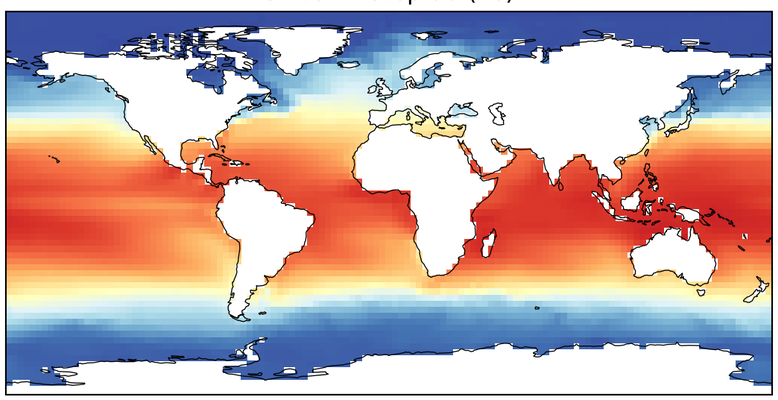
Error — Step 46 (°C)



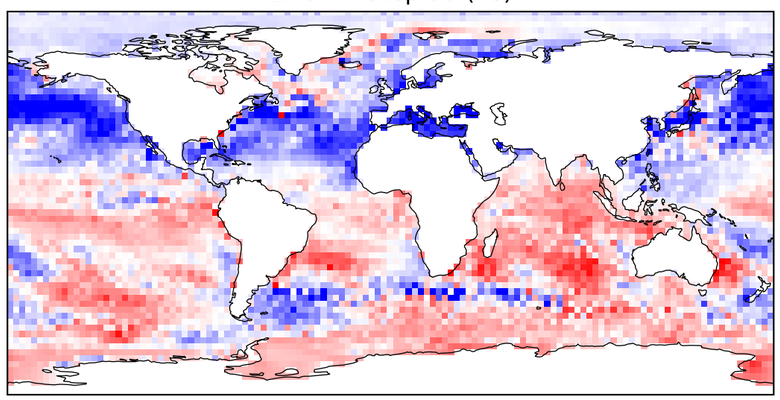
Actual — Step 90 (°C)

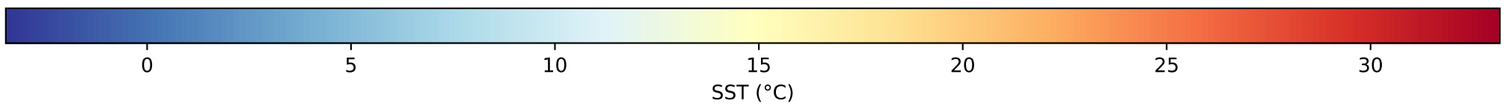


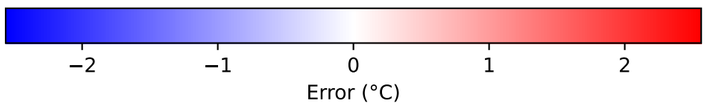
SFNO — Step 90 (°C)



Error — Step 90 (°C)

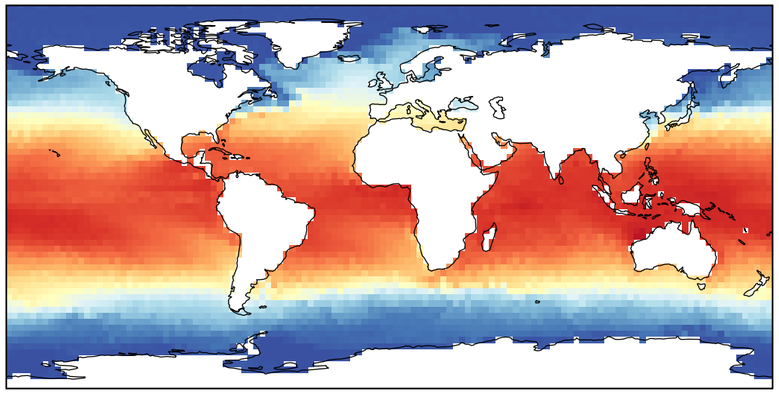




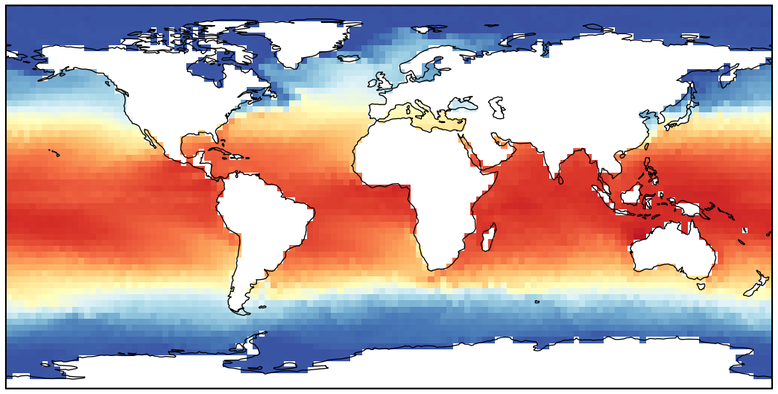


Chain init: 2023-03-27 — Test (°C)

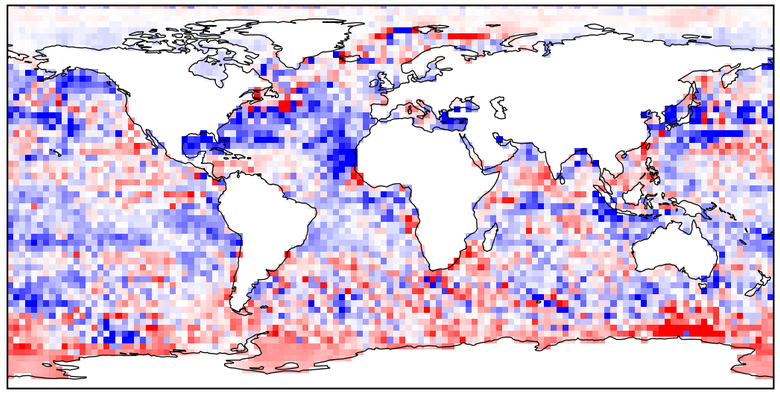
Actual — Step 1 (°C)



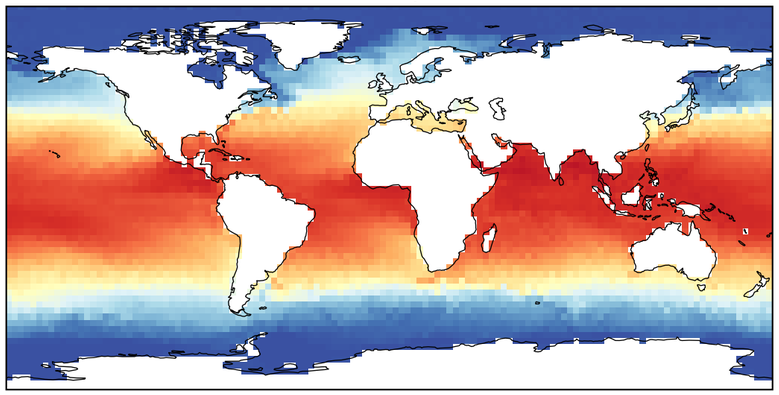
SFNO — Step 1 (°C)



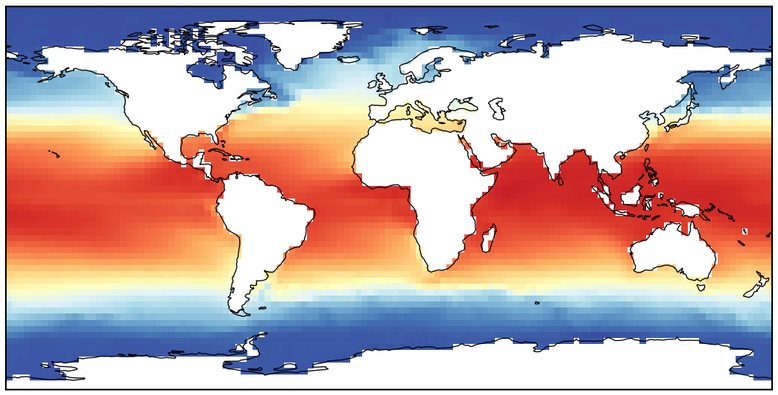
Error — Step 1 (°C)



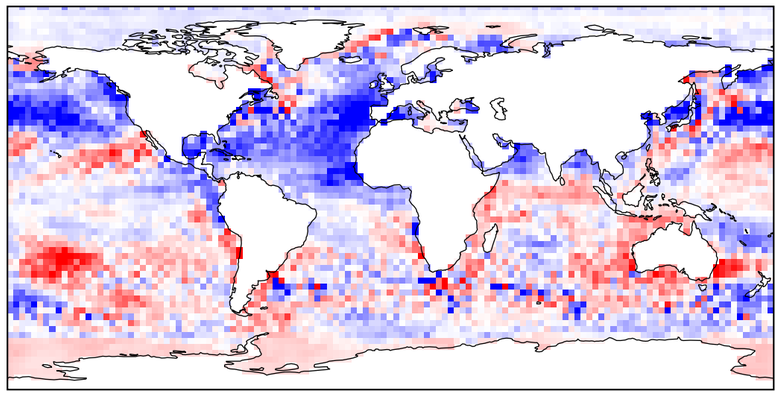
Actual — Step 46 (°C)



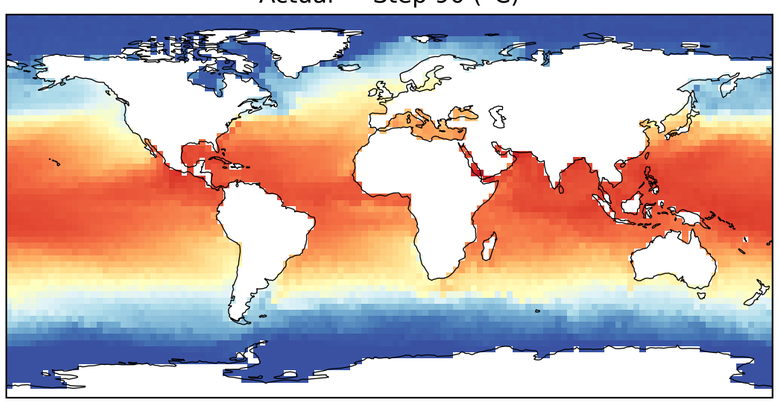
SFNO — Step 46 (°C)



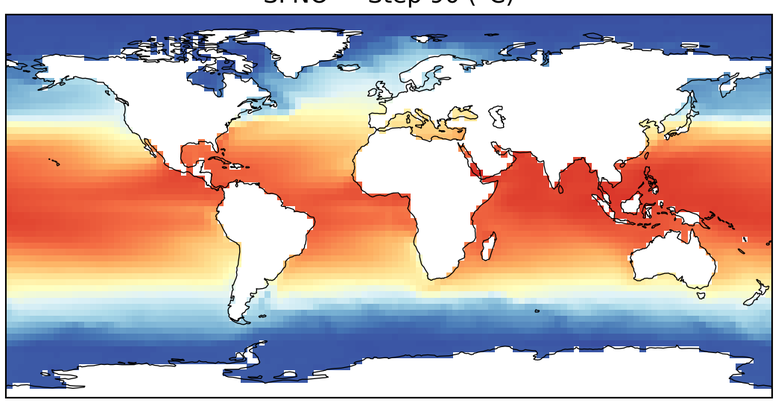
Error — Step 46 (°C)



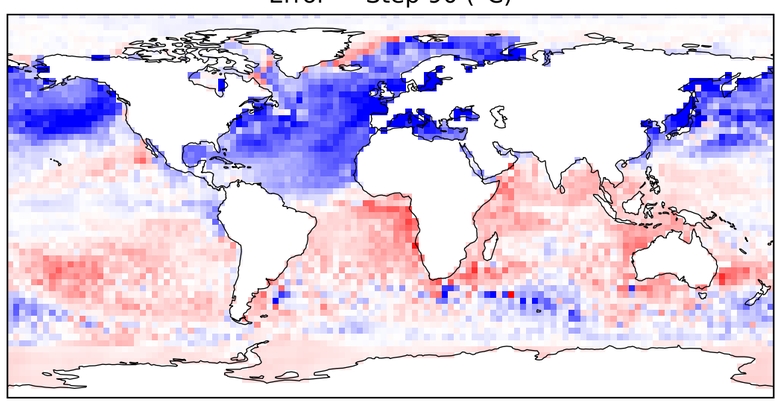
Actual — Step 90 (°C)

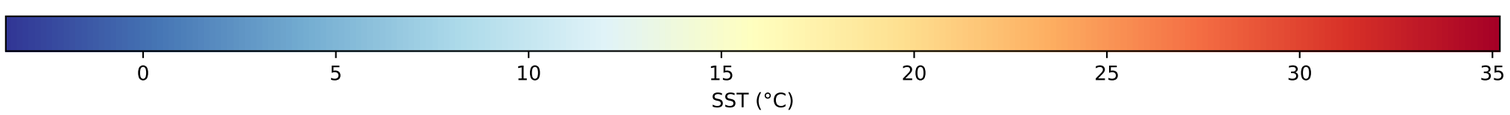


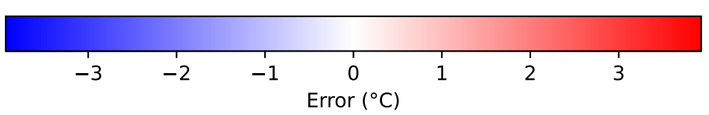
SFNO — Step 90 (°C)



Error — Step 90 (°C)

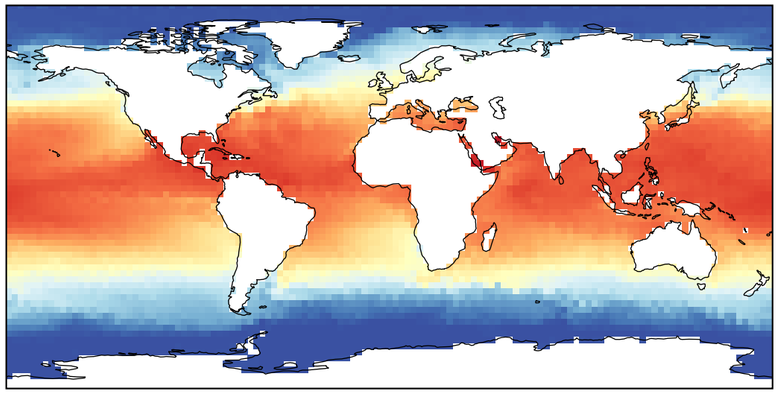




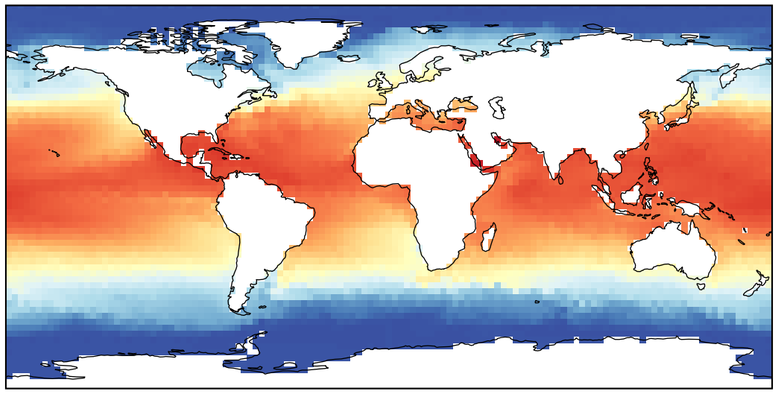


Chain init: 2023-09-23 — Test (°C)

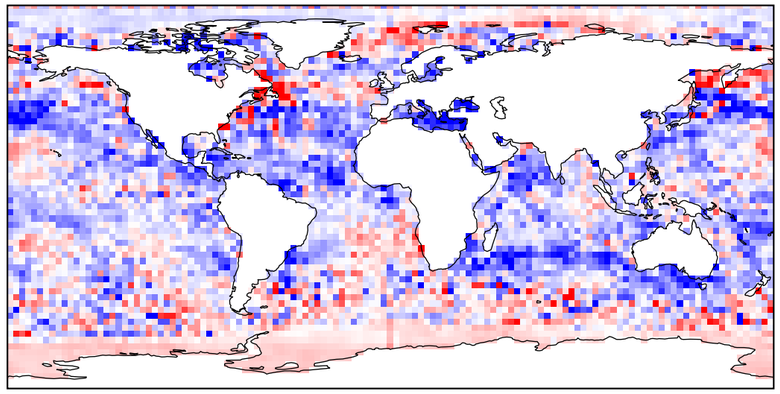
Actual — Step 1 (°C)



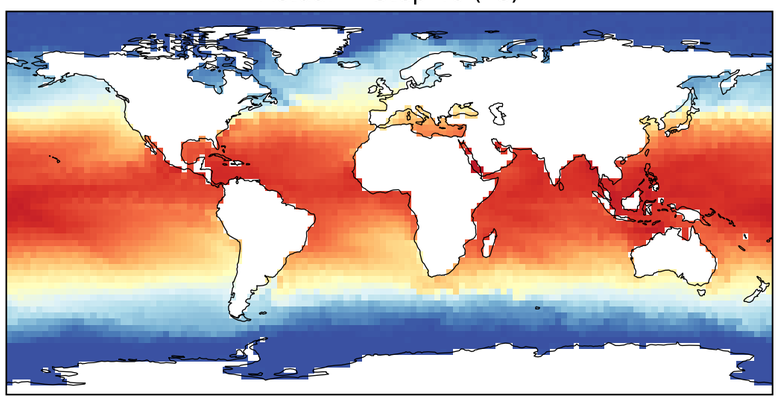
SFNO — Step 1 (°C)



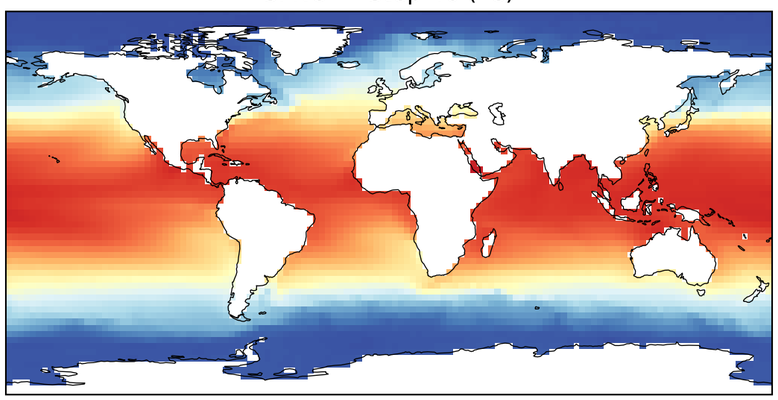
Error — Step 1 (°C)



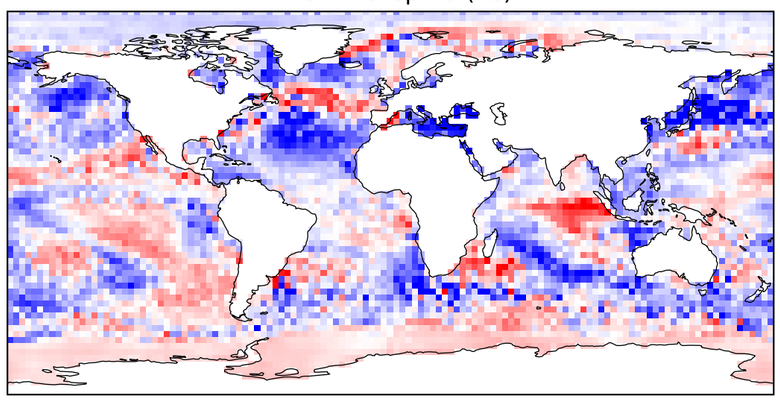
Actual — Step 46 (°C)



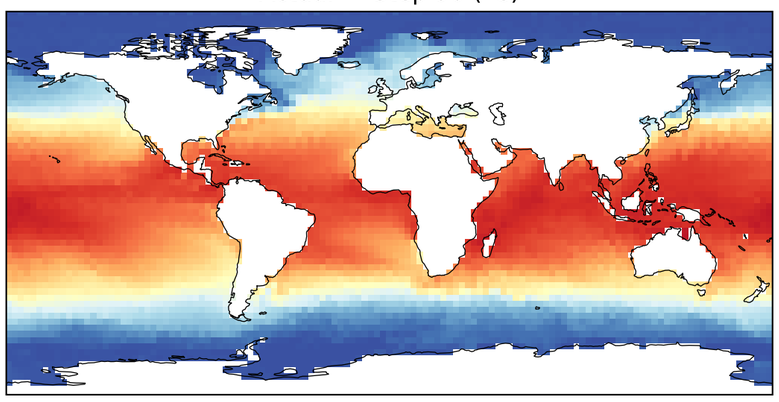
SFNO — Step 46 (°C)



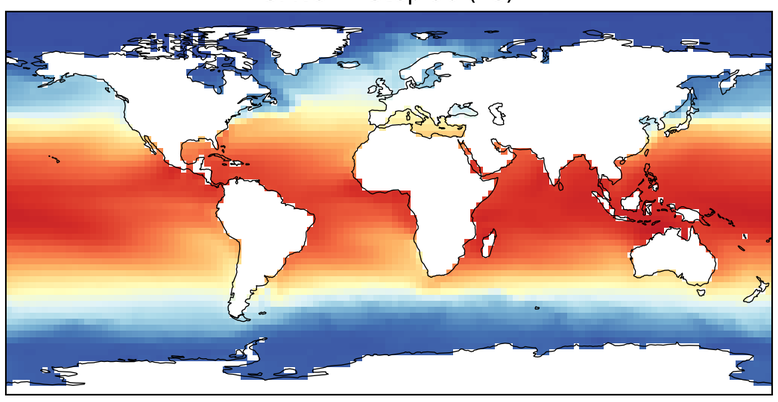
Error — Step 46 (°C)



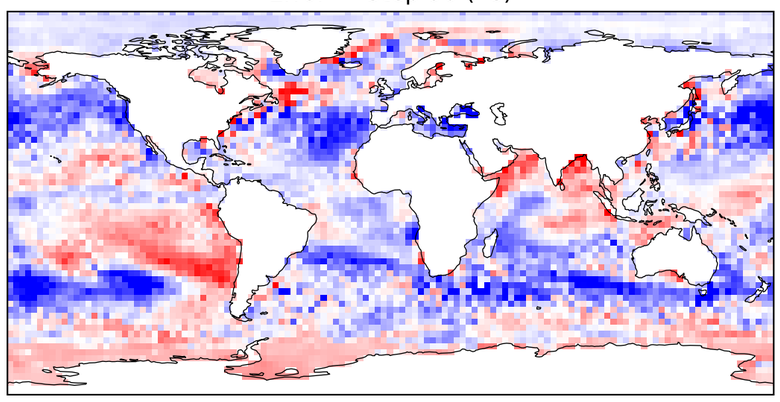
Actual — Step 90 (°C)

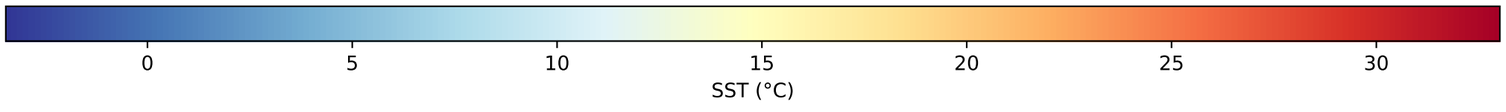


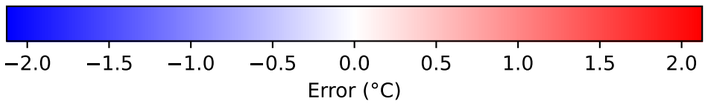
SFNO — Step 90 (°C)



Error — Step 90 (°C)

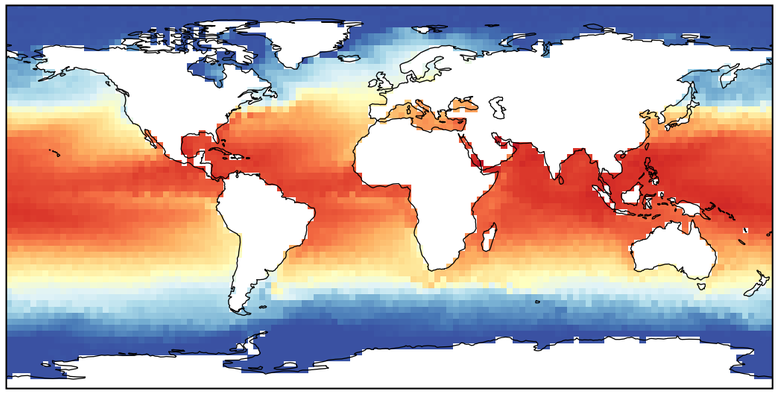




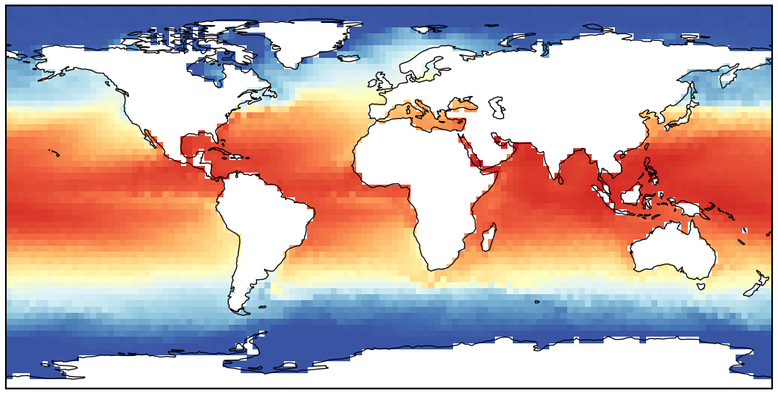


Chain init: 2024-06-19 — Test (°C)

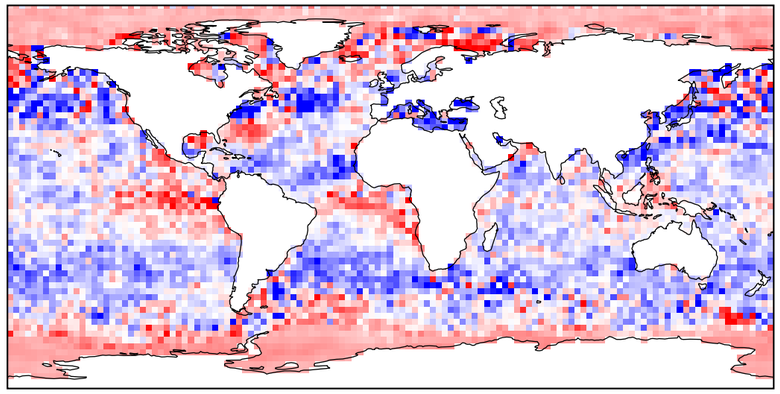
Actual — Step 1 (°C)



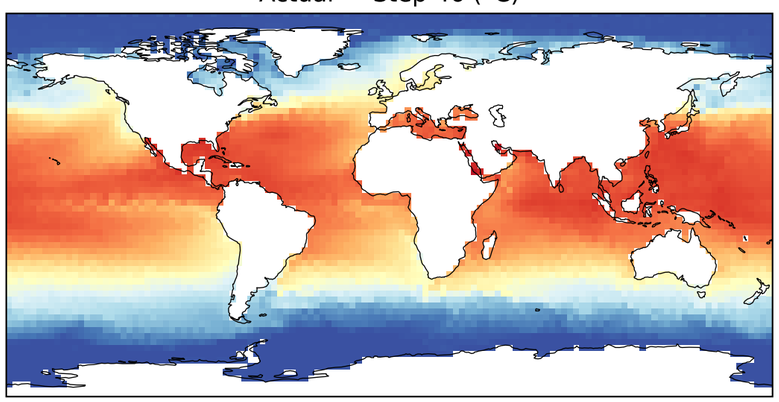
SFNO — Step 1 (°C)



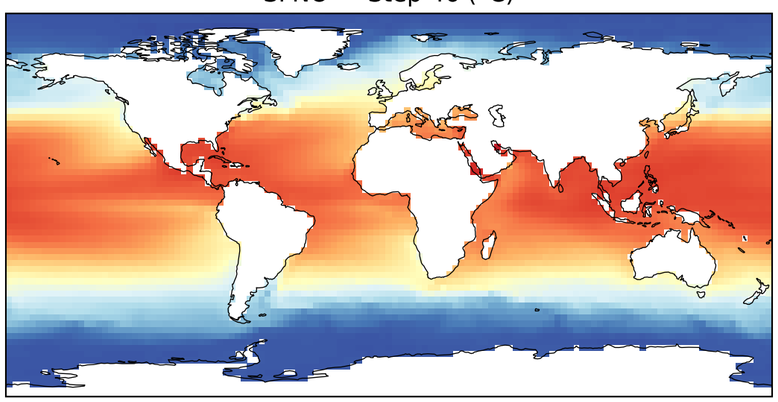
Error — Step 1 (°C)



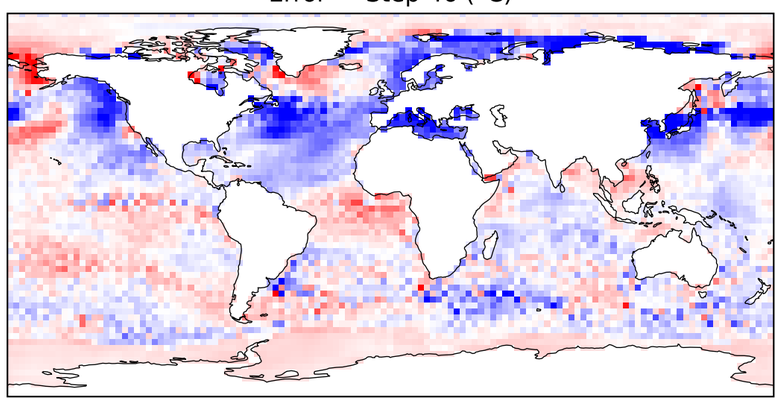
Actual — Step 46 (°C)



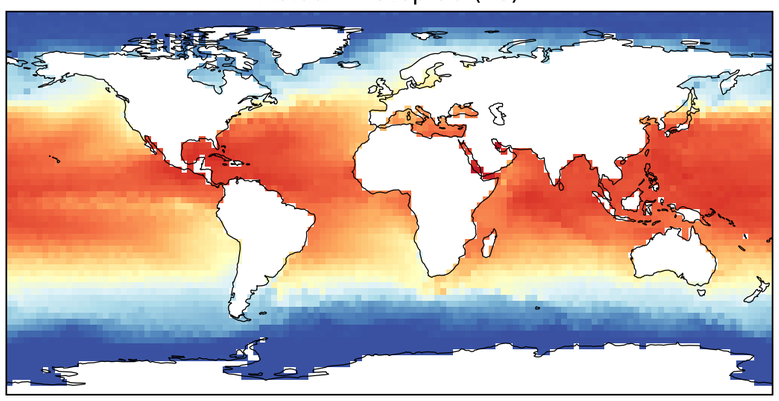
SFNO — Step 46 (°C)



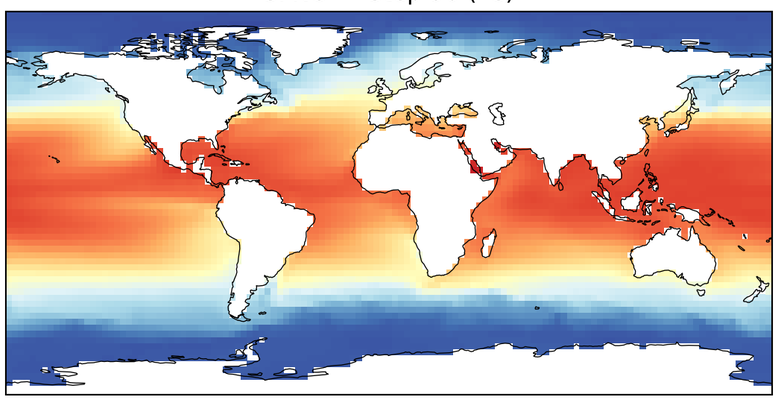
Error — Step 46 (°C)



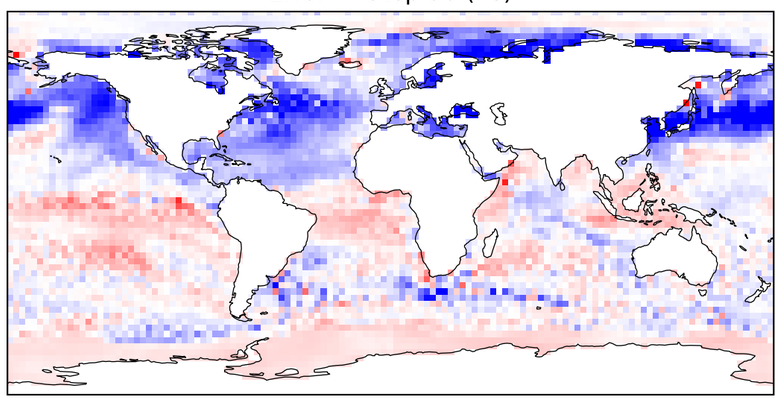
Actual — Step 90 (°C)

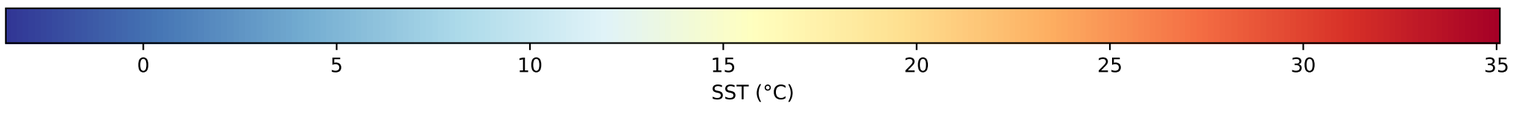


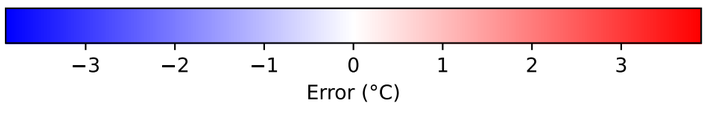
SFNO — Step 90 (°C)



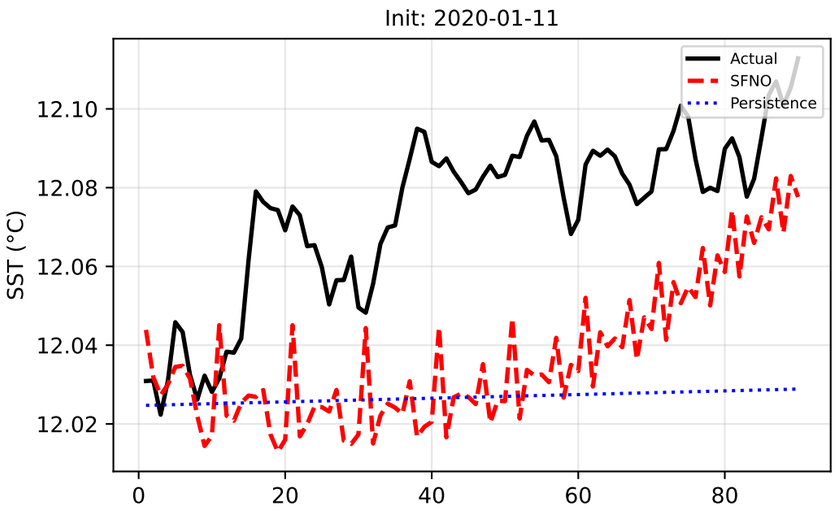
Error — Step 90 (°C)

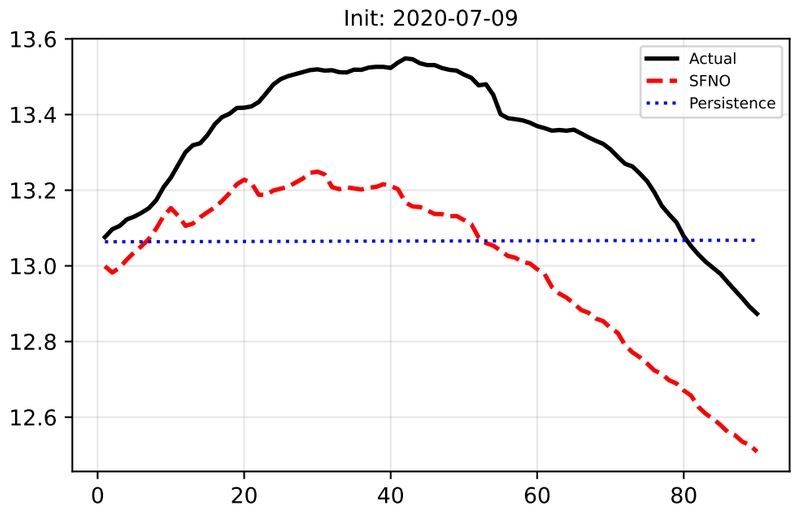


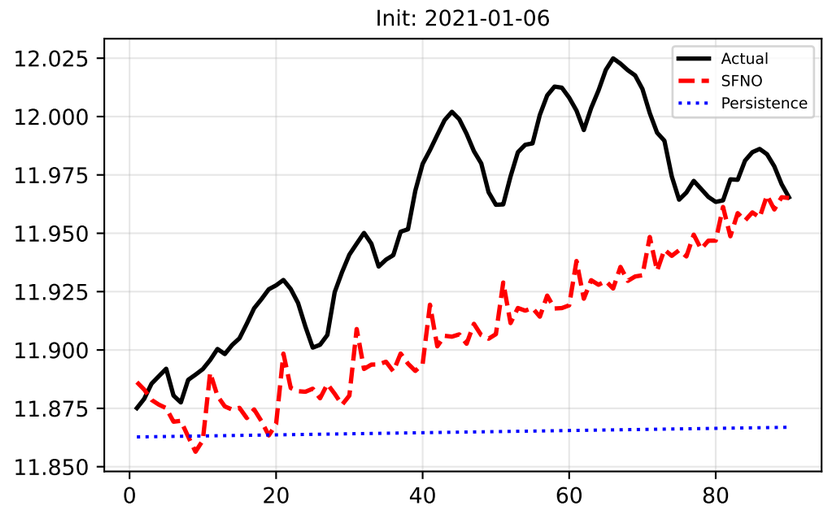


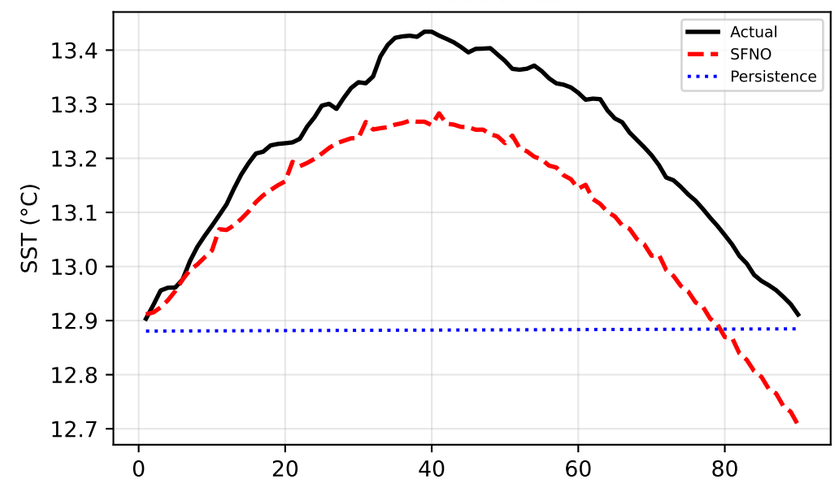


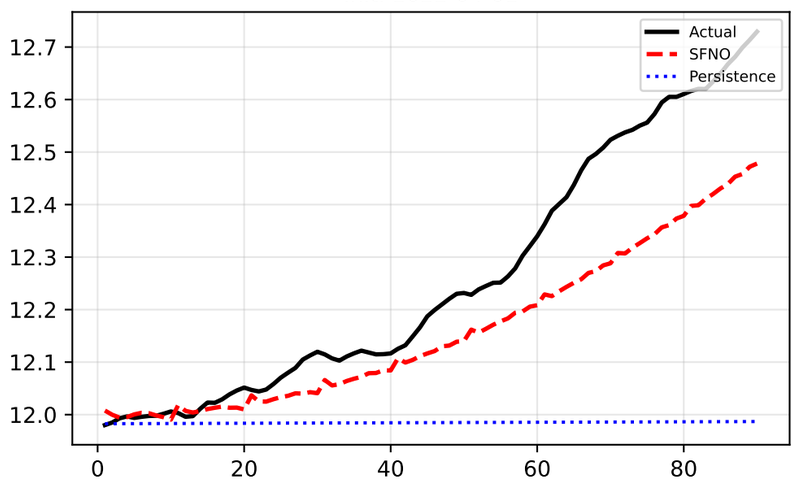
Global-Mean SST — Test (°C)

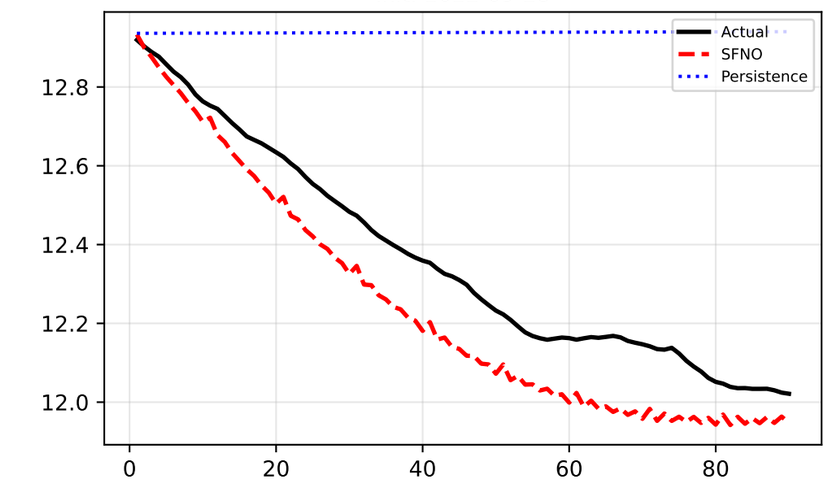


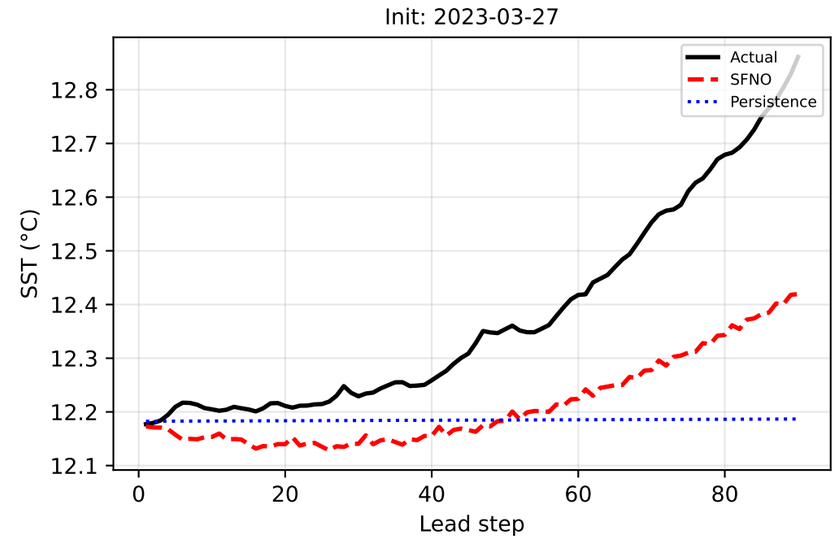


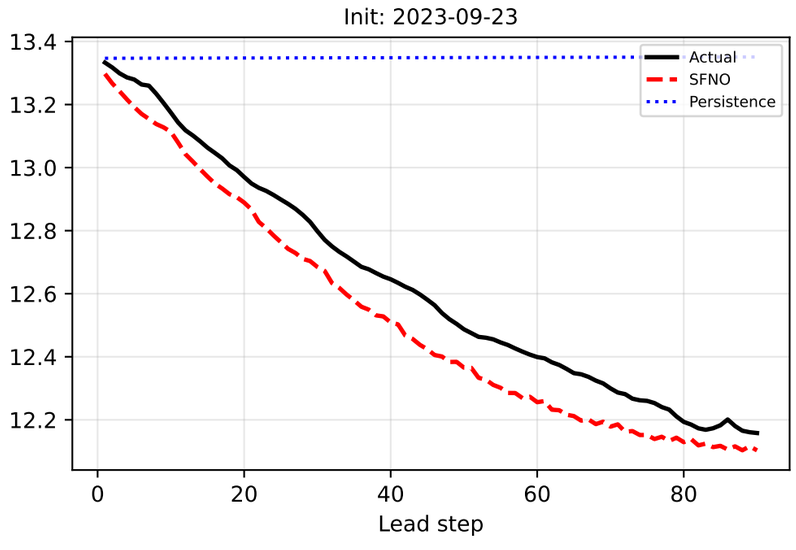


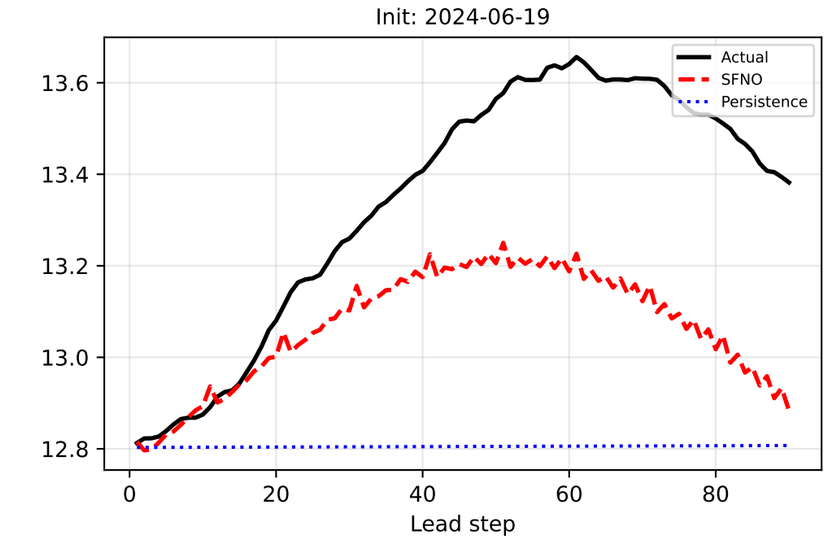






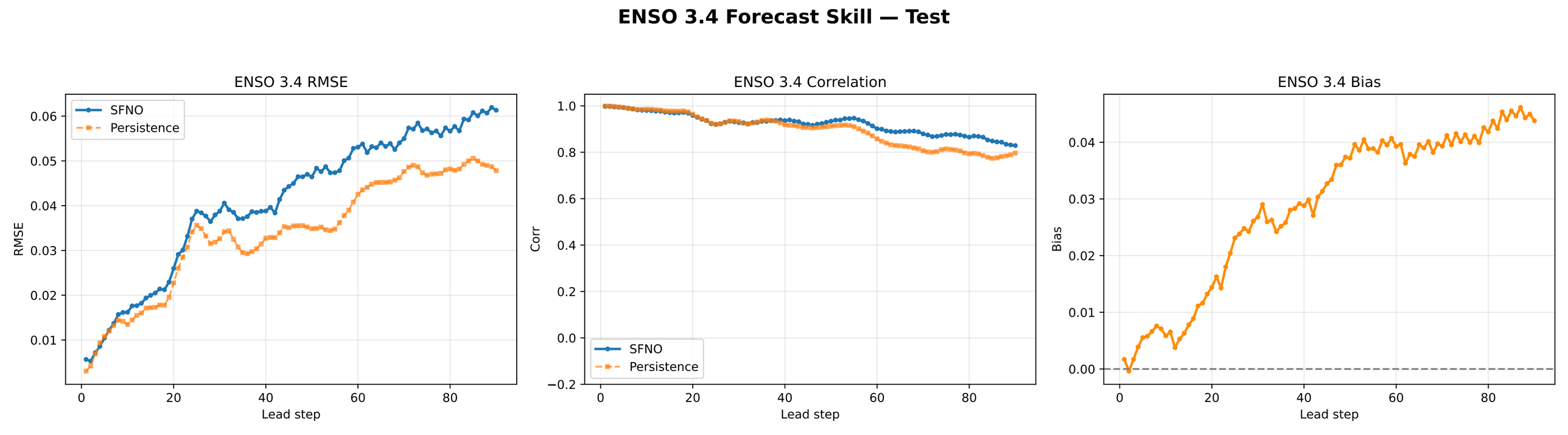




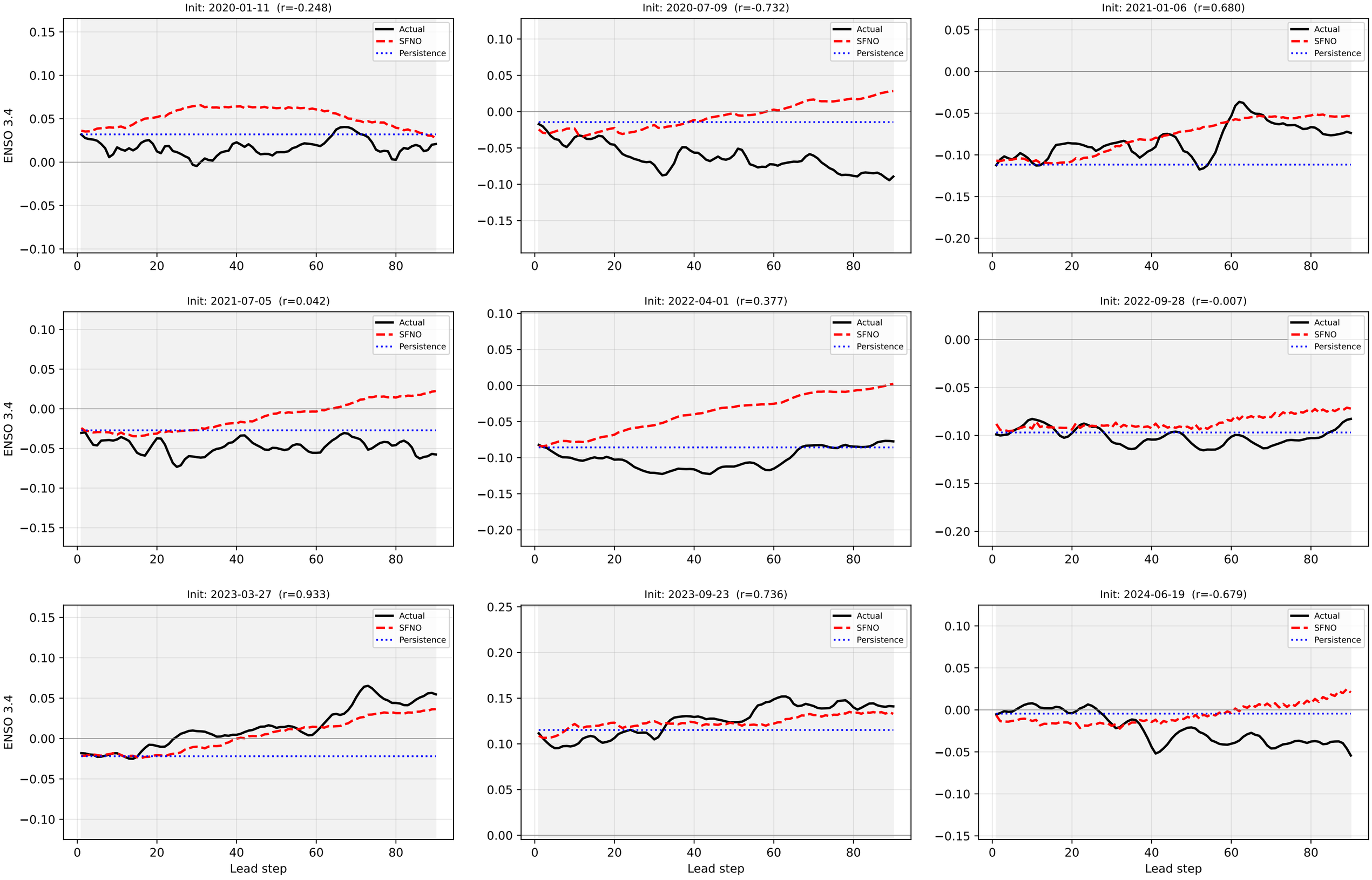


ENSO 3.4 Metrics — Test

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0057	0.0031	0.9978	0.9992	0.0017
2	0.0053	0.0042	0.9975	0.9989	-0.0004
3	0.0072	0.0069	0.9953	0.9972	0.0017
4	0.0086	0.0093	0.9944	0.9950	0.0039
5	0.0105	0.0108	0.9925	0.9927	0.0055
6	0.0122	0.0120	0.9891	0.9896	0.0058
7	0.0137	0.0133	0.9864	0.9867	0.0066
8	0.0157	0.0144	0.9824	0.9842	0.0076
9	0.0161	0.0142	0.9810	0.9843	0.0071
10	0.0162	0.0135	0.9802	0.9857	0.0059
11	0.0176	0.0145	0.9794	0.9850	0.0065
12	0.0177	0.0155	0.9774	0.9833	0.0038
13	0.0182	0.0160	0.9768	0.9817	0.0053
14	0.0194	0.0171	0.9729	0.9784	0.0063
15	0.0200	0.0172	0.9710	0.9772	0.0078
16	0.0205	0.0173	0.9693	0.9774	0.0089
17	0.0214	0.0178	0.9699	0.9770	0.0111
18	0.0212	0.0177	0.9717	0.9777	0.0116
19	0.0229	0.0195	0.9685	0.9736	0.0132
20	0.0260	0.0227	0.9590	0.9644	0.0144
21	0.0291	0.0260	0.9504	0.9526	0.0163
22	0.0301	0.0285	0.9421	0.9440	0.0143
23	0.0332	0.0308	0.9370	0.9369	0.0180
24	0.0370	0.0341	0.9230	0.9251	0.0204
25	0.0388	0.0356	0.9203	0.9197	0.0231
26	0.0384	0.0349	0.9224	0.9226	0.0238
27	0.0377	0.0332	0.9292	0.9287	0.0248
28	0.0364	0.0315	0.9337	0.9351	0.0242
29	0.0379	0.0319	0.9312	0.9350	0.0261
30	0.0388	0.0326	0.9279	0.9322	0.0268
31	0.0406	0.0342	0.9270	0.9251	0.0290
32	0.0391	0.0344	0.9226	0.9217	0.0260
33	0.0385	0.0325	0.9282	0.9261	0.0263
34	0.0371	0.0308	0.9280	0.9305	0.0242
35	0.0371	0.0295	0.9332	0.9367	0.0252
36	0.0376	0.0293	0.9335	0.9386	0.0258
37	0.0387	0.0297	0.9361	0.9370	0.0280
38	0.0385	0.0304	0.9371	0.9336	0.0283
39	0.0388	0.0314	0.9393	0.9265	0.0292
40	0.0388	0.0327	0.9366	0.9182	0.0288
41	0.0396	0.0329	0.9391	0.9161	0.0299
42	0.0384	0.0329	0.9341	0.9151	0.0271
43	0.0414	0.0339	0.9317	0.9113	0.0303
44	0.0435	0.0353	0.9216	0.9066	0.0314
45	0.0443	0.0351	0.9203	0.9069	0.0327
46	0.0450	0.0355	0.9162	0.9049	0.0334
47	0.0465	0.0355	0.9212	0.9067	0.0360
48	0.0465	0.0355	0.9237	0.9080	0.0360
49	0.0470	0.0352	0.9293	0.9099	0.0374
50	0.0464	0.0349	0.9338	0.9127	0.0372
51	0.0484	0.0349	0.9385	0.9148	0.0396
52	0.0476	0.0352	0.9385	0.9157	0.0386
53	0.0487	0.0346	0.9451	0.9176	0.0405
54	0.0473	0.0345	0.9450	0.9155	0.0388
55	0.0474	0.0348	0.9469	0.9104	0.0389
56	0.0478	0.0362	0.9400	0.9010	0.0382
57	0.0501	0.0378	0.9346	0.8912	0.0403
58	0.0506	0.0390	0.9242	0.8831	0.0395
59	0.0528	0.0408	0.9132	0.8706	0.0407
60	0.0531	0.0425	0.9016	0.8575	0.0393
61	0.0538	0.0435	0.8993	0.8473	0.0396
62	0.0519	0.0441	0.8920	0.8398	0.0363
63	0.0532	0.0448	0.8895	0.8324	0.0379
64	0.0529	0.0452	0.8873	0.8290	0.0375
65	0.0540	0.0452	0.8889	0.8279	0.0396
66	0.0532	0.0452	0.8898	0.8256	0.0390
67	0.0539	0.0453	0.8909	0.8230	0.0401
68	0.0525	0.0457	0.8913	0.8187	0.0382
69	0.0540	0.0462	0.8882	0.8146	0.0397
70	0.0550	0.0476	0.8793	0.8065	0.0393
71	0.0574	0.0486	0.8745	0.8014	0.0412
72	0.0571	0.0490	0.8676	0.8005	0.0396
73	0.0585	0.0487	0.8687	0.8030	0.0415
74	0.0568	0.0474	0.8722	0.8113	0.0401
75	0.0571	0.0468	0.8768	0.8146	0.0413
76	0.0563	0.0470	0.8761	0.8124	0.0400
77	0.0567	0.0471	0.8776	0.8097	0.0411
78	0.0556	0.0472	0.8743	0.8062	0.0399
79	0.0574	0.0480	0.8695	0.7976	0.0426
80	0.0566	0.0482	0.8652	0.7935	0.0418
81	0.0577	0.0479	0.8701	0.7956	0.0438
82	0.0567	0.0482	0.8682	0.7931	0.0424
83	0.0594	0.0492	0.8651	0.7855	0.0454
84	0.0592	0.0500	0.8529	0.7786	0.0439
85	0.0608	0.0506	0.8485	0.7738	0.0455
86	0.0601	0.0500	0.8447	0.7761	0.0446
87	0.0612	0.0492	0.8436	0.7818	0.0461
88	0.0607	0.0490	0.8345	0.7847	0.0443
89	0.0619	0.0487	0.8316	0.7888	0.0450
90	0.0613	0.0478	0.8288	0.7967	0.0438


ENSO 3.4 Forecast Skill — Test
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

ENSO 3.4 Index — Test



ENSO 3.4 Scatter — Test